MA'ADEN معادن
Saudi Arabian Mining Company شركة التعدين العربية السعودية

Project **Mansourah Massarah Gold Project**
Customer: **MA'ADEN GOLD AND BASE METALS (MA'ADEN)**

Title: DESIGN CALCULATION 6000 VILLAGE AREA: SWIMMING POOL DESIGN OF PIT & BUILDING STRUCTURE

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DEFINITIONS AND ABBREVIATIONS

Document	:	Information and the medium on which it is contained (ISO 9000:2015)
EMPLOYER	:	'EMPLOYER' means MA'ADEN GOLD AND BASE METALS (MGBM) and the PMC American Bechtel Inc. or their representatives
Contractor	:	Contractor' means consortium composed of LARSEN & TOUBRO LIMITED, Saudi Arabia AND OUTOTEC SAUDI ARABIA LLC, Saudi Arabia (for IK Contract) and LARSEN & TOUBRO LIMITED – ECC Construction Group – (SHJ BR) and OUTOTEC ENGINEERING DMCC, Dubai (for OOK Contract).
'PMT'	:	Ma'aden Project Management Team
'MMGP Project'	:	Mansourah Massarah Gold Processing Project
Attribute	:	Information element that describes a document or tag.
Supplier	:	An entity (supplier, consultant, sub-contractor) under the direct project management of Contractor
Sub-Supplier	:	A sub entity (supplier, consultant, sub-contractor) under the direct project management of Contractor.
QA	:	Quality Assurance
QC	:	Quality Control
Works	:	Full scope of the EPC Contract including the engineering, procurement, construction, commissioning and associated activities
Typ.	:	Typical
Fc'	:	Concrete Compressive Strength
Fy	:	Yield Strength of Reinforcing Steel
ϕ	:	Strength Reduction Factor
SBC	:	Saudi Building Code
SASO	:	Saudi Standards, Metrology and Quality Organization
ACI	:	American Concrete Institute
ASCE	:	American Society of Civil Engineers
PCI	:	Precast/Prestressed Concrete Institute
K_s	:	Coefficient For Active Soil Pressure
As_{tpro}	:	Area Of Steel Provided
As_{Req}	:	Area Of Steel Required

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1. INTENT

This Document is intended to produce the technical needs in Engineering including particular reference to design justification & Engineering validation works. The work described below is related to Design of SWIMMING POOL. The size of the Pool is 25m x 15m x (2.15 max depth, 1.35m min depth)

2. EXECUTIVE SUMMARY

The Saudi Arabian Mining Company (Ma'aden) has announced that its wholly-owned subsidiary, Ma'aden Gold and Base Metals Company, have issued a Notice of Award to Outotec and Larsen & Toubro Consortium to complete the engineering, procurement, construction, pre-commissioning, commissioning and start-up assistance of the Mansourah & Massarah (MMGP) Gold Project.

MMGP project will become Ma'aden's largest ever gold project, producing an average of 250,000 ounces of gold per year over the life of mine. Mansourah-Massarah is situated within the Central Arabian Gold Region (CAGR), approximately 320km northeast of Taif towards Riyadh and about 35 km southeast of Hufairah, which is a small village situated on Jeddah - Riyadh highway. Mansourah - Massarah gold deposits are located 22°38'6.40"N; 42°44'22.90"E, and 22°42'1.40"N; 42°41'58.40"E respectively, approximately 537 km east of Jeddah and 77 km southeast of the town of Zalim.

The plant throughput capacity is 4000000t/a and is required to operate on a continuous basis at nominally 24 hours per day and 7 days a week.

3. SCOPE

The Scope of this document is limited to Complete Analysis & Structural Design Verification of SWIMMING POOL & Pump Room only.

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4. DESIGN METHODOLOGY & ASSUMPTIONS

A swimming pool was modeled and designed using STAAD pro software as shown in figure 1. The Analysis of retaining walls (Cast-In-Situ) has been performed meeting design requirements & keeping in view the Architectural requirements. The loads were applied as per SBC/ASCE/ACI standards & analysis was done to find out the maximum moment applied on the retaining walls due to hydrostatic pressure as well the soil. The base of the pool has been designed to the hydrostatic load (Tank full condition) and the soil pressure (empty condition)

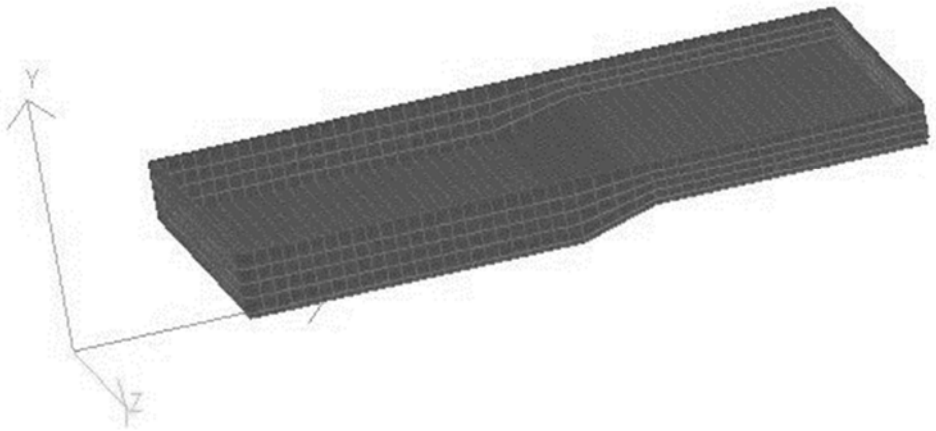


Figure 1 Swimming pool rendering view by STAAD

4.1. Codes

[1] SBC-301-2007-Structural Loading & Forces.

[2] SBC-303-2007-Structural Soil & Foundations.

[3] SBC-304-2007-Structural Concrete Structures.

[4] SASO SSA 2 – Steel Bars for the Reinforcement of Concrete.

[5] SASO SSA 224 – Steel Fabric for Reinforcement of Concrete

[6] ASCE-7-2005 Minimum Design Loads for Buildings and other Structures.

[7] ACI 318-08 Building Code Requirements.

[8] PCI –MNL-120-04 –Design Handbook, 6th edition.

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4.2. Geotechnical investigations

Foundations have been designed based on recommendations of the Soil Investigation Report, Doc # MD-216-3000-EG-CV-CAL-1008-A02.

4.3. Material Properties

Compressive Strength of Concrete at 28th days:

fc' = 35 MPa (5.00 Ksi) for retaining walls.

fc' = 35 MPa (5.00 Ksi) for raft foundations.

Yield Strength of Steel:

fy = 420 MPa (60.91 Ksi) for all reinforcing steel.

Unit Weights:

Concrete =	=	24	kN/m^3
Soil (γ) =	=	18	kN/m^3
Water ($\gamma\omega$) =	=	10	kN/m^3

Allowable S.B.C. as per soil report :

Net allowable Bearing capacity at bottom of foundation = 350 kN/m^2

Modulus of sub grade reaction

As per Bowle's equation = 350 x 60

= 21000 kN/m^3

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5. DESIGN LOADS

5.1. Primary Load Cases

5.1.1. Load Case 1: Dead Loa

- Dead load considered the self-weight of the structural elements such as walls and bottom slab, which taken by using the SELF-WEIGHT command, as shown below figure 2.and the surcharge load a 5 kN/m^2 applied on the walls due to accidental.

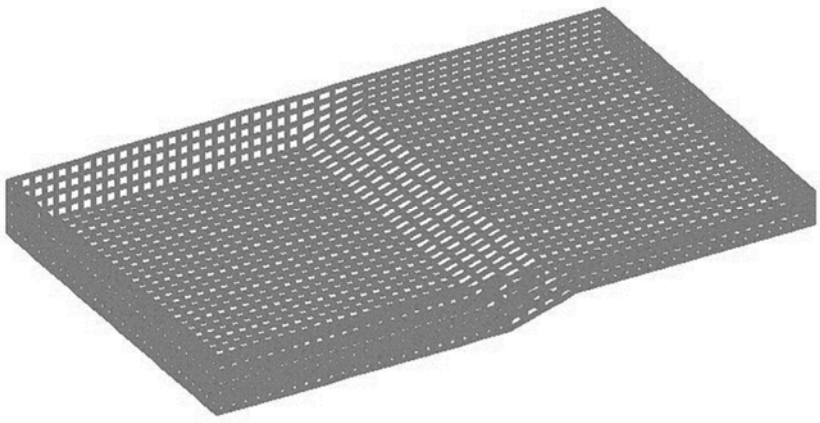


Figure 2 Self weight- Dead Load Applied by STAAD

5.1.2. Load Case 2: Water Pressure On Raft

$$P = \gamma_w H = 10 \times (H)$$

Calculated in help of this formula above and divided into 3 different areas in order to reach the pressure with variant changes in depth. Applied as a pressure on full plate command.

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(i) In maximum depth area as shown below in figure 3.

$$P = \gamma_w H = 10 \times (2.15) = 21.5 \text{ kN/m}^2$$

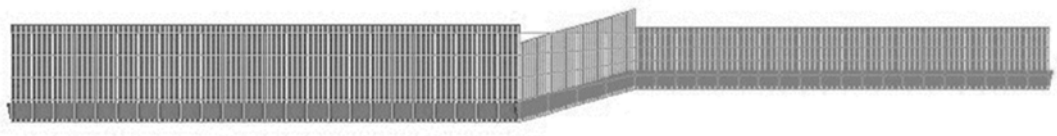


Figure 3 Water Pressure On Raft at 2.15m depth by STAAD

(ii) In mid depth area as shown below in figure 4.

$$P = \gamma_w H = 10 \times (1.75) = 17.5 \text{ kN/m}^2$$

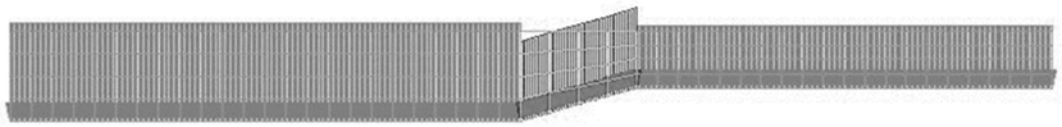


Figure 4 Water Pressure On Raft at 1.75m depth by STAAD

(iii) In minimum depth area as shown below in figure 5.

$$P = \gamma_w H = 10 \times (1.35) = 13.5 \text{ kN/m}^2$$

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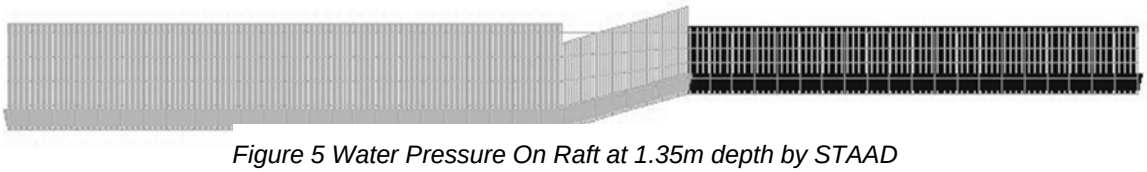


Figure 5 Water Pressure On Raft at 1.35m depth by STAAD

5.1.3. Load Case 3: Hydrostatic Pressure Due To Water On The Walls

$$P = \gamma_w H = 10 \times (H)$$

The maximum pressure in the bottom been calculated in help of the formula above. Applied as a trapezoidal pressure command as shown below figure 6.

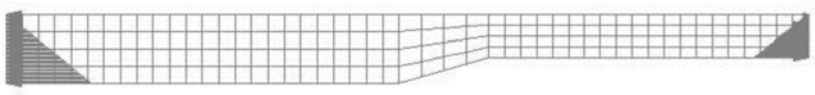


Figure 6 Hydrostatic Pressure due to Water on the walls by STAAD

5.1.4. Load Case 4: Soil Pressure On The Walls

$P = K_s \times \gamma_s (H) =$ kN/m^2 Where K_s is coefficient for active soil pressure 0.33

(i) In maximum depth area as shown below in figure 7

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$$P = K_s \times \gamma_s (H) = 0.33 \times 18 \times 2.15 = 12.77 \text{ kN/m}^2$$

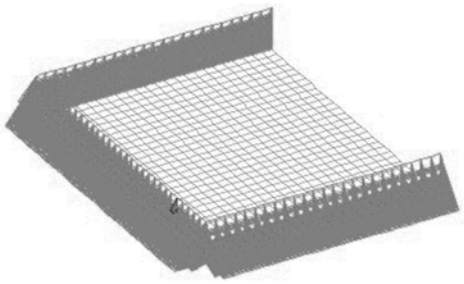


Figure 7 Soil pressure due to soil on the walls at 2.15m depth by STAAD

(ii) In mid depth area as shown below in figure 8

$$P = K_s \times \gamma_s (H) = 0.33 \times 18 \times 1.75 = 10.39 \text{ kN/m}^2$$

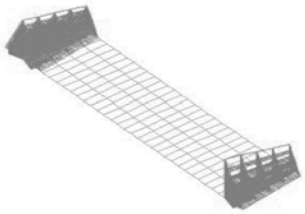


Figure 8 Soil pressure due to Soil on the walls at 1.75m depth by STAAD

(iii) In minimum depth area as shown below in figure 9.

$$P = K_s \times \gamma_s (H) = 0.33 \times 18 \times 1.35 = 8.01 \text{ kN/m}^2$$

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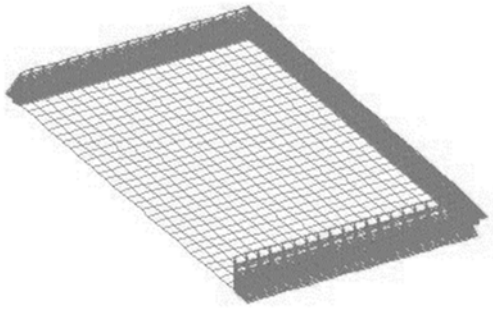


Figure 9 Soil Pressure on the walls at 1.35m depth by STAAD

5.2. Load Combinations Cases

The following critical load combinations are adopted for analysis & further for design
Reference: (PCA – rectangular concrete tank design)

5.2.1. Service Load Combination

- LOAD COMB 1 **Maintenance (pool without water)**
  Adopted as service load case, when Tank with empty of water and live load plus dead load outside plus soil pressure outside
  (DEAD LOAD + SOIL PRESSURE).

- LOAD COMB 2 **Pool full without backfill**
  Adopted as service load case, when Tank is fully filled with water and live plus dead load outside
  (DEAD LOAD + WATER PRESSURE + LOAD ON RAFT) .

- LOAD COMB 3 **Pool empty without backfill**
  Adopted as service load case, when Tank is empty of water and dead load outside while no soil pressure outside
  (DEAD LOAD).

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- LOAD COMB 4 **Running condition**
  Adopted as service load case, when Tank with is fully filled with water and dead load outside with soil pressure outside
  (DEAD LOAD + WATER PRESSURE + SOIL PRESSURE + LOAD ON RAFT)

5.2.2. Factored Load Combination

- LOAD COMB 5 **Maintenance (pool without water)**
  Adopted as service load case, when Tank with empty of water and live load plus dead load outside plus soil pressure outside
  (1.4 DEAD LOAD + 1 SOIL PRESSURE)

- LOAD COMB 6 **Pool full without backfill**
  Adopted as service load case, when Tank is fully filled with water and live plus dead load outside
  (1.4 DEAD LOAD + 1.7 WATER PRESSURE + 1 LOAD ON RAFT)

- LOAD COMB 7 **Pool empty without backfill**
  Adopted as service load case, when Tank is empty of water and dead load outside while no soil pressure outside
  (1.4 DEAD LOAD)

- LOAD COMB 8 **Running condition**
  Adopted as service load case, when Tank with is fully filled with water and dead load outside with soil pressure outside
  (1.4 DEAD LOAD + 1.7 WATER PRESSURE + 1 SOIL PRESSURE + 1 LOAD ON RAFT)

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6. CHECK FOR BASE PRESSURE

The Base Pressure done by getting the service load when the pool full of water as shown below in figure 10.

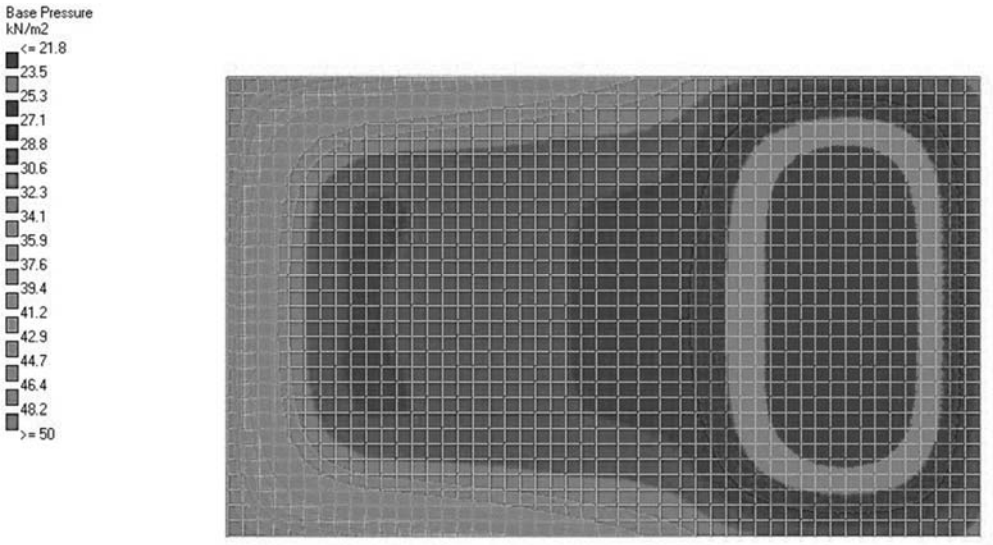


Figure 10 Base Pressure check by STAAD

From the above figure 10, the maximum base pressure at bottom of base slab is 49.96 kN/m^2,which is less than maximum allowable pressure 350 kN/m^2, **Hence Safe**

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7. DESIGN CALCULATIONS FOR WALLS

The maximum moment taken from the MY =22 kn.m as shown below figure 11.

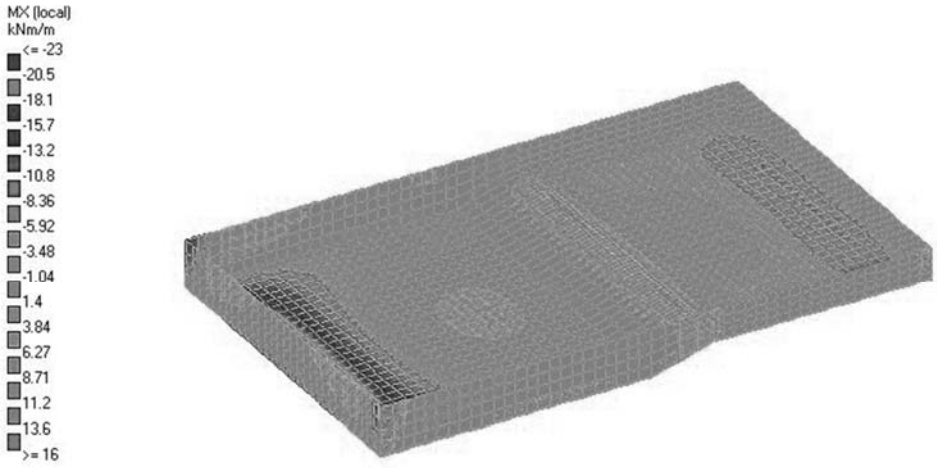


Figure 11 The maximum moment taken from the MY due to the ultimate

The Finite element analysis results of the resultant moment variation for the critical load combinations in terms of contour diagrams above the maximum value is 23 kn.m

7.1 Reinforcement calculation

Providing 250 mm total thickness at section the deepest part and the slope.

$\beta 1 = 0.85$ (ACI 350-01 Clause 10.2.7.3)

$$\rho \text{ bal} = \frac{0.85*\beta 1*fc*600}{fy(600+fy)} = \frac{0.85*0.85*35*600}{420(600+420)} = 0.035$$

ρ max = 0.75 x ρ bal = 0.75 x 0.035 = 0.0265

$\omega = \rho$ max x fy / fc = 0.0265 x 420 / 35 = 0.318

q bal = fc ω (1-ω / 1.7) = 35*0.32 *(1-0.32 /1.7) = 4.48 N/mm^2

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7.2 Check for Thickness

The moment variation in terms of Mx shown in figure 11,
Maximum moment = 22 kNm for Ultimate Load combination.

Mu = 22 kNm

Sanitary factor U= 1.3 (for moment) (PCA – rectangular concrete tank design intro1-5)

(PCA – rectangular concrete tank design)

Mus = Mu * U = 22 * 1.3 = 29 kNm

d_{eff} required = (29 x 1000 / 4.48)^0.5 = 80.45 mm

Effective depth provided = 250 – 75 – 16 – 6 = 153 mm O.K.

7.3 Wall Reinforcement-(Vertical Bars)

$$Mn = As * Fy\left[d - \frac{As*Fy}{0.85*fc*2*b}\right]$$

$$29 \times 10^6 = As * 420 * \left[153 - \frac{As*420}{0.85*35*2*1000}\right] \rightarrow \text{solve for As}$$

As_{Req} = 462 mm2 /m

Provide 16mm Dia bars

$$\text{Spacing of bars} = \frac{1000*\text{Area of bar}}{As_{Req}} = \frac{1000*200}{1067} = 435\text{mm}$$

Provide 12ϕ @ 125 c/c T & B.

As_{tpro} = 904 mm^2/m > As_{Req} Hence OK

7.4 Wall Reinforcement – (Horizontal bars)

Mu = 20 kNm

Sanitary factor U= 1.3 (for moment)

Mus = Mu * U = 20 * 1.3 = 26 kNm

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$$Mn = As * Fy\left[d - \frac{As*Fy}{0.85*fc*2*b}\right]$$

$$26 \times 10^6 = As * 420 * \left[153 - \frac{As*420}{0.85*35*2*1000}\right] \rightarrow \text{solve for As}$$

As_{Req} = 405.3 mm2 /m

Provide 12 mm Día bars

$$\text{Spacing of bars} = \frac{1000*\text{Area of bar}}{As_{Req}} = \frac{1000*113}{405.3} = 278.8 \text{ mm}$$

Provide ϕ12 @ 200 c/c T & B.

As_{tpro} = 678 mm²/m > As_{Req} Hence OK

7.5 Check for Minimum Reinforcement

$$As_{min1} = \frac{3\sqrt{fc'}}{fy} * bw * d = As_{smin} = \frac{3\sqrt{5070}}{60915} * 1000 * 153 = 536.5 \text{ mm2}$$

$$As_{min2} = \frac{200*bw*d}{60000} = \frac{200*1000*153}{60915} = 502.3 \text{ mm2}$$

Take maximum of As_{smin1} & As_{min2}

$As_{min} = As_{min2} = 536.5$ mm2

$As_{tpro} > As_{min}$

Provide 12ϕ @ 300 c/c T & B. – **(Vertical bars) walls 250mm**

Provide 12ϕ @ 200 c/c T & B. – **(Horizontal bars) walls 250mm**

Therefore provide the same spacing & dia in 200mm wall considering the Crack Width Requirements.(Refer Section 9.1 & 9.2 for Detail)

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8. DESIGN CALCULATIONS FOR BASE SLAB

The Finite element analysis results of the resultant moment variation for the critical load combinations in terms of contour diagrams are as follows figure 12.

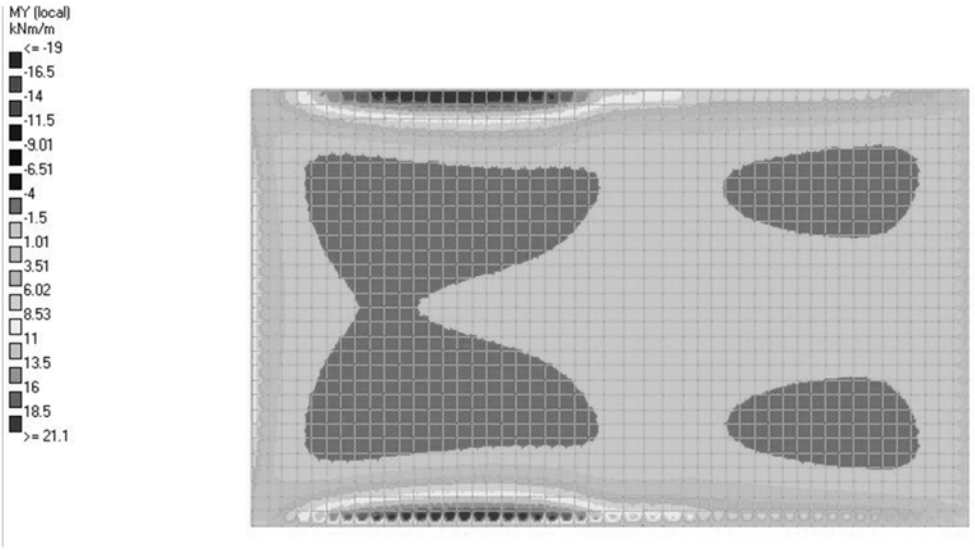


Figure 12 The Finite element analysis shown the moment MY at the base long direction

8.1. Raft Reinforcement Calculation

Providing 350 mm total thickness all around(1.5m wide) below the walls & remaining 200mm thick raft in the Middle area(Refer Drawing for Details).

$\beta 1 = 0.85$ (ACI 350-01 Clause 10.2.7.3)

$$\rho \text{ bal} = \frac{0.85*\beta 1*fc*600}{fy(600+fy)} = \frac{0.85*0.85*35*600}{420(600+420)} = 0.035$$

ρ max = 0.75 x ρ bal = 0.75 x 0.035 = 0.0262

$\omega = \rho$ max x fy / fc = 0.0262 x 420 / 35 = 0.31

q bal = fc ω (1-ω / 1.7) = 35*0.31 *(1-0.31 /1.7) = 4.40 N/mm^2

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8.2. Check for Thickness (Raft)

The moment variation in terms of My shown in figures above,
Maximum moment = 22 kN.m
The moment variation in terms of Mx shown in figures above,
Maximum moment = 20 kN.m

→ Mu = 22 kN.m
Sanitary factor U= 1.3 (for moment)
Mn = Mu * U = 22 * 1.3 = 28.6 KN.m
d_{eff} required = (28.6 x 1000 / 4.40)^0.5 = 80.62 mm
Effective depth = 350 – 80 – 40 = 230 mm **O.K.**

8.3. Raft Reinforcement (Long Direction)

The moment variation in terms of My shown in figures above,
Maximum moment = 22 kN.m

Mn = $As * Fy\left[d - \frac{\text{As}*\text{Fy}}{0.85*\text{fc}*2*\text{b}}\right]$

28.6 x 10^6 = $As * 420 * \left[280 - \frac{\text{As}*420}{0.85*35*2*1000}\right]$ → solve for As

As_{Req} = 243.34 mm2 /m

Provide 12mm Dia bars

Spacing of bars = $\frac{1000*\text{Area of bar}}{\text{As}_{\text{Req}}} = \frac{1000*201}{243.3}$ = 823.77 mm

Provide ϕ12 @ 200 c/c T & B. Both ways for 350mm thick slab considering crack width requirements(Refer Section-9.3.1)

Similarly provide the same ϕ12 @ 200 c/c T & B. Both ways for 200mm thick slab considering crack width requirements (Refer Section-9.3.1)

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9. CHECK FOR CRACK WIDTH :

9.1. For Wall 250mm:

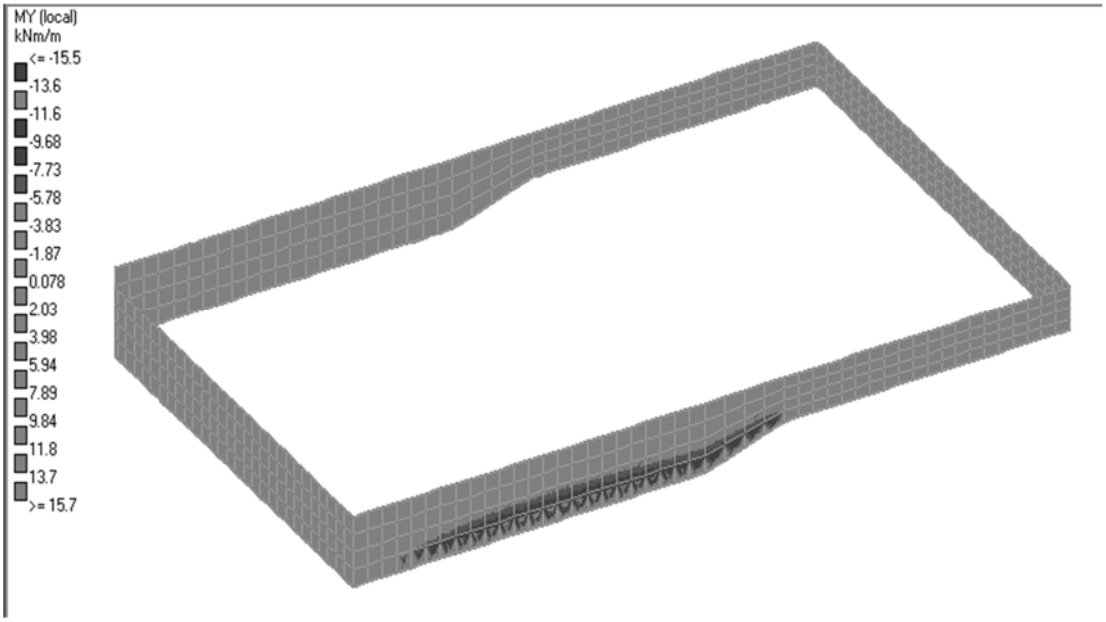


Figure 13 Showing maximum service bending moment in 250mm wall

The finite element analysis result of the resultant moment variation for the critical load combinations in term of contour diagrams as shown below in figure 14.

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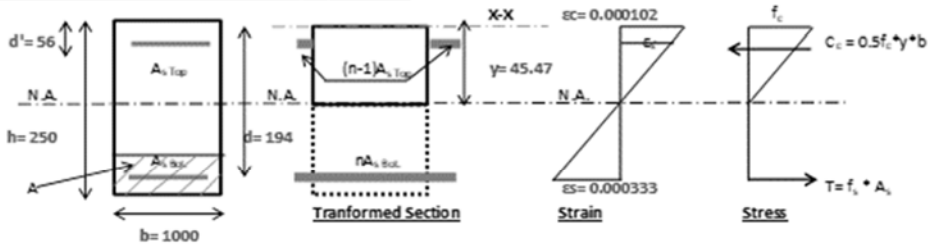
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General Input :-

Width b	1000	mm	Es	200000	N/mm²		
Depth h	250	mm	ε_c	0.003	mm		
cover	50	mm	ε_y	0.0021	mm		
f_y	420	N/mm²	E_c	27806	N/mm2	ACI318 clause 8.5.1	
f_c'	35	N/mm²	d	194	mm		
$M_{d\ service}$	11	KN.m	d'	56	mm		
			n= E_s/E_c =	7.19			

Section RFT.	No of bars	rebar Dia	A_s mm²		
Top Rft.	8	T 12 Top	= 912 mm2	$S_{spacing\ Top}$ = 142.9 mm	
Bot. Rft.	8	T 12 Bot.	= 912 mm2	$S_{spacing\ Bot.}$ = 142.9 mm	
Exposure condition	Water-retaining structures				ACI224R-01,Table 4.1

1- Locate the Neutral axis & Calculate setcion properties:-



Taking moment of areas @ X-X,

$[by+nA_{s\,Bot.}+(n-1)A_{s\,Top}]*y = by*y/2 + nA_{s\,Bot.}*d + (n-1)A_{s\,Top}*d'$

and simplify in form of Quadratic equation ($ay2+by+c=0$),

$(b-b/2)*y2 + (nA_{s\,Bot.}+(n-1)A_{s\,Top})*y + (-nA_{s\,Bot.}*d - (n-1)A_{s\,Top}*d') = 0$

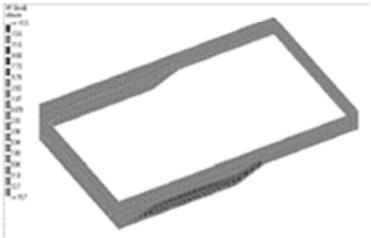
Thus, a = 500.00, b = 12207.67, c = -1588886.89

Thus, $y = \frac{-b \pm \sqrt{b^2-4ac}}{2a}$ = 45.47 mm

$I_g = bh^3/12$ = 1,302,083,333 mm4 $A_g = b*h$ = 250,000 mm2

$A_{cr} = by + nA_{s\,Bot.} + (n-1)A_{s\,Top}$ = 57,678 mm2

$I_{cr} = by^3/12 + by*(y/2)^2 + nA_{s\,Bot.}*(d-y)^2 + (n-1)A_{s\,Top}*(y-d')^2$ = 176,680,550 mm4



2- Check of Concrete & Steel Reinforcement stresses:-

$f_r = 0.62\lambda\sqrt{f_c'}$ = 3.67 N/mm² *ACI 318M-11, Eq 9-10*

$M_{cr} = \frac{f_r I_g}{y_t}$ = 38.21 KN.m *ACI 318M-11, Eq 9-9*

$f_{c\,a} = M_a*y/I_{cr}$ = 2.83 N/mm² = 0.081 fc'

$f_{s\,a} = M_a(y-d)/(I_{cr}/n)$ = 66.51 N/mm² Ok < f s max = 224.03 N/mm², ACI350-06, Eq 10-4 & 10-5

= 0.158 fy

Ok < f s max= 320N/mm2 allowed for rebar Dia 12 mm used ,According ACI224R-04, Table4.2

Ok < f s max= 240N/mm2 allowed for spacing 143 mm used ,According ACI224R-04, Table4.3

3- Calculation of crack width :-

Crack width, $w = 0.011\,\beta f_s \sqrt[3]{d_c A} \times 10^{-3}$ = 0.093 mm *ACI224R-01, Eq 4-2a*

ok Crack Width <= wcrall= 0.100 mm, ACI224R-01,Table 4.1

Where, d_c = 56.000 mm *ACI224R-01, 4.2.1*

β = 1.377 mm *ACI224R-01, 4.2.1 & ACI350-06, Eq 10-6*

A = 14,000 mm2 *ACI224R-01, 4.2.2*

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9.2. For Wall 200mm:

The finite element analysis result of the resultant moment variation for the critical load combinations in term of contour diagrams as shown below in figure 15.

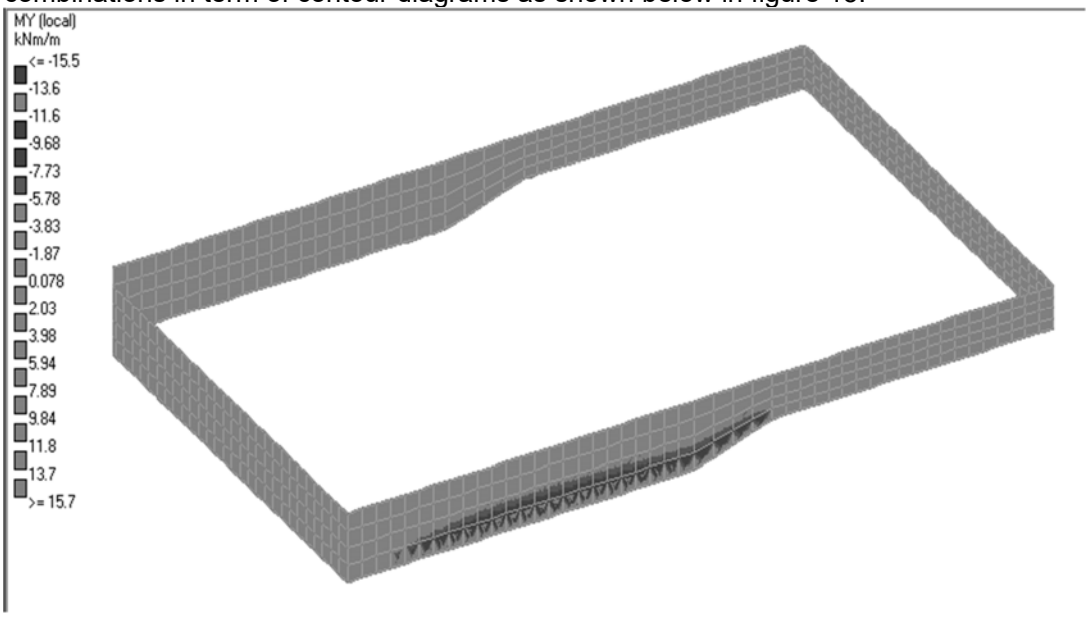


Figure 14 Showing maximum service bending moment in 200mm wall

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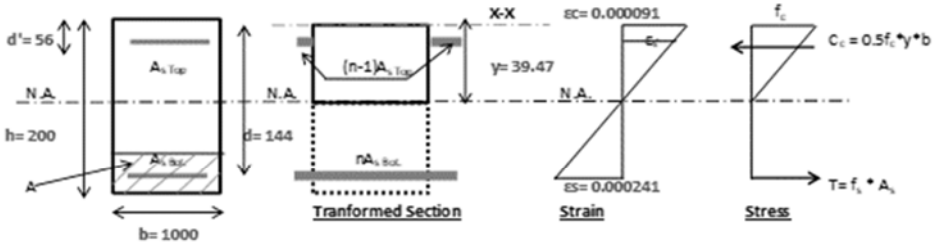
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General Input :-

Width b	1000	mm	Es	200000	N/mm²	
Depth h	200	mm	ε_c	0.003	mm	
cover	50	mm	ε_y	0.0021	mm	
f_y	420	N/mm²	E_c	27806	N/mm2	ACI318 clause 8.5.1
f'_c	35	N/mm²	d	144	mm	
$M_{a\ service}$	6	KN.m	d'	56	mm	
			n= E_s/E_c =	7.19		

Section RFT.	No of bars	rebar Dia	A_s mm²		
Top Rft.	8	T 12 Top	= 912 mm2	$S_{spacing\ Top}$ = 142.9 mm	
Bot. Rft.	8	T 12 Bot.	= 912 mm2	$S_{spacing\ Bot.}$ = 142.9 mm	
Exposure condition	Water-retaining structures				ACI224R-01,Table 4.1

1- Locate the Neutral axis & Calculate setcion properties:-



Taking moment of areas @ X-X,

$[by+nA_{s\ Bot.}+(n-1)A_{s\ Top}]*y = by*y/2 + nA_{s\ Bot.}*d + (n-1)A_{s\ Top}*d'$

and simplifyin form of Quadratic equation (ay2+by+c=0),

$(b-b/2) * y2 + (nA_{s\ Bot.}+(n-1)A_{s\ Top}) * y + (-nA_{s\ Bot.}*d - (n-1)A_{s\ Top}*d') = 0$

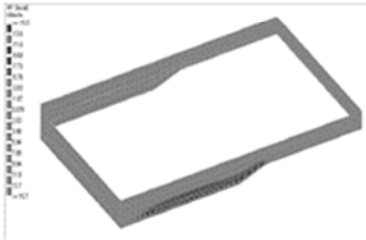
Thus, a = 500.00
b= 12207.67
c= -1260895.12

Thus, $y = \dfrac{-b \pm \sqrt{b^2 - 4ac}}{2a}$ = 39.47 mm

$I_g = bh^3/12$ = 666,666,667 mm4 $A_g = b*h$ = 200,000 mm2

$A_{cr} = by + nA_{s\ Bot.} + (n-1)A_{s\ Top}$ = 51,680 mm2

$I_{cr} = by^3/12 + by*(y/2)^2 + nA_{s\ Bot.}*(d-y)^2 + (n-1)A_{s\ Top}*(y-d')^2$ = 93,715,920 mm4



2- Check of Concrete & Steel Reinforcement stresses:-

$f_r = 0.62\lambda\sqrt{f'_c}$ = 3.67 N/mm² *ACI 318M-11, Eq 9-10*

$M_{cr} = \dfrac{f_r I_g}{y_t}$ = 24.45 KN.m *ACI 318M-11, Eq 9-9*

$f_{c,a} = M_a * y / I_{cr}$ = 2.53 N/mm² = 0.072 fc'

$f_{s,a} = M_a(y-d)/(I_{cr}/n)$ = 48.14 N/mm² Ok < f s max = 200.88 N/mm², ACI350-06, Eq 10-4 & 10-5

= 0.115 fy

Ok < f s max= 320N/mm2 allowed for rebar Dia 12 mm used ,According ACI224R-04, Table4.2

Ok < f s max= 240N/mm2 allowed for spacing 143 mm used ,According ACI224R-04, Table4.3

3- Calculation of crack width :-

Crack width, $w = 0.011\,\beta f_s \sqrt[3]{d_c A} \times 10^{-3}$ = 0.075 mm *ACI224R-01, Eq 4-2a*

ok Crack Width <= wcrall= 0.100 mm, ACI224R-01,Table 4.1

Where, d_c = 56.000 mm *ACI224R-01, 4.2.1*

β = 1.536 mm *ACI224R-01, 4.2.1 & ACI350-06, Eq 10-6*

A = 14,000 mm2 *ACI224R-01, 4.2.2*

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9.3. For Raft:

The finite element analysis result of the resultant moment variation for the critical load combinations in term of contour diagrams as shown below in figure 16.

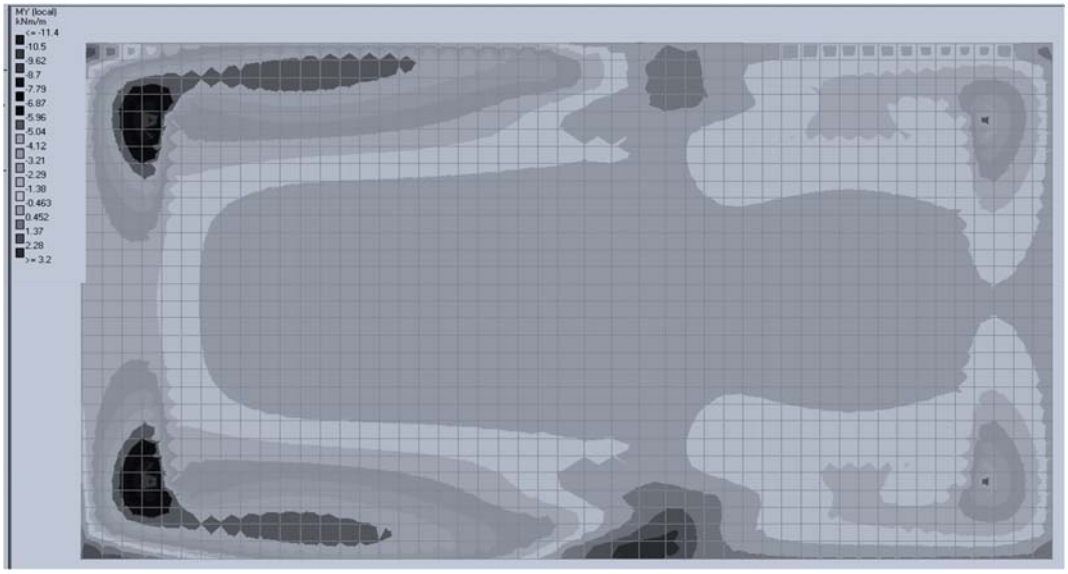


Figure 15 Showing maximum service bending moment in Raft

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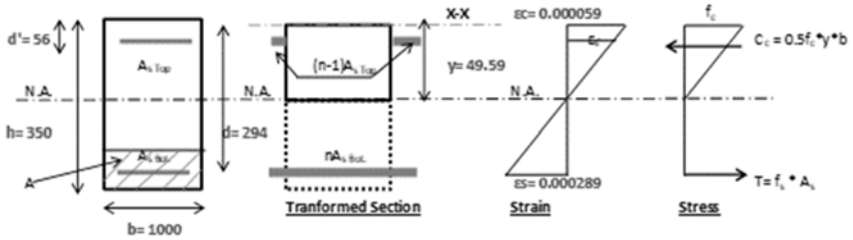
9.3.1. 350mm RAFT

General Input :-

Width b	1000	mm	Es	200000	N/mm²		
Depth h	350	mm	ε_c	0.003	mm		
cover	50	mm	ε_y	0.0021	mm		
f_y	420	N/mm²	E_c	27806	N/mm2	ACI318 clause 8.5.1	
f_c'	35	N/mm²	d	294	mm		
$M_{s\ service}$	11	KN.m	d'	56	mm		
			n= E_s/E_c =	7.19			

Section RFT.	No of bars	rebar Dia	A_s mm²		
Top Rft.	6	T 12 Top	= 684 mm2	$S_{spacing\ top}$ = 200 mm	
Bot. Rft.	6	T 12 Bot.	= 684 mm2	$S_{spacing\ bot.}$ = 200 mm	
Exposure condition	Water-retaining structures				ACI224R-01,Table 4.1

1- Locate the Neutral axis & Calculate setcion properties:-



Taking moment of areas @ X-X,

$[by+nA_{s\,Bot.}+(n-1)A_{s\,Top}]*y = by*y/2 + nA_{s\,Bot.}*d + (n-1)A_{s\,Top}*d'$

and simplifyin form of Quadratic equation (ay2+by+c=0),

$(b-b/2)*y2 + (nA_{s\,Bot.}+(n-1)A_{s\,Top})*y + (-nA_{s\,Bot.}*d - (n-1)A_{s\,Top}*d') = 0$

Thus, a = 500.00, b = 9155.75, c = -1683652.84

Thus, $y = \dfrac{-b \pm \sqrt{b^2-4ac}}{2a}$ = 49.59 mm

$I_g = bh^3/12$ = 3,572,916,667 mm4 $A_g = b*h$ = 350,000 mm2

$A_{tr} = by + nA_{s\,Bot.} + (n-1)A_{s\,Top}$ = 58,746 mm2

$I_{tr} = by^3/12 + by*(y/2)^2 + nA_{s\,Bot.}*(d-y)^2 + (n-1)A_{s\,Top}*(y-d')^2$ = 334,719,067 mm4

2- Check of Concrete & Steel Reinforcement stresses:-

$f_r = 0.62\lambda\sqrt{f_c'}$ = 3.67 N/mm² *ACI 318M-11, Eq 9-10*

$M_{cr} = \dfrac{f_r I_g}{y_t}$ = 74.89 KN.m *ACI 318M-11, Eq 9-9*

$f_{c\,a} = M_a * y / I_{tr}$ = 1.63 N/mm² = 0.047 fc'

$f_{s\,a} = M_a(y-d)/(I_{tr}/n)$ = 57.77 N/mm² Ok < f s max = 198.76 N/mm², ACI350-06, Eq 10-4 & 10-5

= 0.138 fy

Ok < f s max= 320N/mm2 allowed for rebar Dia 12 mm used ,According ACI224R-04, Table4.2

Ok < f s max= 200N/mm2 allowed for spacing 200 mm used ,According ACI224R-04, Table4.3

3- Calculation of crack width :-

Crack width, $w = 0.011\,\beta f_s \sqrt[3]{d_c A} \times 10^{-3}$ = 0.079 mm *ACI224R-01, Eq 4-2a*

ok Crack Width <= wcrall= 0.100 mm, ACI224R-01,Table 4.1

Where, d_c = 56.000 mm *ACI224R-01, 4.2.1*

β = 1.229 mm *ACI224R-01, 4.2.1 & ACI350-06, Eq 10-6*

A = 18,667 mm2 *ACI224R-01, 4.2.2*

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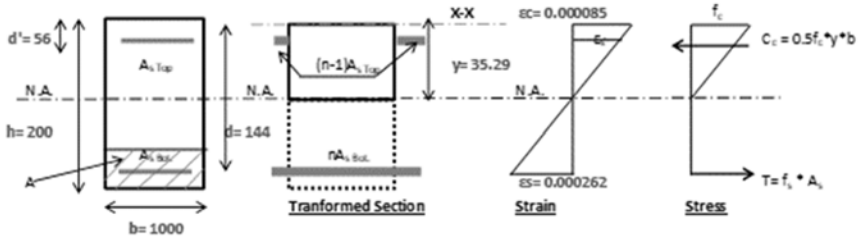
9.3.2. 200mm RAFT

General Input :-

Width b	1000 mm	Es	200000	N/mm²		
Depth h	200 mm	ε_c	0.003	mm		
cover	50 mm	ε_y	0.0021	mm		
f_y	420 N/mm²	E_c	27806	N/mm2	ACI318 clause 8.5.1	
f_c'	35 N/mm²	d	144	mm		
$M_{s\ service}$	5 KN.m	d'	56	mm		
		n= E_s/E_c =	7.19			

Section RFT.	No of bars	rebar Dia	A_s mm²		
Top Rft.	6	T 12 Top	= 684 mm2	$S_{spacing\ Top}$ = 200 mm	
Bot. Rft.	6	T 12 Bot.	= 684 mm2	$S_{spacing\ Bot.}$ = 200 mm	
Exposure condition	Water-retaining structures				ACI224R-01,Table 4.1

1- Locate the Neutral axis & Calculate setcion properties:-



Taking moment of areas @ X-X,

$[by+nA_{s\ Bot.}+(n-1)A_{s\ Top}]*y = by*y/2 + nA_{s\ Bot.}*d + (n-1)A_{s\ Top}*d'$

and simplifyin form of Quadratic equation (ay2+by+c=0),

$(b-b/2)*y2 + (nA_{s\ Bot.}+(n-1)A_{s\ Top})*y + (-nA_{s\ Bot.}*d - (n-1)A_{s\ Top}*d') = 0$

Thus, a = 500.00
b= 9155.75
c = -945671.34

Thus, $y = \dfrac{-b \pm \sqrt{b^2-4ac}}{2a}$ = 35.29 mm

$I_g = bh^3/12$ = 666,666,667 mm4 $A_g = b*h$ = 200,000 mm2

$A_{cr} = by + nA_{s\ Bot.} + (n-1)A_{s\ Top}$ = 44,443 mm2

$I_{cr} = by^3/12 + by*(y/2)^2 + nA_{s\ Bot.}*(d-y)^2 + (n-1)A_{s\ Top}*(y-d')^2$ = 74,609,087 mm4

2- Check of Concrete & Steel Reinforcement stresses:-

$f_r = 0.62\lambda\sqrt{f_c'}$ = 3.67 N/mm² *ACI 318M-11, Eq 9-10*

$M_{cr} = \dfrac{f_r I_g}{y_t}$ = 24.45 KN.m *ACI 318M-11, Eq 9-9*

$f_{c\,s} = M_s*y/I_{cr}$ = 2.36 N/mm² = 0.068 fc'

$f_{s\,s} = M_s(y-d)/(I_{cr}/n)$ = 52.40 N/mm² Ok < f s max = 165.00 N/mm², ACI350-06, Eq 10-4 & 10-5
= 0.125 fy

Ok < f s max= 320N/mm2 allowed for rebar Dia 12 mm used ,According ACI224R-04, Table4.2

Ok < f s max= 200N/mm2 allowed for spacing 200 mm used ,According ACI224R-04, Table4.3

3- Calculation of crack width :-

Crack width, $w = 0.011\,\beta f_s \sqrt[3]{d_c A} \times 10^{-3}$ = 0.089 mm *ACI224R-01, Eq 4-2b*

ok Crack Width <= wcrall= 0.100 mm, ACI224R-01,Table 4.1

Where, d_c = 56.000 mm *ACI224R-01, 4.2.1*

β = 1.515 mm *ACI224R-01, 4.2.1 & ACI350-06, Eq 10-6*

A = 18,667 mm2 *ACI224R-01, 4.2.2*

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10. DESIGN OF PUMP ROOM

It Is a Single-Story Frame Structure with Foundation Being Cast in Situ. The Building Shall Have Insulated Block Walls and Cast in Situ Framing Beams to support the HCS (Hollow Core Slab)

The Sub and superstructure for this Building was modeled and designed using STAAD software. The Analysis of Frames (Cast-In-Situ Columns & Beams) has been performed meeting design requirements & keeping in view the Architectural requirements. The loads were applied as per SBC/ASCE/ACI standards & analysis was done to generate the Column Reactions at the Base that are assumed to be fixed. The Foundation design has been done by importing Column Base Reactions into Foundation Design Software.



STAAD 3D MODEL

10.1. Design Loads

10.1.1. Dead Load(DL)

Roof Dead Loads

- Provision for Electrical Lighting = 0.05 kN/m²

Ceiling

Suspended steel channel system = 0.10 kN/m²

Acoustical Fiber Tile (16mm) = 0.05 kN/m²

Sub-total = **0.2 kN/m²**

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- Hollow Core Slab

  (HCS-150)(To be confirmed by Vendor) b/w Grid Line's (1 & 2) = **2.52 KN/m2**

Prestressed Hollowcore Floor										
Overall Structural Depth (mm)	Self Wt- KN/M2	No. of Cores in Unit	Spans indicated below allow for Characteristic Service Load (Live Load) Plus Self Weight Plus 1.5Kn/m^2 for Finishes Characteristic Service Loads Kn/m^2							
			0.75	1.50	2.00	2.50	3.00	4.00	5.00	10.00
			Clear Span in metres							
100	2.38	0	5.39	5.19	5.08	4.91	4.72	4.40	4.14	3.30
150	2.52	11	7.40	7.40	7.40	7.40	7.28	6.80	6.40	5.12
150 (Dense	2.94	11	7.40	7.40	7.40	7.35	7.08	6.64	6.27	5.06
200	3.10	11	9.40	9.40	9.40	9.40	9.26	8.69	8.21	6.65
250	3.43	6	10.80	10.80	10.80	10.80	10.80	10.80	10.56	8.60
300	3.88	6	13.50	13.50	13.50	13.50	13.18	12.44	11.81	9.70
350	4.23	6	15.00	15.00	15.00	15.00	14.89	14.08	13.39	11.06
400	4.57	6	16.00	16.00	16.00	16.00	15.97	15.13	14.42	11.98
450	4.92	6	18.00	18.00	18.00	17.70	17.19	16.33	15.58	13.02
500	5.27	6	18.00	18.00	18.00	18.00	18.00	17.13	16.38	13.76

- Built-in Water Proofing System

100mm thk Polystyrene Foam

0.01 kN/m² per 25.4mm thickness x (100/25.4mm) = 0.039 kN/m²

Water proofing membrane

Bituminous, smooth surface = 0.07 kN/m²

90 mm thick average cement foam concrete screed Lightweight

Concrete 0.38 kN/m² per 25mm thickness x (90/25mm) = 1.368 kN/m²

75 mm Concrete screed above HCS = 21 x 0.075 =1.6 kN/m²

Sub-total = **3.1 kN/ m²**

Therefore

Total Dead Load b/w Grid's (1& 2) =2.52+0.2+3.1= 5.82 **kN/m²**

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TOTAL DL as UDL ON BEAMS

- UDL on Ledger Beams Along Grid Line-1 & 2 = 5.00 x 5.82/2 = **14.55 Kn/m**

Moment on all the External Beams due to Cantilever Portion of slab and Screed

= (0.15x24+ (0.05+0.2)/2x20)*0.6*0.6/2 = **1 knm/m**

10.1.2. Ground Floor Walls

- 250 tyke. Outer CMU (Partially Grouted Cell@800 c-c)

16 x 0.25 x (2.5) m clear height = **10 kN/m**

10.1.3. Roof Live Loads(Lr)

Live loads are with reference to Clause-4.2.2, MD-216-1000-EG-ST-CRT-1001

Ordinary Flat (Inaccessible) = **0.5 kN/m²**

TOTAL Lr as UDL ON BEAMS

- UDL on Ledger Beams Along Grid Line-1 = 5 x 0.50/2 = **1.25 Kn/m**

Moment on all the External Beams due to Live Load on

Cantilever Portion of slab = 0.5*0.6*0.6/2 = **0.09 kN/m /m**

10.1.4. Wind Load(WL)

Wind Loads have been generated automatically in the Software as per ASCE-07.

10.1.5. Thermal/Temperature Loads (Self-restraining and Thermal Expansion and Contraction Loads)

An ambient temperature ranges from a maximum of 50° C to a low of 0° C. In direct exposure to sunlight, the temperature shall be assumed to rise a minimum of 15° C above ambient. As a minimum, the temperature gradient for thermal loads shall be +25° C and -15° C with the most severe condition governing the design.

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10.2. Design Combinations

Service Loads (Allowable Stress Design) (SBC-301, Section 2.4.1)

D

D + T

D +Lr

D +0.75Lr

D + (W or 0.7E)

D + 0.75W + 0.75Lr

0.6D + W

0.6D + 0.7E

Factored loads (Strength Design) (SBC-301, Section 2.3.2)

1.4D

1.4D+1.4T

1.2(D + T) + 0.5Lr

1.2D + 1.6Lr + 0.8W

1.2D + 1.6W + 0.5Lr

1.2D + 1.0E

0.9D + 1.6W

0.9D + 1.0E

Where:

D = Dead Load

Lr = Live Load

W = Wind Load

E = Seismic Load

T = Temperature Load

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10.3. Cracked Sections

In lieu of a more accurate analysis of the effects of cracking of concrete, provisions of SBC-304-Section 10.11 was used.

For Beams:	I = 0.35Ig
For Columns:	I = 0.70Ig
For Walls:	
Un-cracked	I = 0.70Ig
Cracked	I = 0.35Ig
For Slabs	I = 0.25Ig

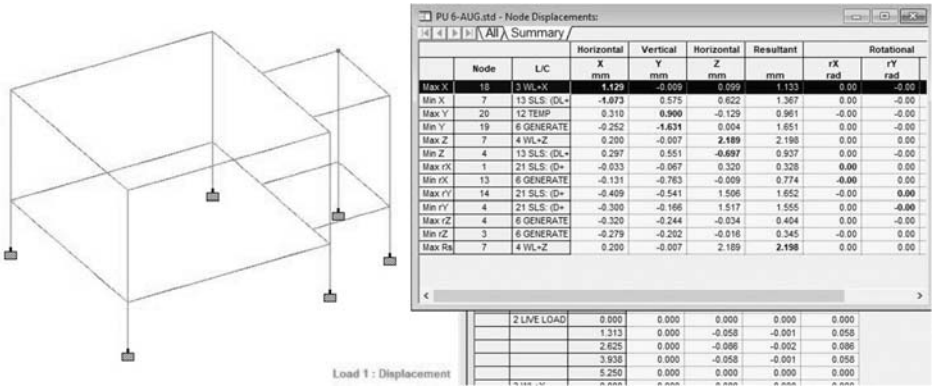
10.4. Deflections

10.4.1. Maximum Allowable Drifting

L/240: for member load with live load plus dead load

L/600: for members supporting masonry under total loads

Where, L = span of member between support



	Node	L/C	Horizontal X mm	Vertical Y mm	Horizontal Z mm	Resultant mm	Rotational rX rad	Rotational rY rad
Max X	18	3 WL+X	**1.129**	-0.009	0.099	1.133	0.00	-0.00
Min X	7	13 SLS: (DL+	**-1.073**	0.575	0.622	1.367	0.00	-0.00
Max Y	20	12 TEMP	0.310	**0.900**	-0.129	0.961	-0.00	-0.00
Min Y	19	6 GENERATE	-0.252	**-1.631**	0.004	1.651	0.00	-0.00
Max Z	7	4 WL+Z	0.200	-0.007	**2.189**	2.198	0.00	0.00
Min Z	4	13 SLS: (DL+	0.297	0.551	**-0.697**	0.937	0.00	-0.00
Max rX	1	21 SLS: (D+	-0.033	-0.067	0.320	0.328	**0.00**	0.00
Min rX	13	6 GENERATE	-0.131	-0.763	-0.009	0.774	**-0.00**	0.00
Max rY	14	21 SLS: (D+	-0.409	-0.541	1.506	1.652	-0.00	**0.00**
Min rY	4	21 SLS: (D+	-0.300	-0.166	1.517	1.555	0.00	**-0.00**
Max rZ	4	6 GENERATE	-0.320	-0.244	-0.034	0.404	0.00	-0.00
Min rZ	3	6 GENERATE	-0.279	-0.202	-0.016	0.345	-0.00	0.00
Max Re	7	4 WL+Z	0.200	-0.007	2.189	**2.198**	0.00	0.00

		2 LIVE LOAD	0.000	0.000	0.000	0.000	0.000
			1.313	0.000	-0.058	-0.001	0.058
			2.625	0.000	-0.086	-0.002	0.086
			3.938	0.000	-0.058	-0.001	0.058
			5.250	0.000	0.000	0.000	0.000

Maximum Drift Attained is 21mm<<<28mm (H/100) .Hence Structure is safe against Drift Check

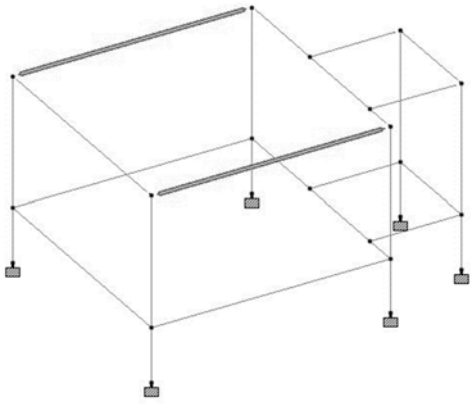
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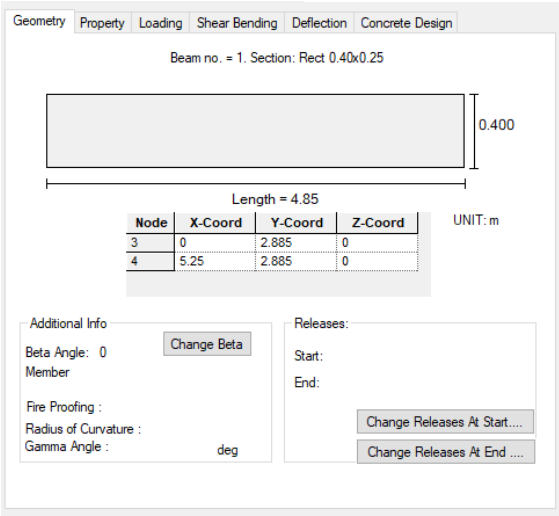
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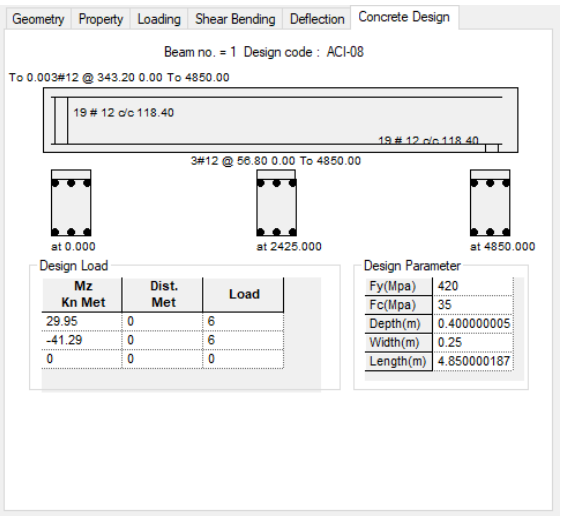
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10.5. Concrete Roof Beam Design

10.5.1. Design of Ledger Beam(LB1)







Design of Ledger Beam (LB1)

Maximum Moment required from Analysis = 41 **Knm (Hogging)**

Therefore Ast.req = $$A_s = 0.85 f_c' \frac{b}{Fy}\left(d - \sqrt{d^2 - \frac{2M_u}{0.85\Phi f_c' b}}\right)$$

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Where b=250mm, Fy=420 Mpa, d=425mm, Fc'=35 Mpa & ϕ =0.9

Ast1.req = **220 mm²**

Ast.min=
(As per clause-10.5-ACI-318)

$$A_{s,min} = \frac{0.25\sqrt{f'_c}}{f_y} b_w d \qquad (10\text{-}3)$$

and not less than $1.4 b_w d / f_y$.

Ast.min= **375mm²**

Maximum Moment required from Analysis = **30 Knm (Sagging)**

Ast2.req = **189 mm²**

Hence Provide minimum 3-16 dia-Full Straight Bars Top & 3-16 dia-Full Straight Bottom

Staad Output For LB1

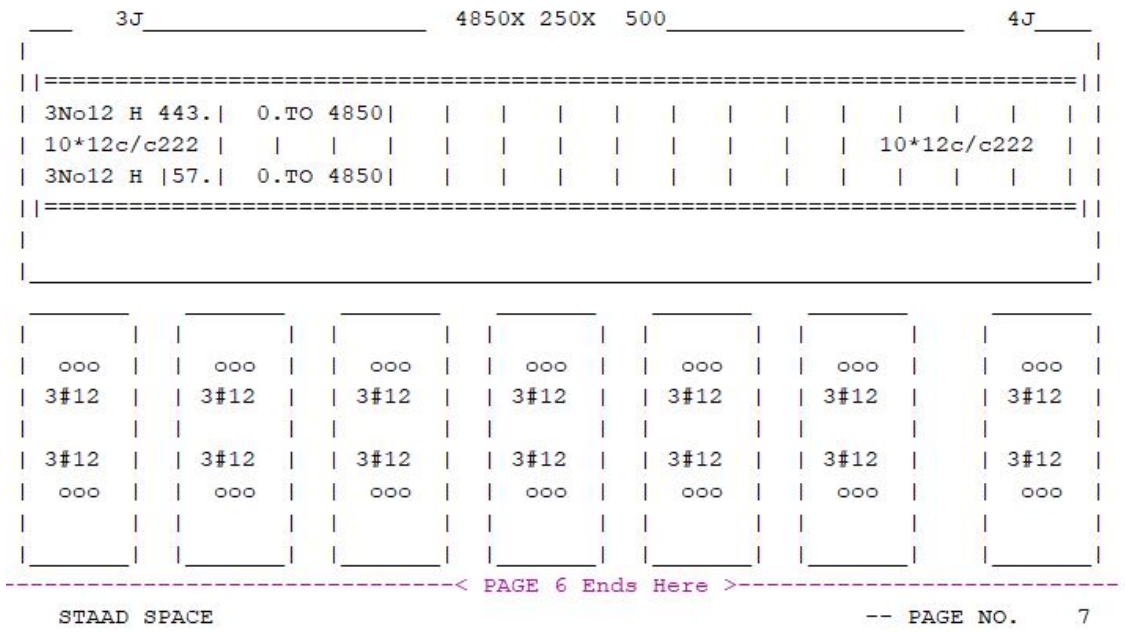
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AT START SUPPORT - Vu=   49.45 KNS  Vc=  115.82 KNS  Vs=    0.00 KNS
Tu=    2.00 KN-MET  Tc=    4.0 KN-MET  Ts=    0.0 KN-MET  LOAD     6
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 222. MM  C/C FOR 1988. MM

AT END   SUPPORT - Vu=   47.96 KNS  Vc=  121.30 KNS  Vs=    0.00 KNS
Tu=    3.56 KN-MET  Tc=    4.0 KN-MET  Ts=    0.0 KN-MET  LOAD     6
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 222. MM  C/C FOR 1988. MM
```


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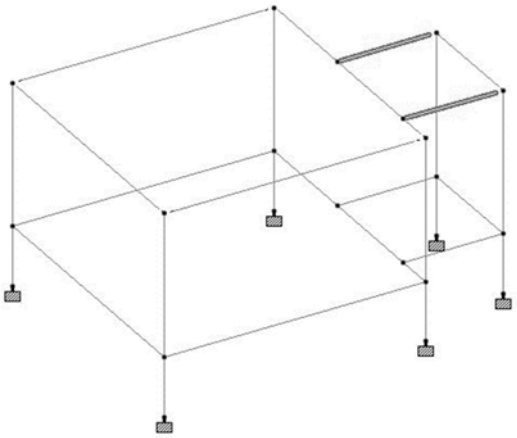
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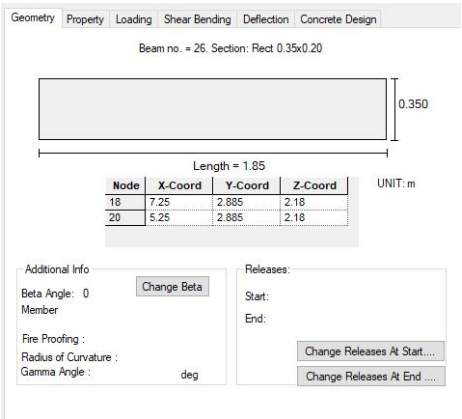
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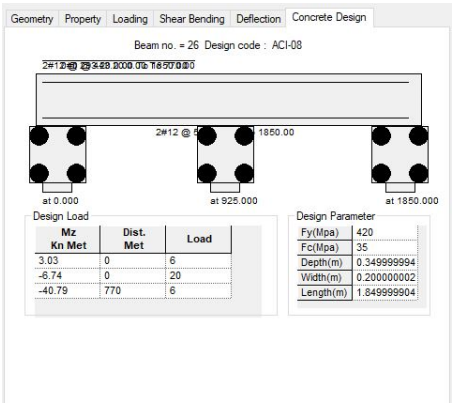
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10.6. Design of Beam-(LB2)







Provide minimum 3-12 dia-Full Straight Bars Top & Bottom

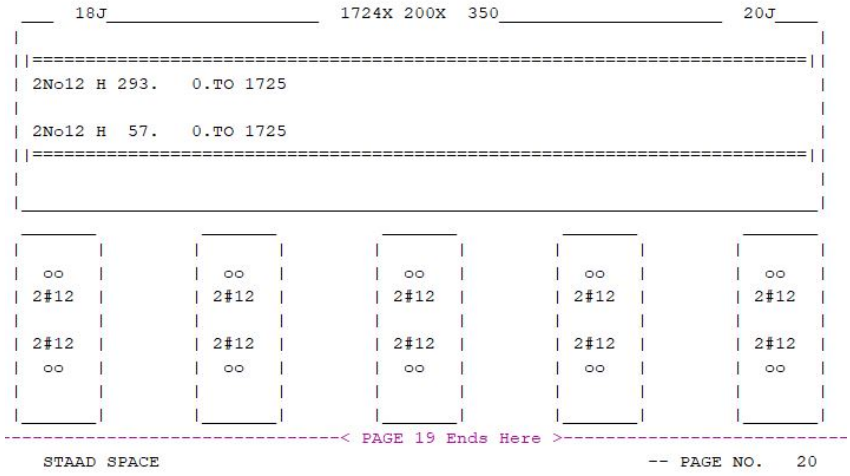
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Staad Output For LB2

```
AT START SUPPORT - Vu=     7.76 KNS  Vc=    59.22 KNS  Vs=     0.00 KNS
Tu=     0.15 KN-MET  Tc=     1.7 KN-MET  Ts=     0.0 KN-MET  LOAD      1
                  STIRRUPS ARE NOT REQUIRED.
AT END   SUPPORT - Vu=     4.02 KNS  Vc=    55.73 KNS  Vs=     0.00 KNS
Tu=     0.15 KN-MET  Tc=     1.7 KN-MET  Ts=     0.0 KN-MET  LOAD      1
                  STIRRUPS ARE NOT REQUIRED.
```



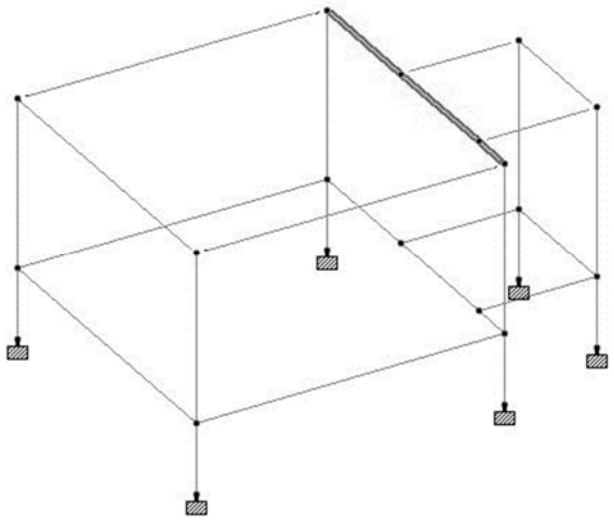

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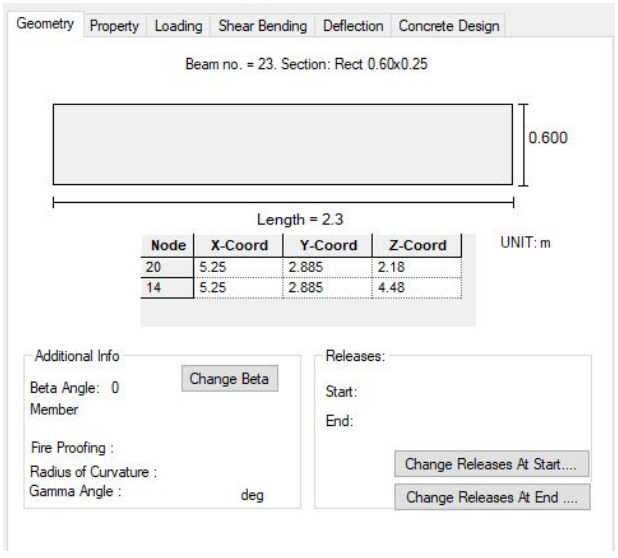

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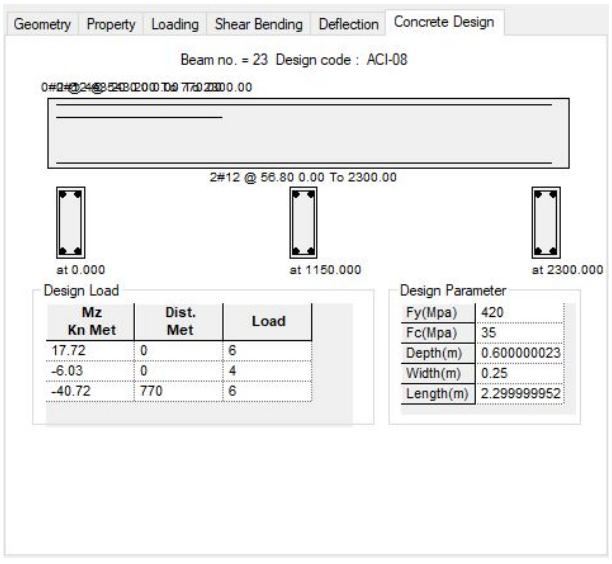
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10.7. Design of Beam-(B1)







Provide minimum 3-16 dia-Full Straight Bars Top and 3-16 dia-Full Straight Bars Top Bottom

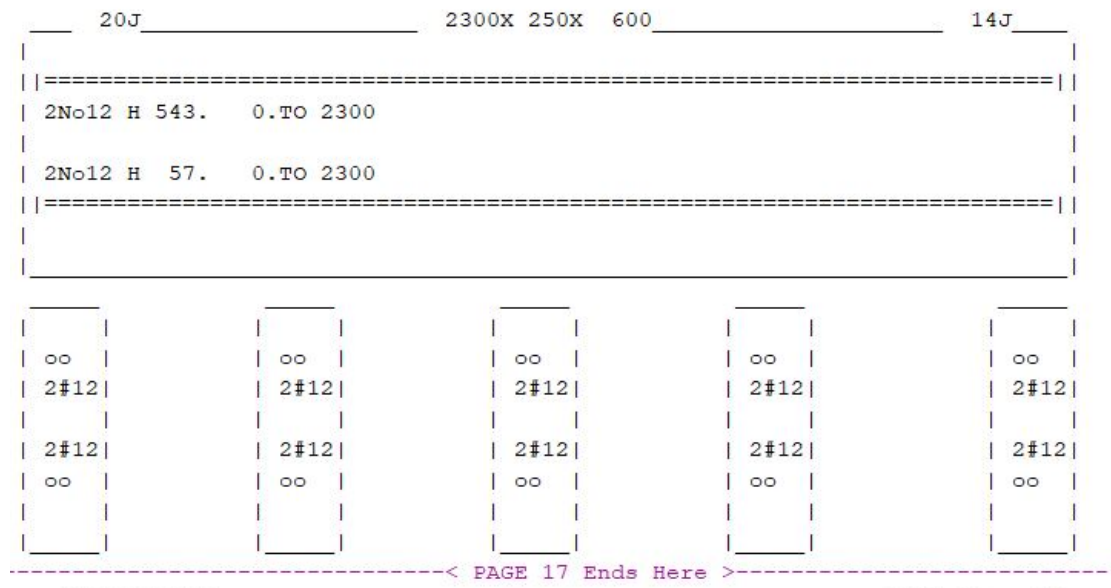
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Staad Output For B1

```
AT START SUPPORT - Vu=     1.81 KNS   Vc=  127.10 KNS   Vs=     0.00 KNS
Tu=     0.34 KN-MET   Tc=     4.9 KN-MET   Ts=     0.0 KN-MET   LOAD      1
                  STIRRUPS ARE NOT REQUIRED.
AT END   SUPPORT - Vu=     6.15 KNS   Vc=  128.05 KNS   Vs=     0.00 KNS
Tu=     0.88 KN-MET   Tc=     4.9 KN-MET   Ts=     0.0 KN-MET   LOAD      1
                  STIRRUPS ARE NOT REQUIRED.
```



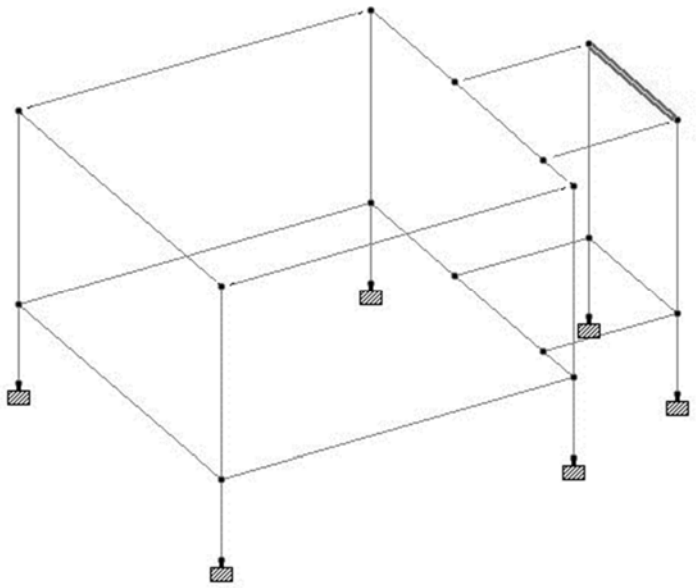

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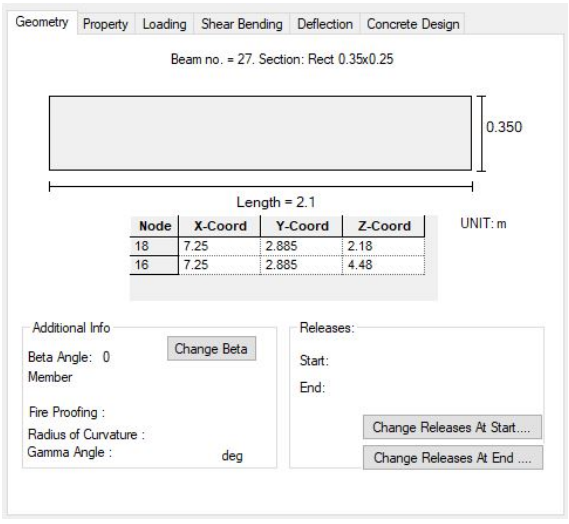

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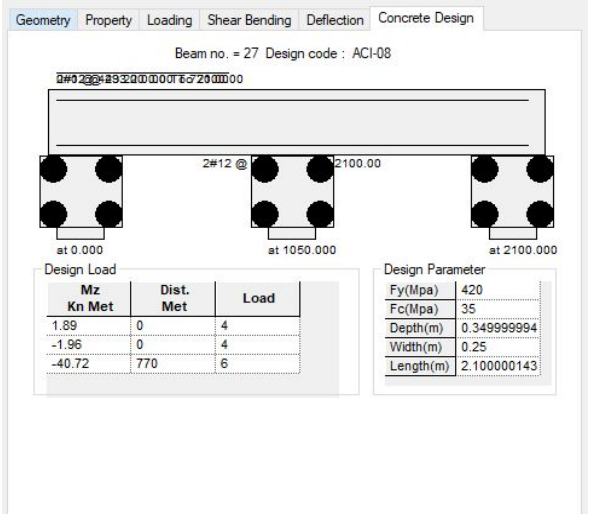
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10.8. Design of Beam-(B2)







Provide minimum 2-12 dia-Full Straight Bars Top and 2-12 dia-Full Straight Bars Top Bottom

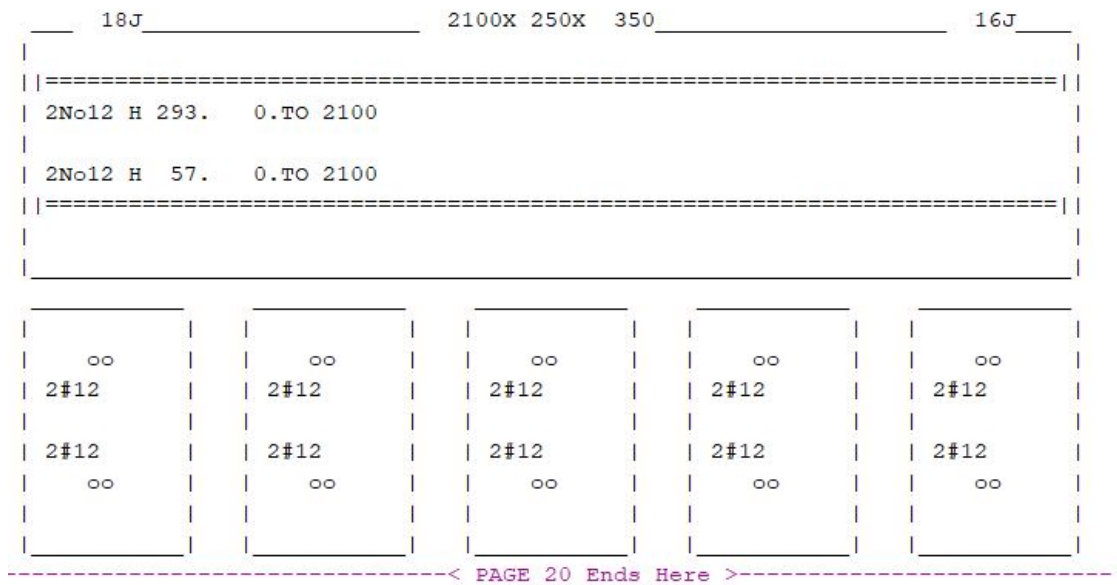
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Staad Output For B2

```
 AT START SUPPORT - Vu=    1.96 KNS  Vc=  126.59 KNS  Vs=    0.00 KNS
 Tu=    0.31 KN-MET  Tc=    2.4 KN-MET  Ts=    0.0 KN-MET  LOAD     1
                    STIRRUPS ARE NOT REQUIRED.
 AT END   SUPPORT - Vu=    1.19 KNS  Vc=   68.97 KNS  Vs=    0.00 KNS
 Tu=    0.31 KN-MET  Tc=    2.4 KN-MET  Ts=    0.0 KN-MET  LOAD     1
                    STIRRUPS ARE NOT REQUIRED.
```




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   STAAD SPACE                                                  -- PAGE NO.    21
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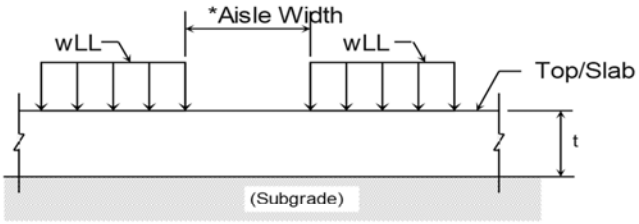
10.9. Concrete Grade Slab Design

CONCRETE SLAB ON GRADE ANALYSIS

For Slab Subjected to Stationary Uniformly Distributed Live Loads

Input Data:

Slab Thickness, t =	3.937	in.	100.000	mm		
Concrete Strength, f 'c =	5000	psi	35.000	kpa		
Subgrade Modulus, k =	39.7866	pci	10800	Kn/m3	(AS PER SECTION-7.7-SOIL REPORT)	
Factor of Safety, FS =	1.500		1.500			
Uniform Live Load, wLL =	104.45	psf	5.00	kpa		



Concrete Slab on Grade with Uniform Loads

Results:

Design Parameters:

Modulus of Rupture, MR =	636.40	psi	4387.79	kpa	MR = 9*SQRT(f 'c)
ow. Bending Stress, Fb =	424.26	psi	2925.19	kpa	Fb = MR/FS
Modulus of Elasticity, Ec =	4030509	psi	27789329	kpa	Ec = 57000*SQRT(f 'c)
Poisson's Ratio, μ =	0.15		0.15		μ = 0.15 (assumed for concrete)
Radius of Stiffness, Lr =	26.94	in.	684.37	mm	Lr = (Ec*t^3/(12*(1-μ^2)*k))^0.25
Critical Aisle Width, Wcr =	4.96	ft.	1511.77	mm	Wcr = (2.209*Lr)/12

Stationary Uniformly Distributed Live Loads:

wLL(allow) =	682.05	psf	wLL(allow) = 257.876*Fb*SQRT(k*t/Ec)
	32.65	kpa	**wLL(allow) >= wLL, O.K.**

Provide Minimum Reinforcement of 0.14%@both faces = 0.14%(h) = 140 mm2/M

Table 24.4.3.2—Minimum ratios of deformed shrinkage and temperature reinforcement area to gross concrete area

Reinforcement type	f_y, psi	Minimum reinforcement ratio	
Deformed bars	< 60,000	0.0020	
Deformed bars or welded wire reinforcement	≥ 60,000	Greater of:	$\frac{0.0018 \times 60,000}{f_y}$
			0.0014

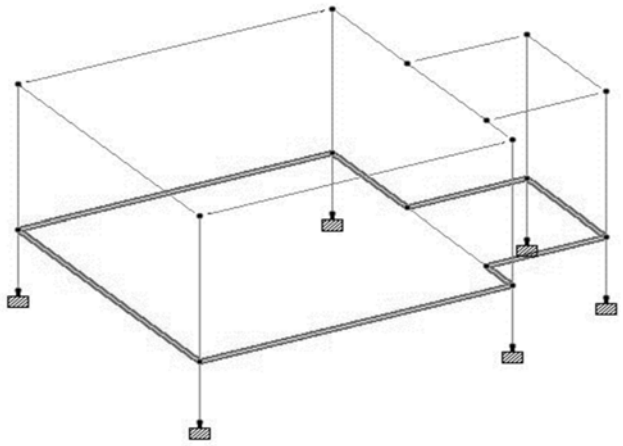
Provide WWF-6mm Dia@200 c-c Both Ways-TOP

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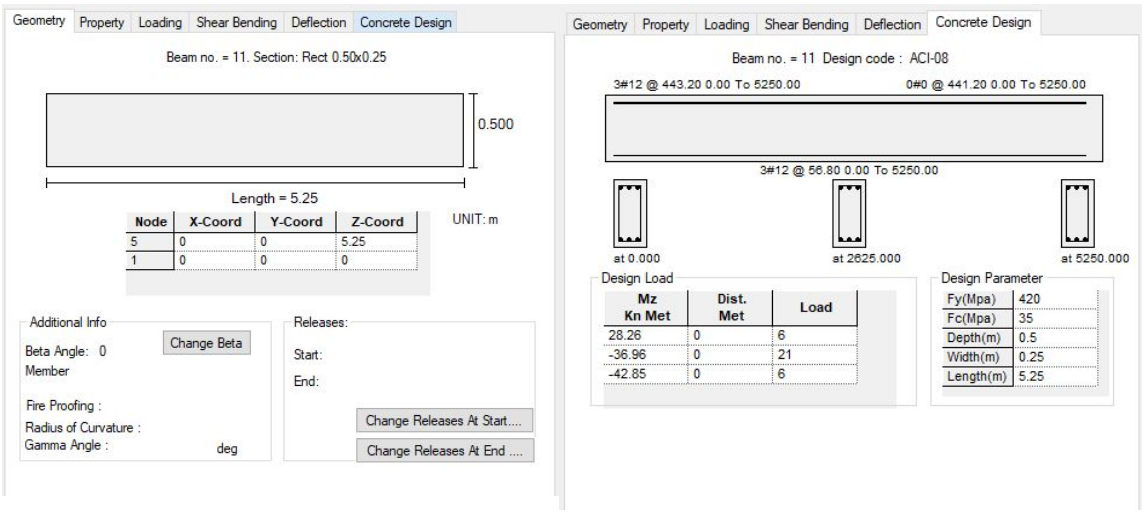
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10.10. Design of Grade Beam



10.10.1. Grade Beam(GB1)

MOMENT ENVELOPE FOR BEAMS GB1



Provide minimum 3-16 dia-Full Straight Bars Top & 3-16 dia-Full Straight Bars Bottom

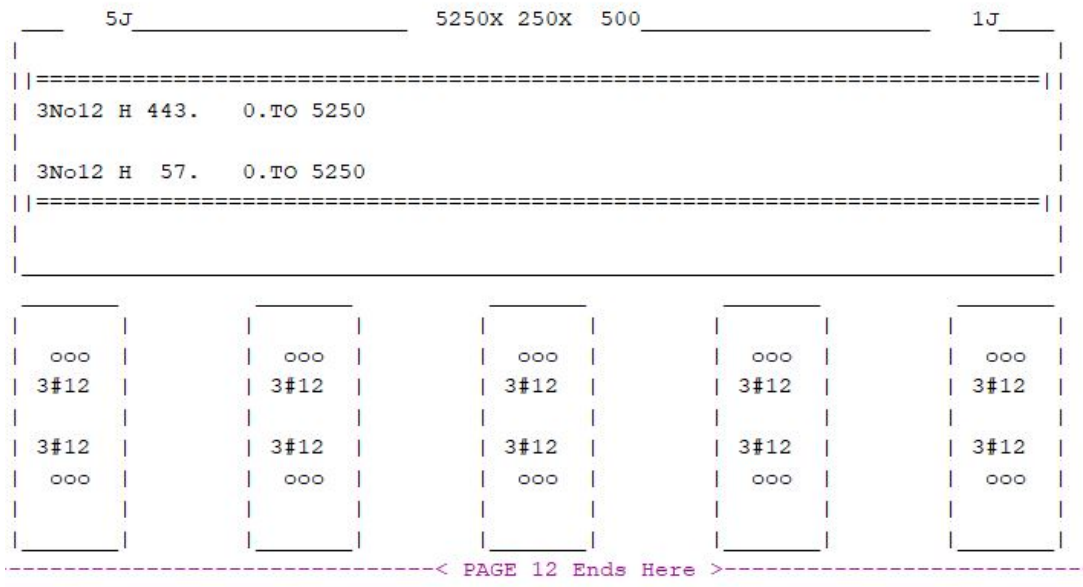
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Staad Output for GB1

```
AT START SUPPORT - Vu=   28.31 KNS  Vc=  110.14 KNS  Vs=    0.00 KNS
Tu=    0.01 KN-MET  Tc=    4.0 KN-MET  Ts=    0.0 KN-MET  LOAD     1
                  STIRRUPS ARE NOT REQUIRED.
AT END   SUPPORT - Vu=   28.35 KNS  Vc=  110.06 KNS  Vs=    0.00 KNS
Tu=    0.01 KN-MET  Tc=    4.0 KN-MET  Ts=    0.0 KN-MET  LOAD     1
                  STIRRUPS ARE NOT REQUIRED.
```




```
-------------------------------< PAGE 12 Ends Here >--------------------------
   STAAD SPACE                                          -- PAGE NO.   13
```

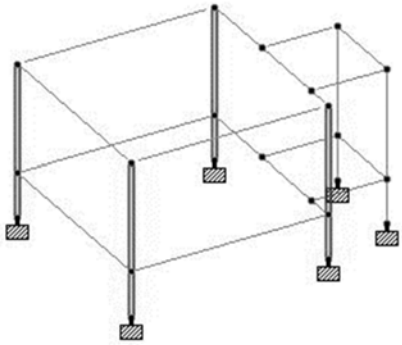

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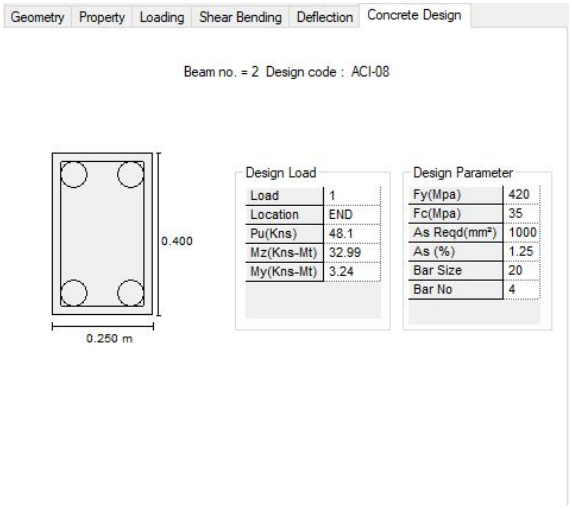
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10.11. Concrete Column Design

10.11.1. Design For Column C1





Column's C1

Provide 4-Dia20 Vertical Reinforcement in Column C1.

Staad Output For Column C1

```
=============================================================================

     COLUMN  NO.     3  DESIGN PER ACI 318-08 - AXIAL + BENDING


FY - 420.0  FC -  35.0 MPA,  RECT SIZE - 250.0 X 400.0 MMS, TIED
        ONLY MINIMUM STEEL IS REQUIRED.
        AREA OF STEEL REQUIRED =  1000.0  SQ. MM

BAR CONFIGURATION       REINF PCT.   LOAD   LOCATION   PHI
-------------------------------------------------------------

  4 - 20 MM              1.257        1      END     0.650
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER   12 SPACING 250.00 MM

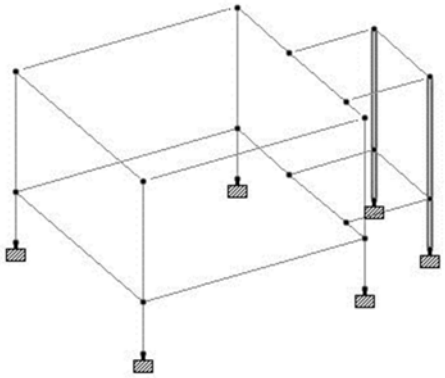

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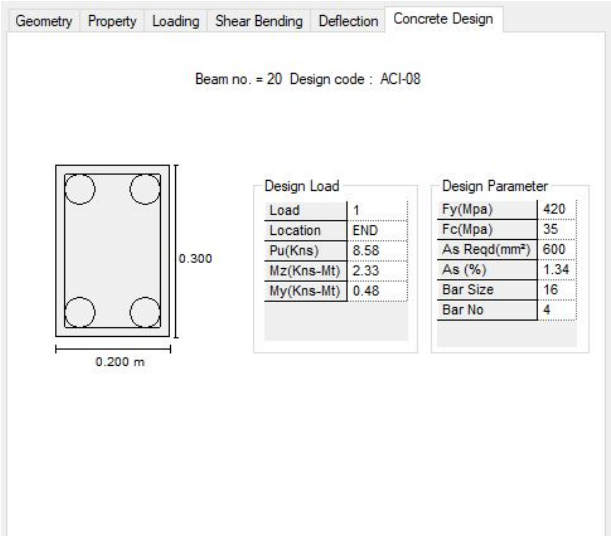

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10.12. Design For Column C2





Column's C2

However provide equivalent 4-Dia16 Vertical Reinforcement in Column C2.

Staad Output For Column C2

```
============================================================================

     COLUMN  NO.    20  DESIGN PER ACI 318-08 - AXIAL + BENDING


FY - 420.0  FC -  35.0 MPA,  RECT SIZE - 200.0 X 300.0 MMS, TIED
        ONLY MINIMUM STEEL IS REQUIRED.
        AREA OF STEEL REQUIRED =   600.0  SQ. MM

BAR CONFIGURATION       REINF PCT.   LOAD   LOCATION   PHI
------------------------------------------------------------

  4 - 16 MM              1.340        1      END      0.650
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER   12 SPACING 200.00 MM


============================================================================
```


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10.13. Cantilever Slab Design

DESIGN OF CANTILEVER SLAB

Geometrical Properties			Material Properties			
Span to center of support "sp_1"	0.8	met	Weight of reinforced concrete "γ_r"	24		kN/m^3
Clear Span	0.6	met	Strength of concrete "f_c"	35		N/mm^2
Clear span + depth of slab "sp_2"	0.75	met	Strength of steel "f_y"	420		N/mm^2
Max. eff span le = min (sp_1;sp_2)	0.8	met	Clear cover to concrete("c")	20		mm
Width of Slab "b"	1000	mm	Flexure strength reduction factor	0.9		
Thickness of slab "t"	150	mm	Shear strength reduction factor	0.75		
Eff. Depth "d"	125		Modulus of Elasticity of steel "Es"	2E+05		N/mm^2
Loading			Modulus of Elasticity of concrete	27806		N/mm^2
Self Weight of slab	3.6	kN/m^2	Modular Ratio "n = Es/Ec"	7.19		
Floor Finish	3.5	kN/m^2	**Provided Reinforcement**			
Wall load	0	kN/m^2	Top layer Nos.	6	Φ 10	mm
Total dead Load "U_{dl}"	7.1	kN/m^2	Bottom layer Nos.	0	Φ 12	mm
Live load	1	kN/m^2	Area of Top steel "$A_{s\,t}$"		468	mm^2
Total Ultimate Load "Uu_{tot}"$_{(1.2D+1.6L)}$	10.12	kN/m^2	Area of Bottom steel "$A_{s\,b}$"		0	mm^2
Moment due to Overhang Point Load(0.125*0.15*24*0.6)	0.27	kNm	"db" = 10 mm			
Calculations						
Shear force Vu = U*le	**8.096**	**kN**				
Neg. Moment Mu = U*le^2/2	**3.51**	**kNm**				

Check for Section capacity								
Depth of comp. block "a" = $A_{s\,t}$*f_y/(0.85*f'_c*b)	6.61	mm						
Section capacity ΦMn = Φ*$A_{s\,t}$*f_y*(d-a/2)	**21.94**	**kNm**	**O.K**	>	**3.51**	**kNm**	(Mu)	
Shear requirements								
Shear capacity ΦVc = Φ*0.17*√f'c*b*d	**94.29**	**kN**	**O.K**	>	**8.096**	**kN**	(Vu)	
Serviceability requirements								
d' = c+d_b/2	26	mm						
B = b/(n*$A_{s\,t}$)	0.297							
r = (n-1)*$A_{s\,b}$/(n*$A_{s\,t}$)	0							
k_d = (√(2*d*B(1+r*d'/d)+(1+r)^2)-(1+r))/B	29.01	mm						
I_{cr} = bkd^3/3+n*$A_{s\,t}$*(d-k_d)^2+(n-1)*$A_{s\,b}$*(k_d-d'	3.915E+07	mm^4						
Calculate Gross moment of inertia								
Area of slab A = t*b	2E+05	mm^2						
Depth of neutral axis "y"	75	mm						
Distance of top fiber from N.A "y_t"	75	mm						
Gross moment of inertia "I_g"	2.813E+08	mm^4						
Allowable stress fr = 0.62√f'c	3.67	mm^2						
Cracking moment M_{cr} = fr*I_g/y_t	13.75	kNm						

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Service moments

Moment due to dead load "M_{dl}"	2.54	kNm	
Moment due to live load "Mll"	0.32	kNm	
Total Moment Ms = $M_{dl}+M_{ll}$	2.86	kNm	
Deflection due to dead + live loads			
R(dl+ll) = M_{cr}/Ms	1.000		
I_e(dl+ll) = $R^3*I_g+(1-R^3)*I_{cr}$	2.813E+08	mm^4	
δ = K*5/48*Ms*le^2/(Ec*I_e)	0.08	mm	**K = 3.2 (conservative)**
Deflection due to dead load			
R(dl) = M_{cr}/Md	5.411		
I_e(dl) = $R^3*I_g+(1-R^3)*I_{cr}$	3.839E+10	mm^4	
δ = K*5/48*Md*le^2/(Ec*I_e)	0.00	mm	
Deflection due to live load			
Live load deflection = dl+ll deflection - dl deflection	0.08	mm	
Allowable immediate deflection due to Live load = per ACI 318 Tab 9.5b I_e/360	**2.22**	**mm**	**> 0.08 mm O.K**
Long term deflection			
% of liveload sustained "SL"	50	%	
M_{sus} = M_{dl}+SL*M_{ll}	2.70	kNm	
R(sus) = M_{cr}/Msus	5.091		
I_e(sus) = $R^3*I_g+(1-R^3)*I_{cr}$	3.198E+10		
δ = K*5/48*Msus*le^2/(Ec*I_e)	0.00	mm	

Magnification factor for creep and shrinkage

$\lambda_\Delta = \xi / (1+50*\rho')$ 2.000

Where:-

ξ = 2 for 5 years or more cl.9.5.2.5 ACI 318

$\rho' = A_{s\ b}/(b*d)$ 0

Total deflection = λ_Δ*sus deflection + ll deflection 0.079 mm

Allowable deflection = I_e/240 **3.333 mm > 0.079 mm O.K**

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10.14. Foundation Design

Foundations have been designed based on the Soil Report Recommendations. The minimum Embedment Depth of Foundations has been considered as 1.2m & applicable SBC of 500 Kpa. (Refer Doc-MD-216-3000-EG-CV-CAL-1008-A01 for Details)

TABLE 4: RECOMMENDED NET ALLOWABLE BEARING CAPACITIES

PRPJECT: MANSOURAH - MANSSARAH GOLD PROJECT
CLIENT: L & T CONSTRUCTION

Profile / Borehole	Foundation Type	Depth of Foundation below FGL	Footing Size (B x L)	Net Allowable Bearing Capacity Based on		Recommended Net Allowable Bearing Capacity	Estimated Total Settlement
				Settlement	Shear Failure		
-	-	(m)	(m)	(kN/m^2)	(kN/m^2)	(kN/m^2)	(mm)
BH-01	Square Isolated	1.20	1.0 x 1.0	1500	570	570	25
		1.50		2000	850	850	25
		1.20	2.0 x 2.0	1000	750	750	25
		1.50		1200	980	980	25
		1.20	3.0 x 3.0	775	900	775	25
		1.50		950	1120	950	25
		1.20	4.0 x 4.0	625	1040	625	25
		1.50		750	1250	750	25
		1.20	5.0 x 5.0	525	1172	525	25
		1.50		600	1342	600	25

Refer Attachment- II for Foundation Design calculations designed by STAAD Pro Foundation

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REFERENCES

1. R.C DETAIL OF SWIMMING POOL — MD-216-6000-EG-CV-DED-1537
2. BUILDING ELEVATIONS & SECTIONS — MD-216-4000-EG-AR-GAD-3026
3. LAYOUT & DETAILS OF FDN, COL & GRADE BEAM — MD-216-6000-EG-CV-DED-1538
4. GEOTECHNICAL REPORT — MD-216-3000-EG-CV-CAL-1008
5. STRUCTURAL DESIGN CRITERIA — MD-216-0000-EG-ST-CRT-0001

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ATTACHMENT 1 STAAD INPUT FILE FOR SWIMMING POOL

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STAAD SPACE
START JOB INFORMATION
ENGINEER DATE 12-Aug-20
END JOB INFORMATION
INPUT WIDTH 79
UNIT METER KN
JOINT COORDINATES
1 4 4 0; 5 16.255 4 0; 9 19.045 4.8 0; 11 29 4.8 0; 13 29 6.15 0; 15 4 6.15 0;
874 4 5.075 0; 1095 29 5.475 0; 3193 16.255 6.15 0; 3194 19.045 6.15 0;
3195 4.51062 6.15 0; 3196 4.51062 5.6125 0; 3197 4 5.6125 0;
3198 5.02125 6.15 0; 3199 5.02125 5.6125 0; 3200 5.53187 6.15 0;
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3273 7.06375 4.5375 0; 3274 7.57438 4.5375 0; 3275 8.085 4.5375 0;
3276 8.59562 4.5375 0; 3277 9.10625 4.5375 0; 3278 9.61687 4.5375 0;
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3311 14.2125 4 0; 3312 14.7231 4 0; 3313 15.2337 4 0; 3314 15.7444 4 0;
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3318 17.65 5.7125 0; 3319 18.3475 6.15 0; 3320 18.3475 5.7625 0;
3321 19.045 5.8125 0; 3322 16.9525 5.175 0; 3323 17.65 5.275 0;
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3327 17.65 4.8375 0; 3328 18.3475 4.9875 0; 3329 19.045 5.1375 0;
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3337 20.5382 6.15 0; 3338 20.5382 5.8125 0; 3339 21.036 6.15 0;
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3343 22.0315 6.15 0; 3344 22.0315 5.8125 0; 3345 22.5292 6.15 0;
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3567 3806 3807 3832 3831; 3568 3807 3808 3833 3832; 3569 3808 3809 3834 3833;

3570 3809 3810 3835 3834; 3571 3810 3811 3836 3835; 3572 3811 3812 3837 3836;

3573 3812 3813 3838 3837; 3574 3813 3814 3839 3838; 3575 3814 3815 3840 3839;

3576 3815 3816 3841 3840; 3577 3816 3817 3842 3841; 3578 3817 3818 3843 3842;

3579 3818 3819 3844 3843; 3580 3819 3820 3845 3844; 3581 3820 3821 3846 3845;

3582 3821 3822 3847 3846; 3583 3822 3823 3848 3847; 3584 3823 3824 3849 3848;

3585 3826 3825 3850 3851; 3586 3825 3827 3852 3850; 3587 3827 3828 3853 3852;

3588 3828 3829 3854 3853; 3589 3829 3830 3855 3854; 3590 3830 3831 3856 3855;

3591 3831 3832 3857 3856; 3592 3832 3833 3858 3857; 3593 3833 3834 3859 3858;

3594 3834 3835 3860 3859; 3595 3835 3836 3861 3860; 3596 3836 3837 3862 3861;

3597 3837 3838 3863 3862; 3598 3838 3839 3864 3863; 3599 3839 3840 3865 3864;

3600 3840 3841 3866 3865; 3601 3841 3842 3867 3866; 3602 3842 3843 3868 3867;

3603 3843 3844 3869 3868; 3604 3844 3845 3870 3869; 3605 3845 3846 3871 3870;

3606 3846 3847 3872 3871; 3607 3847 3848 3873 3872; 3608 3848 3849 3874 3873;

3609 3851 3850 3875 3876; 3610 3850 3852 3877 3875; 3611 3852 3853 3878 3877;

3612 3853 3854 3879 3878; 3613 3854 3855 3880 3879; 3614 3855 3856 3881 3880;

3615 3856 3857 3882 3881; 3616 3857 3858 3883 3882; 3617 3858 3859 3884 3883;

3618 3859 3860 3885 3884; 3619 3860 3861 3886 3885; 3620 3861 3862 3887 3886;

3621 3862 3863 3888 3887; 3622 3863 3864 3889 3888; 3623 3864 3865 3890 3889;

3624 3865 3866 3891 3890; 3625 3866 3867 3892 3891; 3626 3867 3868 3893 3892;

3627 3868 3869 3894 3893; 3628 3869 3870 3895 3894; 3629 3870 3871 3896 3895;

3630 3871 3872 3897 3896; 3631 3872 3873 3898 3897; 3632 3873 3874 3899 3898;

3633 3876 3875 3900 3901; 3634 3875 3877 3902 3900; 3635 3877 3878 3903 3902;

3636 3878 3879 3904 3903; 3637 3879 3880 3905 3904; 3638 3880 3881 3906 3905;

3639 3881 3882 3907 3906; 3640 3882 3883 3908 3907; 3641 3883 3884 3909 3908;

3642 3884 3885 3910 3909; 3643 3885 3886 3911 3910; 3644 3886 3887 3912 3911;

3645 3887 3888 3913 3912; 3646 3888 3889 3914 3913; 3647 3889 3890 3915 3914;

3648 3890 3891 3916 3915; 3649 3891 3892 3917 3916; 3650 3892 3893 3918 3917;

3651 3893 3894 3919 3918; 3652 3894 3895 3920 3919; 3653 3895 3896 3921 3920;

3654 3896 3897 3922 3921; 3655 3897 3898 3923 3922; 3656 3898 3899 3924 3923;

3657 3901 3900 3925 3926; 3658 3900 3902 3927 3925; 3659 3902 3903 3928 3927;

3660 3903 3904 3929 3928; 3661 3904 3905 3930 3929; 3662 3905 3906 3931 3930;

3663 3906 3907 3932 3931; 3664 3907 3908 3933 3932; 3665 3908 3909 3934 3933;

3666 3909 3910 3935 3934; 3667 3910 3911 3936 3935; 3668 3911 3912 3937 3936;

3669 3912 3913 3938 3937; 3670 3913 3914 3939 3938; 3671 3914 3915 3940 3939;

3672 3915 3916 3941 3940; 3673 3916 3917 3942 3941; 3674 3917 3918 3943 3942;

3675 3918 3919 3944 3943; 3676 3919 3920 3945 3944; 3677 3920 3921 3946 3945;

3678 3921 3922 3947 3946; 3679 3922 3923 3948 3947; 3680 3923 3924 3949 3948;

3681 3926 3925 3950 3951; 3682 3925 3927 3952 3950; 3683 3927 3928 3953 3952;

3684 3928 3929 3954 3953; 3685 3929 3930 3955 3954; 3686 3930 3931 3956 3955;

3687 3931 3932 3957 3956; 3688 3932 3933 3958 3957; 3689 3933 3934 3959 3958;

3690 3934 3935 3960 3959; 3691 3935 3936 3961 3960; 3692 3936 3937 3962 3961;

3693 3937 3938 3963 3962; 3694 3938 3939 3964 3963; 3695 3939 3940 3965 3964;

3696 3940 3941 3966 3965; 3697 3941 3942 3967 3966; 3698 3942 3943 3968 3967;

3699 3943 3944 3969 3968; 3700 3944 3945 3970 3969; 3701 3945 3946 3971 3970;

3702 3946 3947 3972 3971; 3703 3947 3948 3973 3972; 3704 3948 3949 3974 3973;

3705 3951 3950 3975 3976; 3706 3950 3952 3977 3975; 3707 3952 3953 3978 3977;

3708 3953 3954 3979 3978; 3709 3954 3955 3980 3979; 3710 3955 3956 3981 3980;

3711 3956 3957 3982 3981; 3712 3957 3958 3983 3982; 3713 3958 3959 3984 3983;

3714 3959 3960 3985 3984; 3715 3960 3961 3986 3985; 3716 3961 3962 3987 3986;

3717 3962 3963 3988 3987; 3718 3963 3964 3989 3988; 3719 3964 3965 3990 3989;

3720 3965 3966 3991 3990; 3721 3966 3967 3992 3991; 3722 3967 3968 3993 3992;

3723 3968 3969 3994 3993; 3724 3969 3970 3995 3994; 3725 3970 3971 3996 3995;

3726 3971 3972 3997 3996; 3727 3972 3973 3998 3997; 3728 3973 3974 3999 3998;

3729 3976 3975 4000 4001; 3730 3975 3977 4002 4000; 3731 3977 3978 4003 4002;

3732 3978 3979 4004 4003; 3733 3979 3980 4005 4004; 3734 3980 3981 4006 4005;

3735 3981 3982 4007 4006; 3736 3982 3983 4008 4007; 3737 3983 3984 4009 4008;

3738 3984 3985 4010 4009; 3739 3985 3986 4011 4010; 3740 3986 3987 4012 4011;

3741 3987 3988 4013 4012; 3742 3988 3989 4014 4013; 3743 3989 3990 4015 4014;

3744 3990 3991 4016 4015; 3745 3991 3992 4017 4016; 3746 3992 3993 4018 4017;

3747 3993 3994 4019 4018; 3748 3994 3995 4020 4019; 3749 3995 3996 4021 4020;

3750 3996 3997 4022 4021; 3751 3997 3998 4023 4022; 3752 3998 3999 4024 4023;

3753 4001 4000 4025 4026; 3754 4000 4002 4027 4025; 3755 4002 4003 4028 4027;

3756 4003 4004 4029 4028; 3757 4004 4005 4030 4029; 3758 4005 4006 4031 4030;

3759 4006 4007 4032 4031; 3760 4007 4008 4033 4032; 3761 4008 4009 4034 4033;

3762 4009 4010 4035 4034; 3763 4010 4011 4036 4035; 3764 4011 4012 4037 4036;

3765 4012 4013 4038 4037; 3766 4013 4014 4039 4038; 3767 4014 4015 4040 4039;

3768 4015 4016 4041 4040; 3769 4016 4017 4042 4041; 3770 4017 4018 4043 4042;

LEMENT PROPERTY

2935 2937 2939 2941 2943 2945 2947 2949 2951 2953 2955 2957 2959 2961 2963 -

2965 2967 2969 2971 2973 2975 2977 2979 2981 TO 3004 3006 3008 TO 3030 3032 -

3034 3036 3038 3040 3042 3044 3046 3048 3050 3052 3054 3056 3058 3060 3062 -

3064 3066 3068 3070 3072 3074 3076 3078 3079 3081 3083 3085 3087 TO 3090 -

3092 TO 3095 3097 3099 3101 3103 3104 3225 TO 3336 4977 TO 5095 -

5096 THICKNESS 0.25

3106 3108 3110 3112 3114 3116 3118 3120 3122 3124 3126 3128 3130 3132 3134 -

3136 3138 3140 3142 3144 TO 3183 3185 3187 3189 3191 3193 3195 3197 3199 -

3201 3203 3205 3207 3209 3211 3213 3215 3217 3219 3221 3223 3224 -

3337 TO 3416 4857 TO 4976 THICKNESS 0.2

3417 TO 4856 THICKNESS 0.4

DEFINE MATERIAL START

ISOTROPIC CONCRETE

E 2.17185e+007

POISSON 0.17

DENSITY 23.5616

ALPHA 1e-005

DAMP 0.05

TYPE CONCRETE

STRENGTH FCU 27579

END DEFINE MATERIAL

CONSTANTS

MATERIAL CONCRETE ALL

SUPPORTS

1 5 9 11 3292 TO 3314 3330 TO 3332 3411 TO 3433 3537 TO 3559 3575 TO 3577 -

3656 TO 5095 ELASTIC MAT DIRECT Y SUBGRADE 21000 PRINT

LOAD 1 LOADTYPE Dead TITLE DEAD LOAD

SELFWEIGHT Y -1

ELEMENT LOAD

4977 TO 5096 PR GX 5

4857 TO 4976 PR GX -5

3225 TO 3416 PR GZ -5

2935 2937 2939 2941 2943 2945 2947 2949 2951 2953 2955 2957 2959 2961 2963 -

2965 2967 2969 2971 2973 2975 2977 2979 2981 TO 3004 3006 3008 TO 3030 3032 -

3034 3036 3038 3040 3042 3044 3046 3048 3050 3052 3054 3056 3058 3060 3062 -

3064 3066 3068 3070 3072 3074 3076 3078 3079 3081 3083 3085 3087 TO 3090 -

3092 TO 3095 3097 3099 3101 3103 3104 3106 3108 3110 3112 3114 3116 3118 -

3120 3122 3124 3126 3128 3130 3132 3134 3136 3138 3140 3142 3144 TO 3183 -

3185 3187 3189 3191 3193 3195 3197 3199 3201 3203 3205 3207 3209 3211 3213 -

3215 3217 3219 3221 3223 3224 PR GZ 5

LOAD 2 LOADTYPE None TITLE WATER PRESSURE ON RAFT

ELEMENT LOAD

3417 TO 4136 PR GY -21.5

4257 TO 4856 PR GY -13.5

4137 TO 4256 PR GY -17.5

LOAD 3 LOADTYPE None TITLE HYDROSTATIC PRESSSURE

ELEMENT LOAD

4858 TRAP JT 0.0010004 0.0010004 3.37575 3.37575
4859 TRAP JT 0.0010004 0.0010004 3.37575 3.37575
4860 TRAP JT 0.0010004 0.0010004 3.37575 3.37575
4861 TRAP JT 0.0010004 0.0010004 3.37575 3.37575
4862 TRAP JT 0.0010004 0.0010004 3.37575 3.37575
4863 TRAP JT 0.0010004 0.0010004 3.37575 3.37575
4864 TRAP JT 0.0010004 0.0010004 3.37575 3.37575
4865 TRAP JT 0.0010004 0.0010004 3.37575 3.37575
4866 TRAP JT 0.0010004 0.0010004 3.37575 3.37575
4867 TRAP JT 0.0010004 0.0010004 3.37575 3.37575
4868 TRAP JT 0.0010004 0.0010004 3.37575 3.37575
4869 TRAP JT 0.0010004 0.0010004 3.37575 3.37575
4870 TRAP JT 0.0010004 0.0010004 3.37575 3.37575
4871 TRAP JT 0.0010004 0.0010004 3.37575 3.37575
4872 TRAP JT 0.0010004 0.0010004 3.37575 3.37575
4873 TRAP JT 0.0010004 0.0010004 3.37575 3.37575
4874 TRAP JT 0.0010004 0.0010004 3.37575 3.37575
4875 TRAP JT 0.0010004 0.0010004 3.37575 3.37575
4876 TRAP JT 0.0010004 0.0010004 3.37575 3.37575
4877 TRAP JT 0.0010004 0.0010004 3.37575 3.37575
4878 TRAP JT 0.0010004 0.0010004 3.37575 3.37575
4879 TRAP JT 0.0010004 0.0010004 3.37575 3.37575
4880 TRAP JT 0.0010004 0.0010004 3.37575 3.37575
4881 TRAP JT 0.0010004 0.0010004 3.37575 3.37575
4882 TRAP JT 0.0010004 0.0010004 3.37575 3.37575
4883 TRAP JT 0.0010004 0.0010004 3.37575 3.37575
4884 TRAP JT 0.0010004 0.0010004 3.37575 3.37575
4885 TRAP JT 0.0010004 0.0010004 3.37575 3.37575
4886 TRAP JT 0.0010004 0.0010004 3.37575 3.37575
4887 TRAP JT 3.37575 3.37575 6.7505 6.7505
4888 TRAP JT 3.37575 3.37575 6.7505 6.7505
4889 TRAP JT 3.37575 3.37575 6.7505 6.7505
4890 TRAP JT 3.37575 3.37575 6.7505 6.7505
4891 TRAP JT 3.37575 3.37575 6.7505 6.7505
4892 TRAP JT 3.37575 3.37575 6.7505 6.7505
4893 TRAP JT 3.37575 3.37575 6.7505 6.7505
4894 TRAP JT 3.37575 3.37575 6.7505 6.7505
4895 TRAP JT 3.37575 3.37575 6.7505 6.7505

LOAD 4 LOADTYPE Soil TITLE SOIL PRESSURE

ELEMENT LOAD

4978 TRAP JT -0.0010004 -0.0010004 -3.19325 -3.19325

4979 TRAP JT -0.0010004 -0.0010004 -3.19325 -3.19325

4980 TRAP JT -0.0010004 -0.0010004 -3.19325 -3.19325

4981 TRAP JT -0.0010004 -0.0010004 -3.19325 -3.19325

4982 TRAP JT -0.0010004 -0.0010004 -3.19325 -3.19325

4983 TRAP JT -0.0010004 -0.0010004 -3.19325 -3.19325

4984 TRAP JT -0.0010004 -0.0010004 -3.19325 -3.19325

4985 TRAP JT -0.0010004 -0.0010004 -3.19325 -3.19325

4986 TRAP JT -0.0010004 -0.0010004 -3.19325 -3.19325

4987 TRAP JT -0.0010004 -0.0010004 -3.19325 -3.19325

4988 TRAP JT -0.0010004 -0.0010004 -3.19325 -3.19325

4989 TRAP JT -0.0010004 -0.0010004 -3.19325 -3.19325

4990 TRAP JT -0.0010004 -0.0010004 -3.19325 -3.19325

4991 TRAP JT -0.0010004 -0.0010004 -3.19325 -3.19325

4992 TRAP JT -0.0010004 -0.0010004 -3.19325 -3.19325

4993 TRAP JT -0.0010004 -0.0010004 -3.19325 -3.19325

4994 TRAP JT -0.0010004 -0.0010004 -3.19325 -3.19325

4995 TRAP JT -0.0010004 -0.0010004 -3.19325 -3.19325

4996 TRAP JT -0.0010004 -0.0010004 -3.19325 -3.19325

LOAD COMB 5 MAINTENENCE

1 1.0 4 1.0

LOAD COMB 6 POOL FULL WITHOUT BACKFILL

1 1.0 2 1.0 3 1.0

LOAD COMB 7 POOL EMPTY WITHOUT BACKFILL

1 1.0

LOAD COMB 8 RUNING CONDITION

1 1.0 2 1.0 3 1.0 4 1.0

LOAD COMB 9 ULT MAINTENENCE

1 1.4 4 1.0

LOAD COMB 10 ULT POOL FULL WITHOUT BACKFILL

1 1.4 3 1.7 2 1.0

LOAD COMB 11 ULT POOL EMPTY WITHOUT BACKFILL

1 1.4

LOAD COMB 12 ULT RUNNING CONDITION

1 1.4 3 1.7 2 1.0 4 1.0

PERFORM ANALYSIS

START CONCRETE DESIGN

CODE ACI

FC 35000 ALL

FYMAIN 420000 ALL

DESIGN ELEMENT 2935 2937 2939 2941 2943 2945 2947 2949 2951 2953 2955 2957 -

2959 2961 2963 2965 2967 2969 2971 2973 2975 2977 2979 2981 TO 3004 3006 -

3008 TO 3030 3032 3034 3036 3038 3040 3042 3044 3046 3048 3050 3052 3054 -

3056 3058 3060 3062 3064 3066 3068 3070 3072 3074 3076 3078 3079 3081 3083 -

3085 3087 TO 3090 3092 TO 3095 3097 3099 3101 3103 3104 3106 3108 3110 3112 -

3114 3116 3118 3120 3122 3124 3126 3128 3130 3132 3134 3136 3138 3140 3142 -

3144 TO 3183 3185 3187 3189 3191 3193 3195 3197 3199 3201 3203 3205 3207 -

3209 3211 3213 3215 3217 3219 3221 3223 TO 5096

END CONCRETE DESIGN

FINISH

ATTACHMENT 2 STAAD INPUT FOR PUMP ROOM

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STAAD SPACE
START JOB INFORMATION
ENGINEER DATE 05-Aug-20
ENGINEER NAME AHMED
END JOB INFORMATION
INPUT WIDTH 79
UNIT METER KN
JOINT COORDINATES
1 0 0 0; 2 5.25 0 0; 3 0 2.885 0; 4 5.25 2.885 0; 5 0 0 5.25; 6 5.25 0 5.25;
7 0 2.885 5.25; 8 5.25 2.885 5.25; 9 0 -1.2 0; 10 5.25 -1.2 0; 11 0 -1.2 5.25;
12 5.25 -1.2 5.25; 13 5.25 0 4.48; 14 5.25 2.885 4.48; 15 7.25 0 4.48;
16 7.25 2.885 4.48; 17 7.25 0 2.18; 18 7.25 2.885 2.18; 19 5.25 0 2.18;
20 5.25 2.885 2.18; 21 7.25 -1.2 4.48; 22 7.25 -1.2 2.18;
MEMBER INCIDENCES
1 3 4; 2 1 3; 3 2 4; 4 7 8; 5 5 7; 6 6 8; 7 3 7; 8 4 20; 9 2 19; 10 6 5;
11 5 1; 12 1 2; 13 1 9; 14 2 10; 15 5 11; 16 6 12; 17 13 6; 18 14 8; 20 15 16;
21 17 18; 22 19 13; 23 20 14; 25 16 14; 26 18 20; 27 18 16; 28 15 21; 29 17 22;
30 15 13; 31 17 19; 32 17 15;
DEFINE MATERIAL START
ISOTROPIC CONCRETE
E 2.17185e+007
POISSON 0.17
DENSITY 23.5616
ALPHA 1e-005
DAMP 0.05
TYPE CONCRETE
STRENGTH FCU 27579
END DEFINE MATERIAL
MEMBER PROPERTY AMERICAN
2 3 5 6 13 TO 16 PRIS YD 0.4 ZD 0.25
1 4 9 TO 12 17 22 30 TO 32 PRIS YD 0.5 ZD 0.25
MEMBER PROPERTY AMERICAN
25 26 PRIS YD 0.35 ZD 0.2
20 21 28 29 PRIS YD 0.3 ZD 0.2
MEMBER PROPERTY AMERICAN
7 27 PRIS YD 0.35 ZD 0.25
8 18 23 PRIS YD 0.6 ZD 0.25
CONSTANTS
MATERIAL CONCRETE ALL
SUPPORTS

9 TO 12 FIXED
21 22 FIXED
MEMBER OFFSET
1 4 START 0.2 0 0
1 4 END -0.2 0 0
7 8 START 0 0 0.125
7 18 END 0 0 -0.125
25 26 START -0.15 0 0
27 START 0 0 0.1
27 END 0 0 -0.1
25 26 END 0.125 0 0
MEMBER CRACKED
1 4 7 TO 12 17 18 22 23 25 TO 27 30 TO 31 -
32 REDUCTION RIX 0.350000 RIY 0.350000 RIZ 0.350000
2 3 5 6 13 TO 16 20 21 28 29 REDUCTION RIX 0.700000 RIY 0.700000 RIZ 0.700000
DEFINE WIND LOAD
TYPE 1 WINDWARD WALL X
<! STAAD PRO GENERATED DATA DO NOT MODIFY !!!
ASCE-7-2010:PARAMS 177.000 KMPH 0 3 1 0 0.000 FT 0.000 FT 0.000 FT 1 -
1 5.000 M 5.000 M 5.000 M 5.000 0.010 0 -
0 0 0 0 1.046 1.000 1.000 0.850 0 -
0 0 0 0.898 0.800 -0.550
!> END GENERATED DATA BLOCK
INT 1.65812 1.65812 1.65929 1.66044 1.66159 1.66274 1.66388 1.66501 1.66613 -
1.66725 1.66836 1.66947 1.67057 1.67166 1.67275 HEIG 0 4.572 4.60492 -
4.63785 4.67077 4.70369 4.73662 4.76954 4.80246 4.83538 4.86831 -
4.90123 4.93415 4.96708 5
TYPE 2 LEEDWARD WALL X
<! STAAD PRO GENERATED DATA DO NOT MODIFY !!!
ASCE-7-2010:PARAMS 177.000 KMPH 0 3 1 0 0.000 FT 0.000 FT 0.000 FT 1 -
1 3.000 M 5.000 M 5.000 M 5.000 0.010 0 -
0 0 0 0 1.030 1.000 1.000 0.850 0 -
0 0 0 0.901 0.800 -0.550
!> END GENERATED DATA BLOCK
INT 1.65037 1.65037 HEIG 0 4.572
TYPE 3 SIDEWALL X
<! STAAD PRO GENERATED DATA DO NOT MODIFY !!!
ASCE-7-2010:PARAMS 177.000 KMPH 0 3 1 0 0.000 FT 0.000 FT 0.000 FT 1 -
1 3.000 M 5.000 M 5.000 M 5.000 0.010 0 -

0 0 0 0 1.030 1.000 1.000 0.850 0 -
0 0 0 0.901 0.800 -0.550
!> END GENERATED DATA BLOCK
INT 1.65037 1.65037 HEIG 0 4.572
TYPE 4 WINDWARD WALL Z
<! STAAD PRO GENERATED DATA DO NOT MODIFY !!!
ASCE-7-2010:PARAMS 177.000 KMPH 0 3 1 0 0.000 FT 0.000 FT 0.000 FT 1 -
1 3.000 M 5.000 M 5.000 M 5.000 0.010 0 -
0 0 0 0 1.030 1.000 1.000 0.850 0 -
0 0 0 0.901 0.800 -0.550
!> END GENERATED DATA BLOCK
INT 1.65037 1.65037 HEIG 0 4.572
TYPE 5 LEEDWARD Z
<! STAAD PRO GENERATED DATA DO NOT MODIFY !!!
ASCE-7-2010:PARAMS 177.000 KMPH 0 3 1 0 0.000 FT 0.000 FT 0.000 FT 1 -
1 3.000 M 5.000 M 5.000 M 5.000 0.010 0 -
0 0 0 0 1.030 1.000 1.000 0.850 0 -
0 0 0 0.901 0.800 -0.550
!> END GENERATED DATA BLOCK
INT 1.65037 1.65037 HEIG 0 4.572
TYPE 6 SIDEWALL Z
<! STAAD PRO GENERATED DATA DO NOT MODIFY !!!
ASCE-7-2010:PARAMS 177.000 KMPH 0 3 1 0 0.000 FT 0.000 FT 0.000 FT 1 -
1 5.000 M 5.000 M 5.000 M 2.000 0.010 2 -
0 0 0 0 1.046 1.000 1.000 0.850 0 -
0 0 0 0.898 -0.700 0.550
!> END GENERATED DATA BLOCK
INT -1.55432 -1.55432 HEIG 0 5
<! STAAD PRO GENERATED DATA DO NOT MODIFY !!!
ASCE-7-2010:PARAMS 177.000 KMPH 0 3 1 0 0.000 FT 0.000 FT 0.000 FT 1 -
1 40.000 FT 30.000 FT 25.000 FT 2.000 0.010 0 -
0 0 0 0 1.222 1.000 1.000 0.850 0 -
0 0 0 0.885 0.800 -0.550
!> END GENERATED DATA BLOCK
<! STAAD PRO GENERATED DATA DO NOT MODIFY !!!
ASCE-7-2010:PARAMS 177.000 KMPH 0 3 1 0 0.000 FT 0.000 FT 0.000 FT 1 -
1 40.000 FT 30.000 FT 25.000 FT 2.000 0.010 0 -
0 0 0 0 1.222 1.000 1.000 0.850 0 -
0 0 0 0.885 0.800 -0.550

```
!> END GENERATED DATA BLOCK
<! STAAD PRO GENERATED DATA DO NOT MODIFY !!!
ASCE-7-2010:PARAMS 177.000 KMPH 0 3 1 0 0.000 FT 0.000 FT 0.000 FT 1 -
1 40.000 FT 30.000 FT 25.000 FT 2.000 0.010 0 -
0 0 0 0 1.222 1.000 1.000 0.850 0 -
0 0 0 0.885 0.800 -0.550
!> END GENERATED DATA BLOCK
<! STAAD PRO GENERATED DATA DO NOT MODIFY !!!
ASCE-7-2010:PARAMS 177.000 KMPH 0 3 1 0 0.000 FT 0.000 FT 0.000 FT 1 -
1 40.000 FT 30.000 FT 25.000 FT 2.000 0.010 0 -
0 0 0 0 1.222 1.000 1.000 0.850 0 -
0 0 0 0.885 0.800 -0.550
!> END GENERATED DATA BLOCK
<! STAAD PRO GENERATED DATA DO NOT MODIFY !!!
ASCE-7-2010:PARAMS 177.000 KMPH 0 3 1 0 0.000 FT 0.000 FT 0.000 FT 1 -
1 5.000 M 5.000 M 5.000 M 2.000 0.010 0 -
0 0 0 0 1.046 1.000 1.000 0.850 0 -
0 0 0 0.898 0.800 -0.550
!> END GENERATED DATA BLOCK
LOAD 1 LOADTYPE Dead  TITLE DEAD LOAD
SELFWEIGHT Y -1
MEMBER LOAD
8 18 27 UMOM GZ -1
7 UMOM GZ 1
1 26 UMOM GX -1
4 25 UMOM GX 1
1 4 UNI GY -14.55
9 TO 12 17 22 30 TO 32 UNI GY -10
25 26 UNI GY -8.58
LOAD 2 LOADTYPE Live REDUCIBLE TITLE LIVE LOAD
MEMBER LOAD
1 4 23 27 UNI GY -1
8 18 27 UMOM GZ -1
7 UMOM GZ 1
1 UMOM GX -1
4 UMOM GX 1
LOAD 3 LOADTYPE Wind  TITLE WL+X
WIND LOAD X 1 TYPE 1 YR 0 3.5
WIND LOAD -X 1 TYPE 2 YR 0 3.5
```

WIND LOAD -Z 1 TYPE 3 YR 0 3.5
WIND LOAD -Z -1 TYPE 3 YR 0 3.5
LOAD 4 LOADTYPE Wind TITLE WL+Z
WIND LOAD Z 1 TYPE 4 YR 0 3.5
WIND LOAD -Z 1 TYPE 5 YR 0 3.5
WIND LOAD X 1 TYPE 6 YR 0 3.5
WIND LOAD X -1 TYPE 6 YR 0 3.5
LOAD 12 LOADTYPE Temperature TITLE TEMP
TEMPERATURE LOAD
1 TO 8 18 20 21 23 25 TO 27 TEMP 25
LOAD COMB 5 SLS: (DL)
1 1.0
LOAD COMB 6 GENERATED ACI TABLE1 1
1 1.4
LOAD COMB 7 GENERATED ACI TABLE1 2
1 1.2 2 1.6
LOAD COMB 8 GENERATED ACI TABLE1 3
1 1.2 2 1.0
LOAD COMB 9 GENERATED ACI TABLE1 4
1 1.2
LOAD COMB 10 GENERATED ACI TABLE1 5
1 0.9
LOAD COMB 11 ULTIMATE
1 1.2 2 1.6
LOAD COMB 13 SLS: (DL+T)
1 1.0 12 1.0
LOAD COMB 14 SLS: (D+LR)
1 1.0 2 1.0
LOAD COMB 15 SLS: (D+0.75LR)
1 1.0 2 0.75
LOAD COMB 16 SLS: (D+0.75WX+0.75LR)
1 1.0 2 0.75 3 0.75
LOAD COMB 17 SLS: (D+0.75WZ+0.75LR)
1 1.0 2 0.75 4 0.75
LOAD COMB 18 SLS: (0.6D+WX)
1 0.6 3 1.0
LOAD COMB 19 SLS: (0.6D+WZ)
1 0.6 4 1.0
LOAD COMB 20 SLS: (D+WX)

1 1.0 3 1.0
LOAD COMB 21 SLS: (D+WZ)
1 1.0 4 1.0
PERFORM ANALYSIS
START CONCRETE DESIGN
CODE ACI
FC 35000 ALL
FYMAIN 420000 ALL
FYSEC 420000 ALL
DESIGN BEAM 1 4 7 TO 12 17 18 22 23 25 TO 27 30 TO 32
DESIGN COLUMN 2 3 5 6 13 TO 16 20 21 28 29
CONCRETE TAKE
MAXMAIN 20 ALL
MINMAIN 10 ALL
CLB 0.04 MEMB 1 4 7 TO 12 17 18 22 23 25 TO 27 30 TO 32
CLS 0.04 MEMB 2 3 5 6 13 TO 16 20 21 28 29
EFACE 0.44 MEMB 1 4 25 26
SFACE 0.44 MEMB 1 4 25 26
EFACE 0.33 MEMB 7 8 18 23 27
SFACE 0.33 MEMB 7 8 18 23 27
END CONCRETE DESIGN
FINISH

ATTACHMENT 3 FOUNDATION DESIGN CALCULATION

PROPRIETARY & CONFIDENTIAL

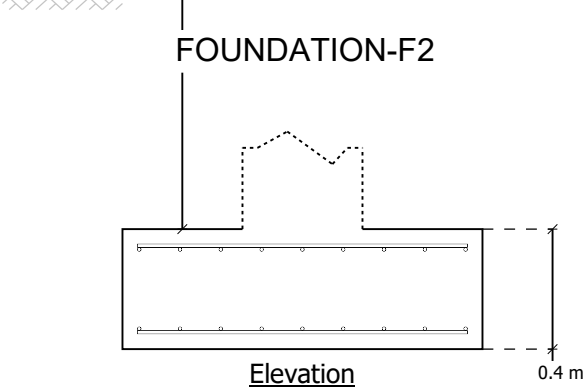
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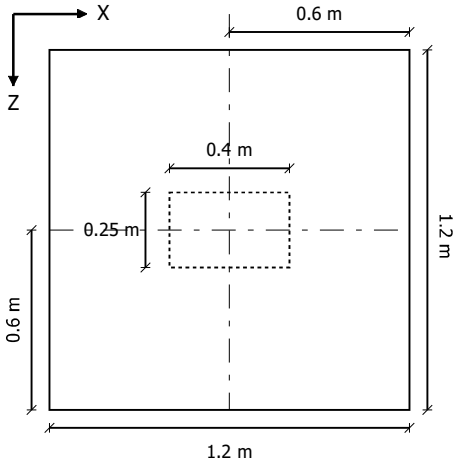
Isolated Footing Design(ACI 318-11)

Footing No.	Group ID	Foundation Geometry		
-	-	Length	Width	Thickness
9	1	1.200m	1.200m	0.400m

Footing No.	Footing Reinforcement				Pedestal Reinforcement	
-	Bottom Reinforcement(M_z)	Bottom Reinforcement(M_x)	Top Reinforcement(M_z)	Top Reinforcement(M_x)	Main Steel	Trans Steel
9	9 - ϕ12	9 - ϕ12	9 - ϕ12	9 - ϕ12	N/A	N/A



Elevation



Plan

Input Values

Footing Geomtery

Design Type : Calculate Dimension
Footing Thickness (Ft) : 400.000mm
Footing Length - X (Fl) : 1200.000mm
Footing Width - Z (Fw) : 1200.000mm
Eccentricity along X (Oxd) : 0.000mm
Eccentricity along Z (Ozd) : 0.000mm

Column Dimensions

Column Shape : Rectangular
Column Length - X (D_{col}) : 0.400m

Column Width - Z (B_{col}) : 0.250m

Pedestal

Include Pedestal? No
Pedestal Shape : N/A
Pedestal Height (Ph) : N/A
Pedestal Length - X (Pl) : N/A
Pedestal Width - Z (Pw) : N/A

Design Parameters

Concrete and Rebar Properties

Unit Weight of Concrete : 24.000kN/m3
Strength of Concrete : 35.000N/mm2
Yield Strength of Steel : 415.000N/mm2
Minimum Bar Size : #12
Maximum Bar Size : #12
Top Footing Minimum Bar Size : #12
Top Footing Maximum Bar Size : #12
Pedestal Minimum Bar Size : #12
Pedestal Maximum Bar Size : #22
Minimum Bar Spacing : 120.000mm
Maximum Bar Spacing : 150.000mm
Pedestal Clear Cover (P, CL) : 50.000mm
Bottom Footing Clear Cover (F, CL) : 50.000mm

Soil Properties

Soil Type : Cohesionless Soil
Unit Weight : 18.000kN/m3
Soil Bearing Capacity : 350.000kN/m2
Soil Bearing Capacity Type: Net Bearing Capacity
Soil Surcharge : 0.000kN/m2
Depth of Soil above Footing : 800.000mm
Type of Depth : Fixed Top
Undrained Shear Strength : 0.000kN/m2
Bearing Capacity Input Method: Fixed Bearing Capacity

Sliding and Overturning

Coefficient of Friction : 0.500
Factor of Safety Against Sliding : 1.500
Factor of Safety Against Overturning : 1.500

Global Settings

Top Reinforcement Option : Always calculate based on self weight
Concrete Design Option : Gross Pressure
Top Reinforcement Factor : 1.000

Design Calculations

Footing Size

Initial Length (L_o) = 1.200m

Initial Width (W_o) = 1.200m

Load Combinations

Load Combination/s- Service Stress Level				
Load Combination Number	Load Combination Title	Load Combination Factor	Soil Bearing Factor	Self Weight Factor
5	SLS: (DL)	1.00	1.00	1.00
6	GENERATED ACI TABLE1 1	1.00	1.00	1.00
7	GENERATED ACI TABLE1 2	1.00	1.00	1.00
8	GENERATED ACI TABLE1 3	1.00	1.00	1.00
9	GENERATED ACI TABLE1 4	1.00	1.00	1.00
10	GENERATED ACI TABLE1 5	1.00	1.00	1.00
11	ULTIMATE	1.00	1.00	1.00
13	SLS: (DL+T)	1.00	1.00	1.00
14	SLS: (D+LR)	1.00	1.00	1.00
15	SLS: (D+0.75LR)	1.00	1.00	1.00
16	SLS: (D+0.75WX+0.75LR)	1.00	1.00	1.00
17	SLS: (D+0.75WZ+0.75LR)	1.00	1.00	1.00
18	SLS: (0.6D+WX)	1.00	1.00	1.00
19	SLS: (0.6D+WZ)	1.00	1.00	1.00
20	SLS: (D+WX)	1.00	1.00	1.00
21	SLS: (D+WZ)	1.00	1.00	1.00

Load Combination/s- Strength Level				
Load Combination Number	Load Combination Title	Load Combination Factor	Soil Bearing Factor	Self Weight Factor
5	SLS: (DL)	1.00	1.00	1.00
6	GENERATED ACI TABLE1 1	1.00	1.00	1.00
7	GENERATED ACI TABLE1 2	1.00	1.00	1.00
8	GENERATED ACI TABLE1 3	1.00	1.00	1.00
9	GENERATED ACI TABLE1 4	1.00	1.00	1.00
10	GENERATED ACI TABLE1 5	1.00	1.00	1.00
11	ULTIMATE	1.00	1.00	1.00
13	SLS: (DL+T)	1.00	1.00	1.00
14	SLS: (D+LR)	1.00	1.00	1.00
15	SLS: (D+0.75LR)	1.00	1.00	1.00
16	SLS: (D+0.75WX+0.75LR)	1.00	1.00	1.00
17	SLS: (D+0.75WZ+0.75LR)	1.00	1.00	1.00
18	SLS: (0.6D+WX)	1.00	1.00	1.00
19	SLS: (0.6D+WZ)	1.00	1.00	1.00
20	SLS: (D+WX)	1.00	1.00	1.00
21	SLS: (D+WZ)	1.00	1.00	1.00

Applied Loads

Applied Loads - Service Stress Level					
LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
5	138.498	-14.837	-14.406	-5.294	5.739
6	193.897	-20.772	-20.168	-7.411	8.035
7	170.429	-17.404	-17.274	-6.348	6.806
8	168.842	-17.554	-17.279	-6.350	6.836
9	166.198	-17.805	-17.287	-6.352	6.887
10	124.648	-13.353	-12.965	-4.764	5.165
11	170.429	-17.404	-17.274	-6.348	6.806
13	139.505	-10.105	-12.467	-4.642	4.820
14	141.143	-14.587	-14.398	-5.291	5.688
15	140.482	-14.649	-14.400	-5.292	5.701
16	135.145	-7.023	-14.579	-5.427	-0.787
17	133.907	-13.852	-5.147	1.235	5.197
18	75.984	1.266	-8.883	-3.357	-5.207
19	74.333	-7.840	3.693	5.526	2.772
20	131.383	-4.669	-14.645	-5.474	-2.911
21	129.732	-13.775	-2.069	3.408	5.068

Applied Loads - Strength Level					
LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
5	138.498	-14.837	-14.406	-5.294	5.739
6	193.897	-20.772	-20.168	-7.411	8.035
7	170.429	-17.404	-17.274	-6.348	6.806
8	168.842	-17.554	-17.279	-6.350	6.836
9	166.198	-17.805	-17.287	-6.352	6.887
10	124.648	-13.353	-12.965	-4.764	5.165
11	170.429	-17.404	-17.274	-6.348	6.806
13	139.505	-10.105	-12.467	-4.642	4.820
14	141.143	-14.587	-14.398	-5.291	5.688
15	140.482	-14.649	-14.400	-5.292	5.701
16	135.145	-7.023	-14.579	-5.427	-0.787
17	133.907	-13.852	-5.147	1.235	5.197
18	75.984	1.266	-8.883	-3.357	-5.207
19	74.333	-7.840	3.693	5.526	2.772
20	131.383	-4.669	-14.645	-5.474	-2.911
21	129.732	-13.775	-2.069	3.408	5.068

Reduction of force due to buoyancy = 0.000kN

Effect due to adhesion = 0.000kN

Area from initial length and width, $A_o = L_o \times W_o = 1.440 m^2$

Min. area required from bearing pressure, $A_{min} = P / q_{max} = 0.654 m^2$

Note: A_{min} is an initial estimation.
P = Critical Factored Axial Load(without self weight/buoyancy/soil).
q_{max} = Respective Factored Bearing Capacity.

Final Footing Size

Length (L_2) = 1.200 m Governing Load Case : # 5

Width (W_2) = 1.200 m Governing Load Case : # 5

Depth (D_2) = 0.400 m Governing Load Case : # 6

Depth is governed by Ultimate Load Case

(Service check is performed with footing thickness requirements from concrete check)

Area (A_2) = 1.440 m^2

Final Soil Height = 0.800 m

Footing Self Weight = 13.824 kN

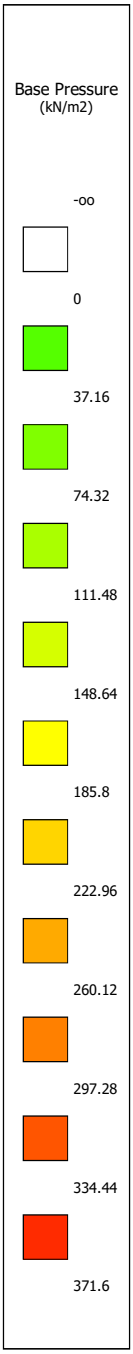
Gross Soil Bearing Capacity = 371.60 kN/m2

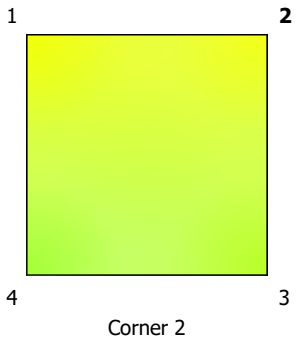
Soil Weight On Top Of Footing = 19.295 kN

Pressures at 4 Corners

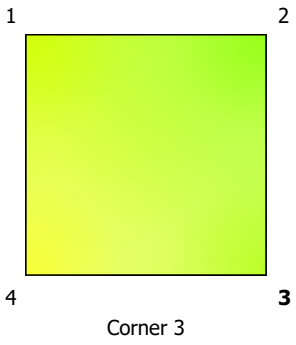
Please note that pressures values displayed in tables below are calculated after dividing by soil bearing factor

Governing Base Pressure Contour for Each Corner

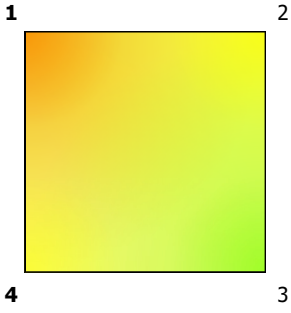




Corner 2

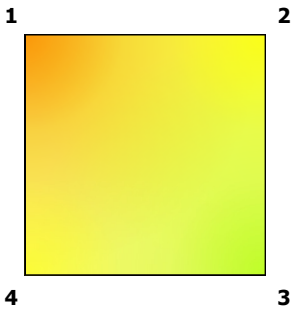


Corner 3



Corner 1, 4

Governing Base Pressure Contour



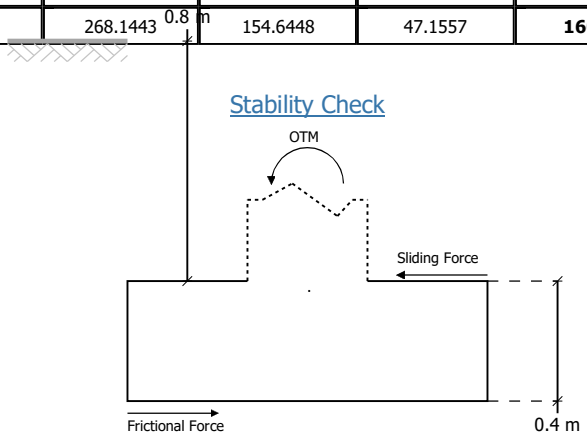
Corner 1, 2, 3, 4

Load Case	Pressure at corner 1 (q_1) (kN/m2)	Pressure at corner 2 (q_2) (kN/m2)	Pressure at corner 3 (q_3) (kN/m2)	Pressure at corner 4 (q_4) (kN/m2)	Area of footing in uplift (A_u) (m^2)
6	**268.1443**	154.6448	47.1557	160.6552	0.000
20	149.9628	**157.2080**	78.5116	71.2664	0.000
21	140.8583	67.4024	**85.3236**	158.7794	0.000
6	268.1443	154.6448	47.1557	**160.6552**	0.000

If A_u is zero, there is no uplift and no pressure adjustment is necessary. Otherwise, to account for uplift, areas of negative pressure will be set to zero and the pressure will be redistributed to remaining corners.

Summary of Adjusted Pressures at 4 corners Four Corners

Load Case	Pressure at corner 1 (q_1) (kN/m2)	Pressure at corner 2 (q_2) (kN/m2)	Pressure at corner 3 (q_3) (kN/m2)	Pressure at corner 4 (q_4) (kN/m2)
6	**268.1443**	154.6448	47.1557	160.6552
20	149.9628	**157.2080**	78.5116	71.2664
21	140.8583	67.4024	**85.3236**	158.7794
6	268.1443	154.6448	47.1557	**160.6552**



Stability Check

-	Factor of safety against sliding			Factor of safety against overturning			
Load Case No.	Along X-Direction	Along Z-Direction	Resultant	Required FOS	About X-Direction	About Z-Direction	Required FOS
5	5.783	5.956	5.868	1.500	9.367	8.835	1.500
6	5.464	5.628	5.544	1.500	8.838	8.345	1.500
7	5.848	5.892	5.870	1.500	9.257	8.884	1.500
8	5.753	5.844	5.798	1.500	9.182	8.757	1.500
9	5.597	5.765	5.679	1.500	9.059	8.549	1.500
10	5.907	6.084	5.994	1.500	9.573	9.026	1.500
11	5.848	5.892	5.870	1.500	9.257	8.884	1.500
13	8.541	6.923	7.606	1.500	10.818	11.707	1.500
14	5.973	6.052	6.012	1.500	9.516	9.089	1.500
15	5.925	6.028	5.976	1.500	9.479	9.025	1.500
16	11.979	5.771	7.352	1.500	9.020	49.999	1.500
17	6.029	16.225	7.992	1.500	122.318	9.349	1.500
18	43.092	6.141	8.598	1.500	9.560	11.489	1.500
19	6.853	14.548	8.768	1.500	9.291	10.942	1.500
20	17.617	5.616	7.567	1.500	8.762	94.769	1.500
21	5.911	39.347	8.267	1.500	38.093	9.254	1.500

Critical Load Case And The Governing Factor Of Safety For Overturning And Sliding - X Direction

Critical Load Case for Sliding along X-Direction : 6
Governing Disturbing Force : -20.772kN
Governing Restoring Force : 113.508kN
Minimum Sliding Ratio for the Critical Load Case : 5.464
Critical Load Case for Overturning about X-Direction : 20
Governing Overturning Moment : -11.332kNm
Governing Resisting Moment : 99.293kNm
Minimum Overturning Ratio for the Critical Load Case : 8.762

Critical Load Case And The Governing Factor Of Safety For Overturning And Sliding - Z Direction

Critical Load Case for Sliding along Z-Direction :	20
Governing Disturbing Force :	-14.645kN
Governing Restoring Force :	82.251kN
Minimum Sliding Ratio for the Critical Load Case :	5.616
Critical Load Case for Overturning about Z-Direction :	6
Governing Overturning Moment :	16.344kNm
Governing Resisting Moment :	136.380kNm
Minimum Overturning Ratio for the Critical Load Case :	8.345

Critical Load Case And The Governing Factor Of Safety For Sliding Along Resultant Direction

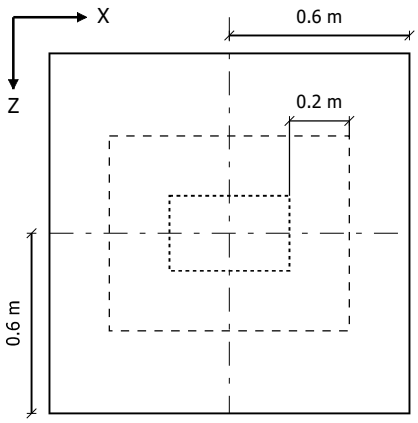
Critical Load Case for Sliding along Resultant Direction :	6
Governing Disturbing Force :	28.952kN
Governing Restoring Force :	160.525kN
Minimum Sliding Ratio for the Critical Load Case :	5.544

Compression Development Length Check

Development length skipped as column reinforcement is not specified in input (Column Dimnesion Task Pane)

Shear Calculation

Punching Shear Check



Plan

Total Footing Depth, D = 0.400m

Calculated Effective Depth, d_{eff} = D - C_{cover} - 0.5 * d_b = 0.344m

For rectangular column, β_c = B_{col} / D_{col} = 1.600

Effective depth, d_{eff}, increased until 0.75XV_c $\geq$ Punching Shear Force

Punching Shear Force, Vu = 157.345kN, Load Case # 6

From ACI Cl.11.11.2, b_o for column= $2 \times (B_{col} + D_{col} + 2 \times d_{eff})$ = 2.676m

Equation 11-31, V_{c1} = $0.17 \times \left(1 + \frac{2}{\beta}\right) \times \lambda \times \sqrt{f_c'} \times b_o \times d_{eff}$ 2034.936kN

Equation 11-32, V_{c2} = $0.083 \times \left(\frac{\alpha_s \times d}{b_o} + 2\right) \times \lambda \times \left(\sqrt{f_c'} \times b_o \times d_{eff}\right)$ 3229.670kN

Equation 11-33, V_{c3} = $0.33 \times \lambda \times \sqrt{f_c'} \times b_o \times d_{eff}$ 1808.832kN

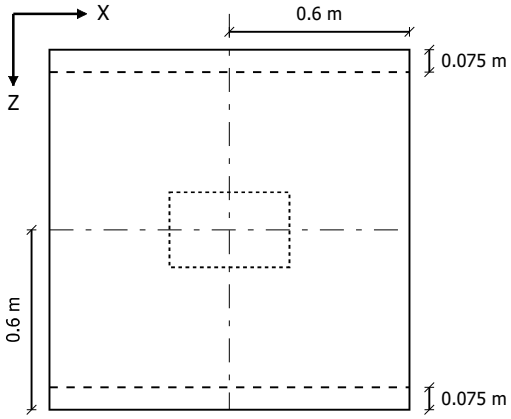
Punching shear strength, V_c = 0.75 X minimum of (V_{c1}, V_{c2}, V_{c3}) = 1356.624kN

0.75 X V_c > V_u hence, OK

One-Way Shear Check

One-Way Shear Along X

(Shear Plane Parallel to Global X Axis)



Plan

From ACI Cl.11.2.1.1, V_c = $2 \times L \times d_{eff} \times \sqrt{1000 \times F_c'}$ = 405.568kN

Distance along X to design for shear, D_x = $0.5 \times (W \pm D_{col}) - d_{eff} + O_{zd}$ = 0.131m

Check that 0.75 X V_c > V_{ux} where V_{ux} is the shear force for the critical load cases at a distance d_{eff} from the face of the column caused by bending about the X axis.

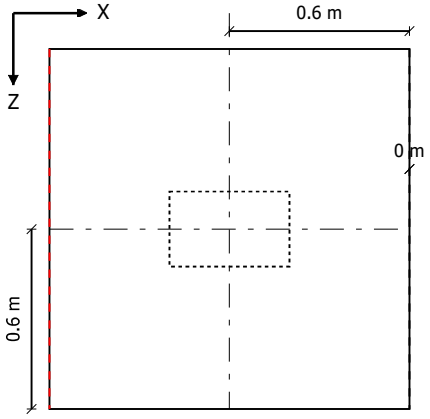
From above calculations, 0.75 X V_c = 304.176 kN

Critical load case for V_{ux} is # 6 $V_{ux} = V_{ux}|_{x=D_x}$ = 32.309 kN

0.75 X V_c > V_{ux} hence, OK

One-Way Shear Along Z

(Shear Plane Parallel to Global Z Axis)



Plan

From ACI Cl.11.2.1.1, V_c =

$2 \times W \times d_{eff} \times \sqrt{1000 \times F_c'}$ = 405.568 kN

Distance along X to design for shear, D_z = $0.5 \times (L \pm B_{col}) - d_{eff} + O_{xd}$ = 0.056 m

Check that **0.75 X V_c > V_{uz}** where V_{uz} is the shear force for the critical load cases at a distance d_{eff} from the face of the column caused by bending about the Z axis.

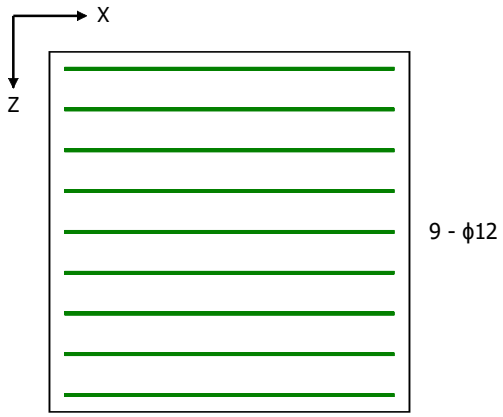
From above calculations, 0.75 X V_c = 304.176 kN

Critical load case for V_{uz} is # 6 $V_{uz} = V_{uz}|_{z=D_z}$ = 14.230 kN

0.75 X V_c > V_{uz} hence, OK

Flexure About Z-Axis

Design For Bottom Reinforcement Parallel to X Axis



Calculate the flexural reinforcement along the X direction of the footing. Find the area of steel required, A, as per Section 3.8 of Reinforced Concrete Design (5th ed.) by Salmon and Wang (Ref. 1)

Critical Load Case # 6

The strength values of steel and concrete used in the formulae are in ksi

Bars parallel to X Direction are placed at bottom

Effective Depth d_{eff} =		0.344 m
Factor β_1 from ACI Cl.10.2.7.3 =		0.796
From ACI Cl. 10.3.2, ρ_{bal} =	$0.85 \times \beta_1 \times f_c' \times \left(\frac{600}{600 + f_y}\right)$	0.03374
From ACI Cl. 10.3.3, ρ_{max} =	$0.75 \times \rho_{bal}$ =	0.02530
From ACI Cl. 7.12.2, ρ_{min} =		0.00194
From Ref. 1, Eq. 3.8.4a, constant m =	$\frac{F_y}{\left(0.85 \times F_c'\right)}$ =	13.950

Calculate reinforcement ratio ρ for critical load case

Design for flexure about Z axis is performed at the face of the column at a distance, D_x =	$0.5 \times L \pm 0.5 \times D_{col} + O_{xd}$ =	0.400 m
Ultimate moment,	$M_u\big\|_{z=D_z}$ =	19.370 kNm
Nominal moment capacity, M_n =	$\frac{M_u}{\Phi}$ =	21.523 kNm
(Based on effective depth) Required ρ =	$\frac{1}{m} \times \left[1 - \sqrt{1 - 2 \times m \times \frac{M_n}{\left(F_y \times W \times d_{eff}^2\right)}}\right]$ =	0.00037
	(Based on gross depth) ρ x d_{eff} / Depth =	0.00031
Since	ρ= ρ_{min}	ρ_{min} Governs
Area of Steel Required, A_s =	$\rho \times W \times d_{eff}$ =	929.531 mm2

Selected bar Size = #12

Minimum spacing allowed (S_{min}) = = 120.000mm

Selected spacing (S) = 136.000mm

S_{min}<= S <= S_{max} and selected bar size < selected maximum bar size...

The reinforcement is accepted.

Based on spacing reinforcement increment; provided reinforcement is

#12 @ 135mm o.c.

Required development length for bars =	$\frac{d_b \times f_y}{20 \times \lambda \times \sqrt{f_c}}$ =	=0.406	m
Available development length for bars, D_L =	$0.5 \times (L - D_{col}) \cdot - C_{cover}$ =	0.350	m
Try bar size # 12	Area of one bar =	113.097	mm2
Number of bars required, N_{bar} =	$\frac{A_s}{A_{bar}}$ =	9	

Because the number of bars is rounded up, make sure new reinforcement ratio < ρ_{max}

Total reinforcement area, A_{s_total} =	N_{bar} X (Area of one bar) =	1017.869	mm2
d_{eff} =	D - C_{cover} - 0.5 X (dia. of one bar) =	0.344	m
Reinforcement ratio, ρ =	$\frac{A_{s_total}}{(d_{eff} \times W)}$ =	0.00247	

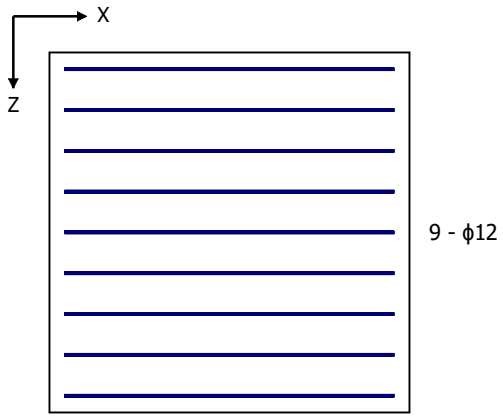
From ACI Cl.7.6.1, minimum req'd clear distance between bars

C_d = max (Diameter of one bar, 1.0" (25.4mm), Min. User Spacing) = 120.000mm

Provided Steel Area / Required Steel Area = 1.095

Check to see if width is sufficient to accomodate bars

Design For Top Reinforcement Parallel to X Axis



Top reinforcement is calculated based on self weight of footing and soil

Calculate the flexural reinforcement for M_z. Find the area of steel required

The strength values of steel and concrete used in the formulae are in ksi

Bars parallel to X Direction are placed at bottom

Effective Depth d_{eff} =		0.344	m
Factor β_1 from ACI Cl.10.2.7.3 =		0.796	
From ACI Cl. 10.3.2, ρ_{bal} =	$0.85 \times \beta_1 \times f_c' \times \left(\frac{600}{600 + f_y}\right)$	0.03374	
From ACI Cl. 10.3.3, ρ_{max} =	$0.75 \times \rho_{bal}$ =	0.02530	
From ACI Cl.7.12.2, ρ_{min} =		0.00000	
From Ref. 1, Eq. 3.8.4a, constant m =	$\frac{F_y}{(0.85 \times F_c')}$ =	13.950	

Calculate reinforcement ratio ρ for critical load case

$0.5 \times L \pm 0.5 \times D_{col} + O_{xd}$ =	0.400	m

Design for flexure about Z axis is performed at the face of the column at a distance, D_x =			
Ultimate moment,	$M_u\|_{x=D_x}$ =	2.304	kNm
Nominal moment capacity, M_n =	$\frac{M_u}{\Phi}$ =	2.560	kNm
(Based on effective depth) Required ρ =	$\frac{1}{m} \times \left[1 - \sqrt{1 - 2 \times m \times \frac{M_n}{\left(F_y \times W \times d_{eff}^2\right)}}\right]$ =	0.000043	
	(Based on gross depth) ρ x d_{eff} / Depth =	0.000037	
Since	$\rho_{min} < \rho < \rho_{max}$	OK	
Area of Steel Required, A_s =	$\rho \times W \times d_{eff}$ =	17.937	mm2
Total reinforcement area, A_{s_total} =	N_{bar} X (Area of one bar) =	1017.876	mm2

Provided Steel Area / Required Steel Area = 56.747

Selected bar Size = #12

Minimum spacing allowed (S_{min}) = 120.000mm

Selected spacing (S) = 150.000mm

S_{min}<= S <= S_{max} and selected bar size < selected maximum bar size...

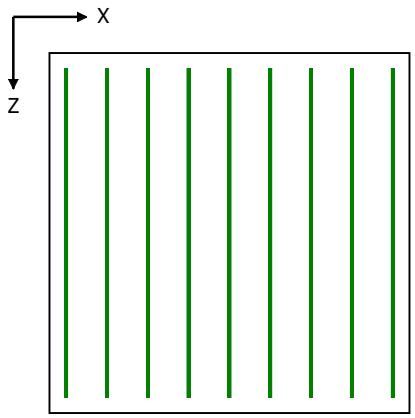
The reinforcement is accepted.

Based on spacing reinforcement increment; provided reinforcement is

#12 @ 150mm o.c.

Flexure About X-Axis

Design For Bottom Reinforcement Parallel to Z Axis



9 - ϕ12

Calculate the flexural reinforcement along the Z direction of the footing. Find the area of steel required, A, as per Section 3.8 of Reinforced Concrete Design (5th ed.) by Salmon and Wang (Ref. 1)

Critical Load Case # 6

The strength values of steel and concrete used in the formulae are in ksi

Bars parallel to X Direction are placed at bottom

Effective Depth d_{eff} =		0.332	m
Factor β_1 from ACI Cl.10.2.7.3 =		0.796	
From ACI Cl. 10.3.2, ρ_{bal} =	$0.85 \times \beta_1 \times f_c' \times \left(\frac{600}{600 + f_y}\right)$	0.03374	
From ACI Cl. 10.3.3, ρ_{max} =	$0.75 \times \rho_{bal}$ =	0.02530	

From ACI Cl.7.12.2, ρ_{min} = 0.00196

From Ref. 1, Eq. 3.8.4a, constant m = $\frac{F_y}{\left(0.85 \times F_c'\right)}$ = 13.950

Calculate reinforcement ratio ρ for critical load case

Design for flexure about X axis is performed at the face of the column at a distance, D_z = $0.5 \times L \pm 0.5 \times B_{col} + O_{zd}$ = 0.475 m

Ultimate moment, $M_u|_{x=D_x}$ = 26.696 kNm

Nominal moment capacity, M_n = $\frac{M_u}{\Phi}$ = 29.662 kNm

(Based on effective depth) Required ρ = $\frac{1}{m} \times \left[1 - \sqrt{1 - 2 \times m \times \frac{M_n}{\left(F_y \times W \times d_{eff}^2\right)}}\right]$ = 0.00054

(Based on gross depth) ρ x d_{eff} / Depth = 0.00045

Since ρ= ρ_{min} ρ_{min} Governs

Area of Steel Required, A_s = $\rho \times W \times d_{eff}$ = 939.852 mm2

Selected Bar Size = #12

Minimum spacing allowed (S_{min}) = 120.000mm

Selected spacing (S) = 136.000mm

S_{min}<= S <= S_{max} and selected bar size < selected maximum bar size...

The reinforcement is accepted.

Based on spacing reinforcement increment; provided reinforcement is

#12 @ 135mm o.c.

Required development length for bars = $\frac{d_b \times f_y}{20 \times \lambda \times \sqrt{f_c}}$ = =0.406 m

Available development length for bars, D_L = $0.5 \times \left(L - D_{col}\right) - C_{cover}$ = 0.425 m

Try bar size # 12 Area of one bar = 113.097 mm2

Number of bars required, N_{bar} = $\frac{A_s}{A_{bar}}$ = 9

Because the number of bars is rounded up, make sure new reinforcement ratio < ρ_{max}

Total reinforcement area, A_{s_total} = N_{bar} X (Area of one bar) = 1017.869 mm2

d_{eff} = D - C_{cover} - 1.5 X (dia. of one bar) = 0.332 m

Reinforcement ratio, ρ = $\frac{A_{s_total}}{\left(d_{eff} \times W\right)}$ = 0.00255

From ACI Cl.7.6.1, minimum req'd clear distance between bars

C_d = max (Diameter of one bar, 1.0" (25.4mm), Min. User Spacing) = 120.000mm

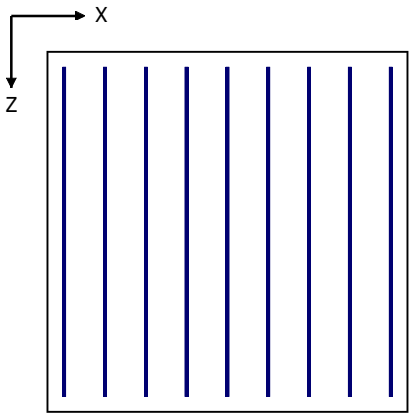
Provided Steel Area / Required Steel Area = 1.083

Check to see if width is sufficient to accomodate bars

Bending moment for uplift cases will be calculated based solely on selfweight, soil depth and surcharge loading.

As the footing size has already been determined based on all servicebility load cases, and design moment calculation is based on selfweight, soil depth and surcharge only, top reinforcement value for all pure uplift load cases will be the same.

Design For Top Reinforcement Parallel to Z Axis



9 - ϕ12

Top reinforcement is calculated based on self weight of footing and soil

Calculate the flexural reinforcement for M_x. Find the area of steel required

The strength values of steel and concrete used in the formulae are in ksi

Bars parallel to X Direction are placed at bottom

Effective Depth d_{eff} =		0.332	m
Factor β_1 from ACI Cl.10.2.7.3 =		0.796	
From ACI Cl. 10.3.2, ρ_{bal} =	$0.85 \times \beta_1 \times f_c' \times \left(\frac{600}{600 + f_y}\right)$	0.03374	
From ACI Cl. 10.3.3, ρ_{max} =	$0.75 \times \rho_{bal}$ =	0.02530	
From ACI Cl. 7.12.2, ρ_{min} =		0.00000	
From Ref. 1, Eq. 3.8.4a, constant m =	$\frac{F_y}{\left(0.85 \times F_c'\right)}$ =	13.950	

Calculate reinforcement ratio ρ for critical load case

Design for flexure about X axis is performed at the face of the column at a distance, D_x =	$0.5 \times L \pm 0.5 \times D_{col} + O_{xd}$ =	0.475	m
Ultimate moment,	$M_u\big\vert_{z=D_z}$ =	3.249	kNm
Nominal moment capacity, M_n =	$\frac{M_u}{\Phi}$ =	3.610	kNm
(Based on effective depth) Required ρ =	$\frac{1}{m} \times \left[1 - \sqrt{1 - 2 \times m \times \frac{M_n}{\left(F_y \times W \times d_{eff}^2\right)}}\right]$ =	0.00007	
	(Based on gross depth) ρ x d_{eff} / Depth =	0.00005	
Since	$\rho_{min} < \rho < \rho_{max}$	OK	
Area of Steel Required, A_s =	$\rho \times W \times d_{eff}$ =	26.212	mm2
Total reinforcement area, A_{s_total} =	N_{bar} X (Area of one bar) =	1017.876	mm2

Provided Steel Area / Required Steel Area = 38.832

Selected bar Size = #12

Minimum spacing allowed (S_{min}) = 120.000mm

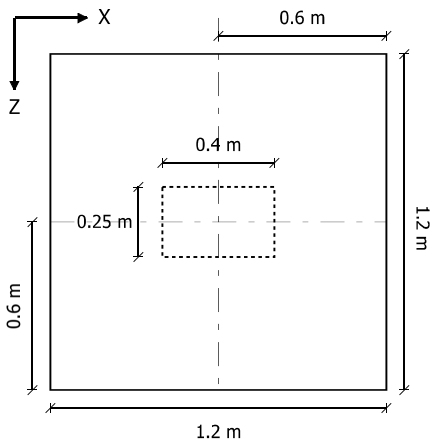
Selected spacing (S) = 150.000mm

S_{min}<= S <= S_{max} and selected bar size < selected maximum bar size...

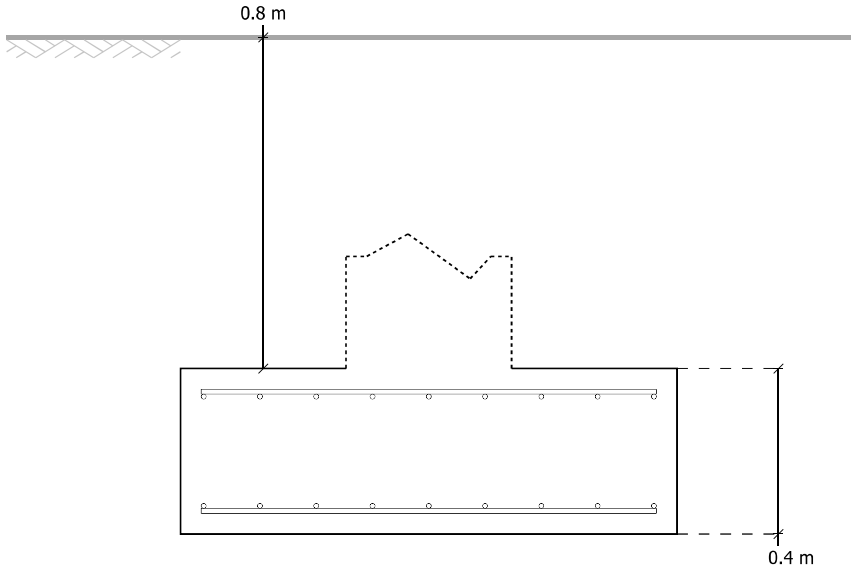
The reinforcement is accepted.

Based on spacing reinforcement increment; provided reinforcement is

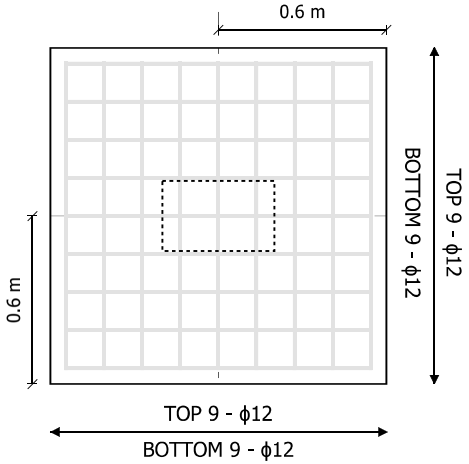
#12 @ 150mm o.c.



Plan



Elevation



Plan

	Dimensions (m)			Reinforcement			
	L	B	D	AstX(T)	AstZ(T)	AstX(B)	AstZ(B)
Foundation	1.2 m	1.2 m	0.4 m	9 - ϕ12	9 - ϕ12	9 - ϕ12	9 - ϕ12
	l	b	h	Main Reinforcement		Trans Reinforcement	
Pedestal	N/A	N/A	N/A	N/A		N/A	

Print Calculation Sheet

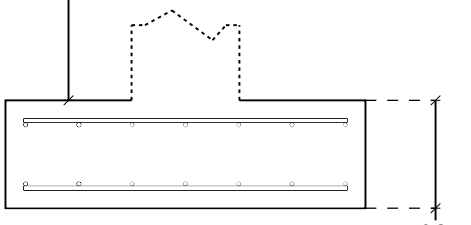
file:///C:/Staad%20Foundation%20Advanced%20SS3/CalcXsl/footing.xml

Isolated Footing Design(ACI 318-11)

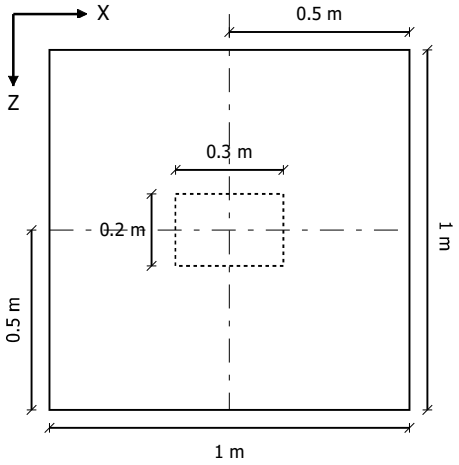
Footing No.	Group ID	Foundation Geometry		
-	-	Length	Width	Thickness
21	1	1.000m	1.000m	0.300m

Footing No.	Footing Reinforcement				Pedestal Reinforcement	
-	Bottom Reinforcement(M_z)	Bottom Reinforcement(M_x)	Top Reinforcement(M_z)	Top Reinforcement(M_x)	Main Steel	Trans Steel
21	7 - ϕ12	7 - ϕ12	7 - ϕ12	7 - ϕ12	N/A	N/A

FOUNDATION-F4



Elevation



Plan

Input Values

Footing Geomtery

Design Type : Calculate Dimension
Footing Thickness (Ft) : 300.000mm
Footing Length - X (Fl) : 1000.000mm
Footing Width - Z (Fw) : 1000.000mm
Eccentricity along X (Oxd) : 0.000mm
Eccentricity along Z (Ozd) : 0.000mm

Column Dimensions

Column Shape : Rectangular
Column Length - X (D_{col}) : 0.300m

Column Width - Z (B_{col}) : 0.200m

Pedestal

Include Pedestal? No
Pedestal Shape : N/A
Pedestal Height (Ph) : N/A
Pedestal Length - X (Pl) : N/A
Pedestal Width - Z (Pw) : N/A

Design Parameters

Concrete and Rebar Properties

Unit Weight of Concrete : 24.000kN/m3
Strength of Concrete : 35.000N/mm2
Yield Strength of Steel : 415.000N/mm2
Minimum Bar Size : #12
Maximum Bar Size : #12
Top Footing Minimum Bar Size : #12
Top Footing Maximum Bar Size : #12
Pedestal Minimum Bar Size : #12
Pedestal Maximum Bar Size : #22
Minimum Bar Spacing : 120.000mm
Maximum Bar Spacing : 150.000mm
Pedestal Clear Cover (P, CL) : 50.000mm
Bottom Footing Clear Cover (F, CL) : 50.000mm

Soil Properties

Soil Type : Cohesionless Soil
Unit Weight : 18.000kN/m3
Soil Bearing Capacity : 350.000kN/m2
Soil Bearing Capacity Type: Net Bearing Capacity
Soil Surcharge : 0.000kN/m2
Depth of Soil above Footing : 900.000mm
Type of Depth : Fixed Top
Undrained Shear Strength : 0.000kN/m2
Bearing Capacity Input Method: Fixed Bearing Capacity

Sliding and Overturning

Coefficient of Friction : 0.500
Factor of Safety Against Sliding : 1.500
Factor of Safety Against Overturning : 1.500

Global Settings

Top Reinforcement Option : Always calculate based on self weight
Concrete Design Option : Gross Pressure
Top Reinforcement Factor : 1.000

Design Calculations

Footing Size

Initial Length (L_o) = 1.000m

Initial Width (W_o) = 1.000m

Load Combinations

Load Combination/s- Service Stress Level				
Load Combination Number	Load Combination Title	Load Combination Factor	Soil Bearing Factor	Self Weight Factor
5	SLS: (DL)	1.00	1.00	1.00
6	GENERATED ACI TABLE1 1	1.00	1.00	1.00
7	GENERATED ACI TABLE1 2	1.00	1.00	1.00
8	GENERATED ACI TABLE1 3	1.00	1.00	1.00
9	GENERATED ACI TABLE1 4	1.00	1.00	1.00
10	GENERATED ACI TABLE1 5	1.00	1.00	1.00
11	ULTIMATE	1.00	1.00	1.00
13	SLS: (DL+T)	1.00	1.00	1.00
14	SLS: (D+LR)	1.00	1.00	1.00
15	SLS: (D+0.75LR)	1.00	1.00	1.00
16	SLS: (D+0.75WX+0.75LR)	1.00	1.00	1.00
17	SLS: (D+0.75WZ+0.75LR)	1.00	1.00	1.00
18	SLS: (0.6D+WX)	1.00	1.00	1.00
19	SLS: (0.6D+WZ)	1.00	1.00	1.00
20	SLS: (D+WX)	1.00	1.00	1.00
21	SLS: (D+WZ)	1.00	1.00	1.00

Load Combination/s- Strength Level				
Load Combination Number	Load Combination Title	Load Combination Factor	Soil Bearing Factor	Self Weight Factor
5	SLS: (DL)	1.00	1.00	1.00
6	GENERATED ACI TABLE1 1	1.00	1.00	1.00
7	GENERATED ACI TABLE1 2	1.00	1.00	1.00
8	GENERATED ACI TABLE1 3	1.00	1.00	1.00
9	GENERATED ACI TABLE1 4	1.00	1.00	1.00
10	GENERATED ACI TABLE1 5	1.00	1.00	1.00
11	ULTIMATE	1.00	1.00	1.00
13	SLS: (DL+T)	1.00	1.00	1.00
14	SLS: (D+LR)	1.00	1.00	1.00
15	SLS: (D+0.75LR)	1.00	1.00	1.00
16	SLS: (D+0.75WX+0.75LR)	1.00	1.00	1.00
17	SLS: (D+0.75WZ+0.75LR)	1.00	1.00	1.00
18	SLS: (0.6D+WX)	1.00	1.00	1.00
19	SLS: (0.6D+WZ)	1.00	1.00	1.00
20	SLS: (D+WX)	1.00	1.00	1.00
21	SLS: (D+WZ)	1.00	1.00	1.00

Applied Loads

Applied Loads - Service Stress Level					
LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
5	44.406	2.776	2.277	0.756	-0.027
6	62.169	3.886	3.188	1.059	-0.037
7	55.011	3.368	2.683	0.888	-0.060
8	54.365	3.354	2.702	0.895	-0.049
9	53.288	3.331	2.733	0.908	-0.032
10	39.966	2.498	2.050	0.681	-0.024
11	55.011	3.368	2.683	0.888	-0.060
13	48.888	2.070	2.091	0.762	-0.480
14	45.483	2.799	2.246	0.744	-0.044
15	45.214	2.793	2.254	0.747	-0.040
16	49.011	5.970	2.442	0.889	-2.815
17	47.165	1.455	3.411	1.625	0.895
18	31.706	5.901	1.617	0.642	-3.717
19	29.245	-0.118	2.908	1.625	1.230
20	49.469	7.011	2.528	0.945	-3.728
21	47.007	0.992	3.819	1.927	1.220

Applied Loads - Strength Level					
LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
5	44.406	2.776	2.277	0.756	-0.027
6	62.169	3.886	3.188	1.059	-0.037
7	55.011	3.368	2.683	0.888	-0.060
8	54.365	3.354	2.702	0.895	-0.049
9	53.288	3.331	2.733	0.908	-0.032
10	39.966	2.498	2.050	0.681	-0.024
11	55.011	3.368	2.683	0.888	-0.060
13	48.888	2.070	2.091	0.762	-0.480
14	45.483	2.799	2.246	0.744	-0.044
15	45.214	2.793	2.254	0.747	-0.040
16	49.011	5.970	2.442	0.889	-2.815
17	47.165	1.455	3.411	1.625	0.895
18	31.706	5.901	1.617	0.642	-3.717
19	29.245	-0.118	2.908	1.625	1.230
20	49.469	7.011	2.528	0.945	-3.728
21	47.007	0.992	3.819	1.927	1.220

Reduction of force due to buoyancy = 0.000kN

Effect due to adhesion = 0.000kN

Area from initial length and width, $A_o = L_o \times W_o = 1.000 m^2$

Min. area required from bearing pressure, $A_{min} = P / q_{max} = 0.232 m^2$

Note: A_{min} is an initial estimation.
P = Critical Factored Axial Load(without self weight/buoyancy/soil).
q_{max} = Respective Factored Bearing Capacity.

Final Footing Size

Length (L_2) = 1.000 m Governing Load Case : # 5

Width (W_2) = 1.000 m Governing Load Case : # 5

Depth (D_2) = 0.300 m Governing Load Case : # 20

Depth is governed by Ultimate Load Case

(Service check is performed with footing thickness requirements from concrete check)

Area (A_2) = 1.000 m^2

Final Soil Height = 0.900 m

Footing Self Weight = 7.200 kN

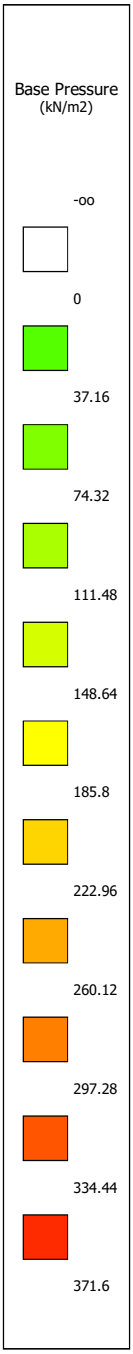
Gross Soil Bearing Capacity = 371.60 kN/m2

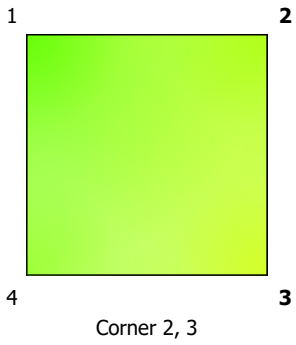
Soil Weight On Top Of Footing = 15.227 kN

Pressures at 4 Corners

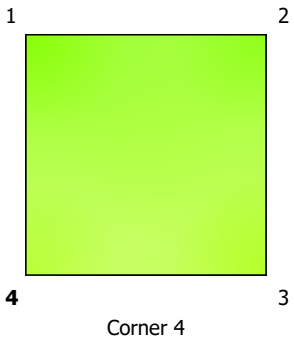
Please note that pressures values displayed in tables below are calculated after dividing by soil bearing factor

Governing Base Pressure Contour for Each Corner

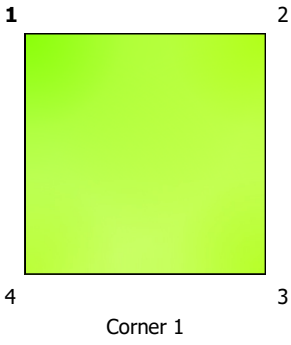




Corner 2, 3

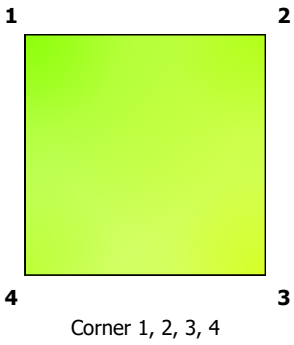


Corner 4



Corner 1

Governing Base Pressure Contour



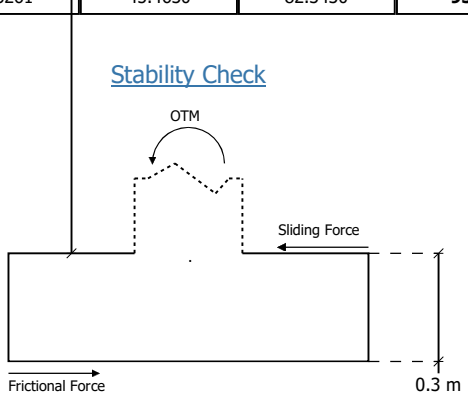
Corner 1, 2, 3, 4

Load Case	Pressure at corner 1 (q_1) (kN/m2)	Pressure at corner 2 (q_2) (kN/m2)	Pressure at corner 3 (q_3) (kN/m2)	Pressure at corner 4 (q_4) (kN/m2)	Area of footing in uplift (A_u) (m^2)
6	**65.2841**	79.7210	103.9079	89.4710	0.000
20	26.6877	**96.6641**	117.1039	47.1275	0.000
20	26.6877	96.6641	**117.1039**	47.1275	0.000
21	56.5261	45.4630	82.3430	**93.4061**	0.000

If A_u is zero, there is no uplift and no pressure adjustment is necessary. Otherwise, to account for uplift, areas of negative pressure will be set to zero and the pressure will be redistributed to remaining corners.

Summary of Adjusted Pressures at 4 corners Four Corners

Load Case	Pressure at corner 1 (q_1) (kN/m2)	Pressure at corner 2 (q_2) (kN/m2)	Pressure at corner 3 (q_3) (kN/m2)	Pressure at corner 4 (q_4) (kN/m2)
6	**65.2841**	79.7210	103.9079	89.4710
20	26.6877	**96.6641**	117.1039	47.1275
20	26.6877	96.6641	**117.1039**	47.1275
21	56.5261	45.4630	82.3430	**93.4061**



-	Factor of safety against sliding			Factor of safety against overturning			
Load Case No.	Along X-Direction	Along Z-Direction	Resultant	Required FOS	About X-Direction	About Z-Direction	Required FOS
5	12.040	14.673	13.163	1.500	23.436	39.028	1.500
6	10.885	13.266	11.901	1.500	21.146	35.259	1.500
7	11.496	14.431	12.716	1.500	23.062	36.302	1.500
8	11.448	14.212	12.608	1.500	22.697	36.495	1.500
9	11.366	13.853	12.427	1.500	22.100	36.830	1.500
10	12.488	15.220	13.654	1.500	24.326	40.494	1.500
11	11.496	14.431	12.716	1.500	23.062	36.302	1.500
13	17.229	17.050	17.139	1.500	25.893	32.498	1.500
14	12.132	15.117	13.381	1.500	24.174	38.569	1.500
15	12.109	15.005	13.326	1.500	23.988	38.681	1.500
16	5.983	14.626	7.832	1.500	22.233	7.780	1.500
17	23.907	10.203	13.271	1.500	13.260	76.163	1.500
18	4.587	16.736	6.256	1.500	24.295	4.954	1.500
19	218.788	8.883	12.552	1.500	10.475	20.509	1.500
20	5.127	14.219	6.821	1.500	21.295	6.185	1.500
21	34.992	9.090	12.442	1.500	11.402	37.789	1.500

Critical Load Case And The Governing Factor Of Safety For Overturning And Sliding - X Direction

Critical Load Case for Sliding along X-Direction : 18
Governing Disturbing Force : 5.901kN
Governing Restoring Force : 27.067kN
Minimum Sliding Ratio for the Critical Load Case : 4.587
Critical Load Case for Overturning about X-Direction : 19
Governing Overturning Moment : 2.497kNm
Governing Resisting Moment : 26.160kNm
Minimum Overturning Ratio for the Critical Load Case : 10.475

Critical Load Case And The Governing Factor Of Safety For Overturning And Sliding - Z Direction

Critical Load Case for Sliding along Z-Direction : 19
Governing Disturbing Force : 2.908kN
Governing Restoring Force : 25.836kN
Minimum Sliding Ratio for the Critical Load Case : 8.883
Critical Load Case for Overturning about Z-Direction : 18
Governing Overturning Moment : -5.488kNm
Governing Resisting Moment : 27.188kNm
Minimum Overturning Ratio for the Critical Load Case : 4.954

Critical Load Case And The Governing Factor Of Safety For Sliding Along Resultant Direction

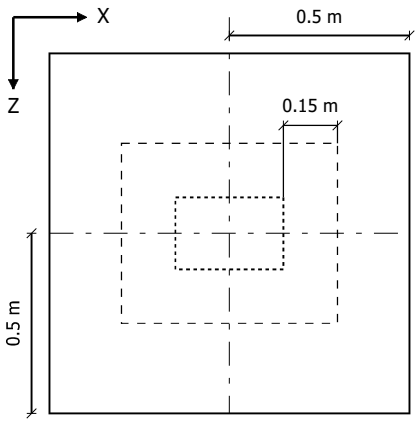
Critical Load Case for Sliding along Resultant Direction : 18
Governing Disturbing Force : 6.119kN
Governing Restoring Force : 38.278kN
Minimum Sliding Ratio for the Critical Load Case : 6.256

Compression Development Length Check

Development length skipped as column reinforcement is not specified in input (Column Dimnesion Task Pane)

Shear Calculation

Punching Shear Check



Plan

Total Footing Depth, D = 0.300m

Calculated Effective Depth, d_{eff} = D - C_{cover} - 0.5 * d_b = 0.244m

For rectangular column, β_c = B_{col} / D_{col} = 1.500

Effective depth, d_{eff}, increased until $0.75XV_c \geq$ Punching Shear Force

Punching Shear Force, Vu = 64.163kN, Load Case # 6

From ACI Cl.11.11.2, b_o for column= $2 \times (B_{col} + D_{col} + 2 \times d_{eff})$ = 1.976m

Equation 11-31, V_{c1} = $0.17 \times \left(1 + \frac{2}{\beta}\right) \times \lambda \times \sqrt{f_c'} \times b_o \times d_{eff}$ 1105.292kN

Equation 11-32, V_{c2} = $0.083 \times \left(\frac{\alpha_s \times d}{b_o} + 2\right) \times \lambda \times \left(\sqrt{f_c'} \times b_o \times d_{eff}\right)$ 1643.555kN

Equation 11-33, V_{c3} = $0.33 \times \lambda \times \sqrt{f_c'} \times b_o \times d_{eff}$ 947.393kN

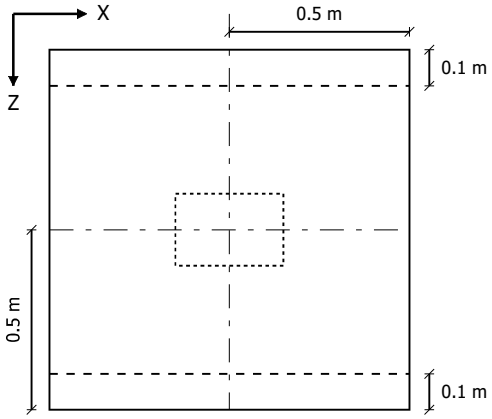
Punching shear strength, V_c = 0.75 X minimum of (V_{c1}, V_{c2}, V_{c3}) = 710.545kN

0.75 X V_c > V_u hence, OK

One-Way Shear Check

One-Way Shear Along X

(Shear Plane Parallel to Global X Axis)



Plan

From ACI Cl.11.2.1.1, V_c = $2 \times L \times d_{eff} \times \sqrt{1000 \times F_c'}$ = 239.725kN

Distance along X to design for shear, D_x = $0.5 \times (W \pm D_{col}) - d_{eff} + O_{zd}$ = 0.156m

Check that 0.75 X V_c > V_{ux} where V_{ux} is the shear force for the critical load cases at a distance d_{eff} from the face of the column caused by bending about the X axis.

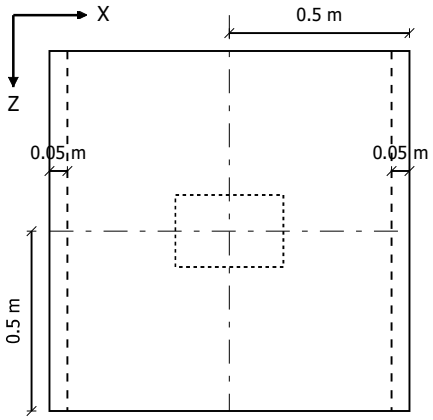
From above calculations, 0.75 X V_c = 179.794 kN

Critical load case for V_{ux} is # 6 $V_{ux} = V_{ux}|_{x=D_x}$ = 14.789 kN

0.75 X V_c > V_{ux} hence, OK

One-Way Shear Along Z

(Shear Plane Parallel to Global Z Axis)



Plan

From ACI Cl.11.2.1.1, V_c =

$2 \times W \times d_{eff} \times \sqrt{1000 \times F_c'}$ = 239.725 kN

Distance along X to design for shear, D_z = $0.5 \times (L \pm B_{col}) - d_{eff} + O_{xd}$ = 0.106 m

Check that **0.75 X V_c > V_{uz}** where V_{uz} is the shear force for the critical load cases at a distance d_{eff} from the face of the column caused by bending about the Z axis.

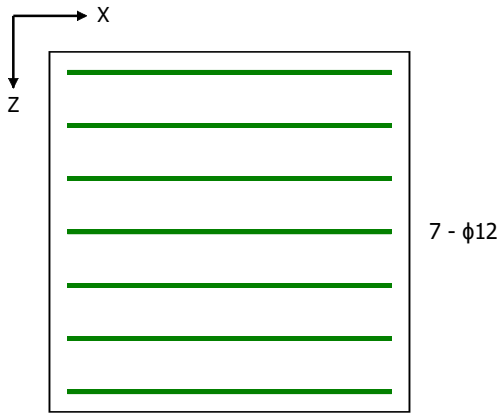
From above calculations, 0.75 X V_c = 179.794 kN

Critical load case for V_{uz} is # 20 $V_{uz} = V_{uz}|_{z=D_z}$ = 10.937 kN

0.75 X V_c > V_{uz} hence, OK

Flexure About Z-Axis

Design For Bottom Reinforcement Parallel to X Axis



Calculate the flexural reinforcement along the X direction of the footing. Find the area of steel required, A, as per Section 3.8 of Reinforced Concrete Design (5th ed.) by Salmon and Wang (Ref. 1)

Critical Load Case # 20

The strength values of steel and concrete used in the formulae are in ksi

Bars parallel to X Direction are placed at bottom

Effective Depth d_{eff} =		0.244 m
Factor β_1 from ACI Cl.10.2.7.3 =		0.796
From ACI Cl. 10.3.2, ρ_{bal} =	$0.85 \times \beta_1 \times f_c' \times \left(\frac{600}{600 + f_y}\right)$	0.03374
From ACI Cl. 10.3.3, ρ_{max} =	$0.75 \times \rho_{bal}$ =	0.02530
From ACI Cl. 7.12.2, ρ_{min} =		0.00191
From Ref. 1, Eq. 3.8.4a, constant m =	$\frac{F_y}{\left(0.85 \times F_c'\right)}$ =	13.950

Calculate reinforcement ratio ρ for critical load case

Design for flexure about Z axis is performed at the face of the column at a distance, D_x =	$0.5 \times L \pm 0.5 \times D_{col} + O_{xd}$ =	0.650 m
Ultimate moment,	$M_u\big\|_{z=D_z}$ =	6.049 kNm
Nominal moment capacity, M_n =	$\frac{M_u}{\Phi}$ =	6.721 kNm
(Based on effective depth) Required ρ =	$\frac{1}{m} \times \left[1 - \sqrt{1 - 2 \times m \times \frac{M_n}{\left(F_y \times W \times d_{eff}^2\right)}}\right]$ =	0.00027
	(Based on gross depth) ρ x d_{eff} / Depth =	0.00022
Since	ρ= ρ_{min}	ρ_{min} Governs
Area of Steel Required, A_s =	$\rho \times W \times d_{eff}$ =	573.440 mm2

Selected bar Size = #12

Minimum spacing allowed (S_{min}) = = 120.000mm

Selected spacing (S) = 150.000mm

S_{min}<= S <= S_{max} and selected bar size < selected maximum bar size...

The reinforcement is accepted.

Based on spacing reinforcement increment; provided reinforcement is

#12 @ 150mm o.c.

Required development length for bars =	$\frac{d_b \times f_y}{20 \times \lambda \times \sqrt{f_c}}$ =	=0.406	m
Available development length for bars, D_L =	$0.5 \times (L - D_{col}) - C_{cover}$ =	0.300	m
Try bar size # 12	Area of one bar =	113.097	mm2
Number of bars required, N_{bar} =	$\frac{A_s}{A_{bar}}$ =	7	

Because the number of bars is rounded up, make sure new reinforcement ratio < ρ_{max}

Total reinforcement area, A_{s_total} =	N_{bar} X (Area of one bar) =	791.676	mm2
d_{eff} =	D - C_{cover} - 0.5 X (dia. of one bar) =	0.244	m
Reinforcement ratio, ρ =	$\frac{A_{s_total}}{(d_{eff} \times W)}$ =	0.00324	

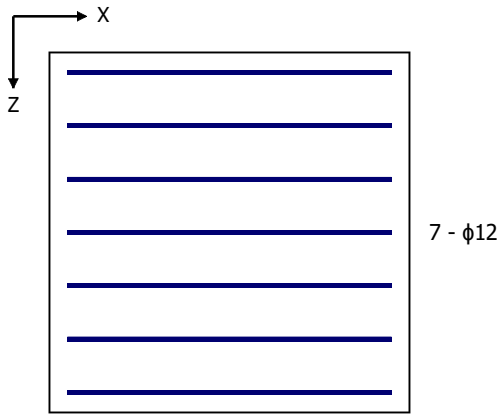
From ACI Cl.7.6.1, minimum req'd clear distance between bars

C_d = max (Diameter of one bar, 1.0" (25.4mm), Min. User Spacing) = 120.000mm

Provided Steel Area / Required Steel Area = 1.381

Check to see if width is sufficient to accomodate bars

Design For Top Reinforcement Parallel to X Axis



Top reinforcement is calculated based on self weight of footing and soil

Calculate the flexural reinforcement for M_z. Find the area of steel required

The strength values of steel and concrete used in the formulae are in ksi

Bars parallel to X Direction are placed at bottom

Effective Depth d_{eff} =		0.244	m
Factor β_1 from ACI Cl.10.2.7.3 =		0.796	
From ACI Cl. 10.3.2, ρ_{bal} =	$0.85 \times \beta_1 \times f_c' \times \left(\frac{600}{600 + f_y}\right)$	0.03374	
From ACI Cl. 10.3.3, ρ_{max} =	$0.75 \times \rho_{bal}$ =	0.02530	
From ACI Cl.7.12.2, ρ_{min} =		0.00000	
From Ref. 1, Eq. 3.8.4a, constant m =	$\frac{F_y}{(0.85 \times F_c')}$ =	13.950	

Calculate reinforcement ratio ρ for critical load case

	$0.5 \times L \pm 0.5 \times D_{col} + O_{xd}$ =	0.350	m

Design for flexure about Z axis is performed at the face of the column at a distance, D_x =			
Ultimate moment,	$M_u\|_{x=D_x}$ =	1.433	kNm
Nominal moment capacity, M_n =	$\frac{M_u}{\Phi}$ =	1.592	kNm
(Based on effective depth) Required ρ =	$\frac{1}{m} \times \left[1 - \sqrt{1 - 2 \times m \times \frac{M_n}{\left(F_y \times W \times d_{eff}^2\right)}}\right]$ =	0.000064	
	(Based on gross depth) ρ x d_{eff} / Depth =	0.000052	
Since	$\rho_{min} < \rho < \rho_{max}$	OK	
Area of Steel Required, A_s =	$\rho \times W \times d_{eff}$ =	15.733	mm2
Total reinforcement area, A_{s_total} =	N_{bar} X (Area of one bar) =	791.682	mm2

Provided Steel Area / Required Steel Area = 50.319

Selected bar Size = #12

Minimum spacing allowed (S_{min}) = 120.000mm

Selected spacing (S) = 150.000mm

S_{min}<= S <= S_{max} and selected bar size < selected maximum bar size...

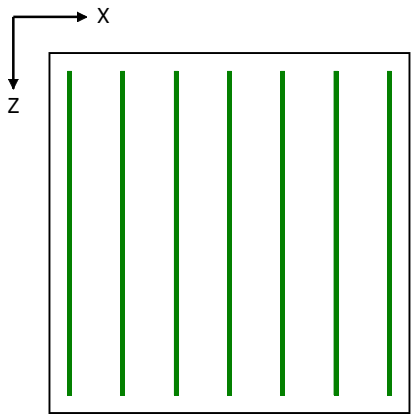
The reinforcement is accepted.

Based on spacing reinforcement increment; provided reinforcement is

#12 @ 150mm o.c.

Flexure About X-Axis

Design For Bottom Reinforcement Parallel to Z Axis



7 - ϕ12

Calculate the flexural reinforcement along the Z direction of the footing. Find the area of steel required, A, as per Section 3.8 of Reinforced Concrete Design (5th ed.) by Salmon and Wang (Ref. 1)

Critical Load Case # 6

The strength values of steel and concrete used in the formulae are in ksi

Bars parallel to X Direction are placed at bottom

Effective Depth d_{eff} =		0.232	m
Factor β_1 from ACI Cl.10.2.7.3 =		0.796	
From ACI Cl. 10.3.2, ρ_{bal} =	$0.85 \times \beta_1 \times f_c' \times \left(\frac{600}{600 + f_y}\right)$	0.03374	
From ACI Cl. 10.3.3, ρ_{max} =	$0.75 \times \rho_{bal}$ =	0.02530	

From ACI Cl.7.12.2, ρ_{min} = 0.00194

From Ref. 1, Eq. 3.8.4a, constant m = $\frac{F_y}{\left(0.85 \times F_c'\right)}$ = 13.950

Calculate reinforcement ratio ρ for critical load case

Design for flexure about X axis is performed at the face of the column at a distance, D_z = $0.5 \times L \pm 0.5 \times B_{col} + O_{zd}$ = 0.600 m

Ultimate moment, $M_u|_{x=D_x}$ = 7.478 kNm

Nominal moment capacity, M_n = $\frac{M_u}{\Phi}$ = 8.309 kNm

(Based on effective depth) Required ρ = $\frac{1}{m} \times \left[1 - \sqrt{1 - 2 \times m \times \frac{M_n}{\left(F_y \times W \times d_{eff}^2\right)}}\right]$ = 0.00037

(Based on gross depth) ρ x d_{eff} / Depth = 0.00029

Since ρ= ρ_{min} ρ_{min} Governs

Area of Steel Required, A_s = $\rho \times W \times d_{eff}$ = 581.574 mm2

Selected Bar Size = #12

Minimum spacing allowed (S_{min}) = 120.000mm

Selected spacing (S) = 150.000mm

S_{min}<= S <= S_{max} and selected bar size < selected maximum bar size...

The reinforcement is accepted.

Based on spacing reinforcement increment; provided reinforcement is

#12 @ 150mm o.c.

Required development length for bars = $\frac{d_b \times f_y}{20 \times \lambda \times \sqrt{f_c}}$ = =0.406 m

Available development length for bars, D_L = $0.5 \times \left(L - D_{col}\right) - C_{cover}$ = 0.350 m

Try bar size # 12 Area of one bar = 113.097 mm2

Number of bars required, N_{bar} = $\frac{A_s}{A_{bar}}$ = 7

Because the number of bars is rounded up, make sure new reinforcement ratio < ρ_{max}

Total reinforcement area, A_{s_total} = N_{bar} X (Area of one bar) = 791.676 mm2

d_{eff} = D - C_{cover} - 1.5 X (dia. of one bar) = 0.232 m

Reinforcement ratio, ρ = $\frac{A_{s_total}}{\left(d_{eff} \times W\right)}$ = 0.00341

From ACI Cl.7.6.1, minimum req'd clear distance between bars

C_d = max (Diameter of one bar, 1.0" (25.4mm), Min. User Spacing) = 120.000mm

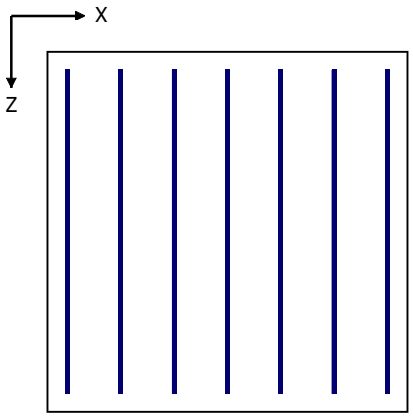
Provided Steel Area / Required Steel Area = 1.361

Check to see if width is sufficient to accomodate bars

Bending moment for uplift cases will be calculated based solely on selfweight, soil depth and surcharge loading.

As the footing size has already been determined based on all servicebility load cases, and design moment calculation is based on selfweight, soil depth and surcharge only, top reinforcement value for all pure uplift load cases will be the same.

Design For Top Reinforcement Parallel to Z Axis



7 - ϕ12

Top reinforcement is calculated based on self weight of footing and soil

Calculate the flexural reinforcement for M_x. Find the area of steel required

The strength values of steel and concrete used in the formulae are in ksi

Bars parallel to X Direction are placed at bottom

Effective Depth d_{eff} =		0.232	m
Factor β_1 from ACI Cl.10.2.7.3 =		0.796	
From ACI Cl. 10.3.2, ρ_{bal} =	$0.85 \times \beta_1 \times f_c' \times \left(\frac{600}{600 + f_y}\right)$	0.03374	
From ACI Cl. 10.3.3, ρ_{max} =	$0.75 \times \rho_{bal}$ =	0.02530	
From ACI Cl. 7.12.2, ρ_{min} =		0.00000	
From Ref. 1, Eq. 3.8.4a, constant m =	$\frac{F_y}{\left(0.85 \times F_c'\right)}$ =	13.950	

Calculate reinforcement ratio ρ for critical load case

Design for flexure about X axis is performed at the face of the column at a distance, D_x =	$0.5 \times L \pm 0.5 \times D_{col} + O_{xd}$ =	0.600	m
Ultimate moment,	$M_u\big\vert_{z=D_z}$ =	1.872	kNm
Nominal moment capacity, M_n =	$\frac{M_u}{\Phi}$ =	2.080	kNm
(Based on effective depth) Required ρ =	$\frac{1}{m} \times \left[1 - \sqrt{1 - 2 \times m \times \frac{M_n}{\left(F_y \times W \times d_{eff}^2\right)}}\right]$ =	0.00009	
	(Based on gross depth) ρ x d_{eff} / Depth =	0.00007	
Since	$\rho_{min} < \rho < \rho_{max}$	OK	
Area of Steel Required, A_s =	$\rho \times W \times d_{eff}$ =	21.617	mm2
Total reinforcement area, A_{s_total} =	N_{bar} X (Area of one bar) =	791.682	mm2

Provided Steel Area / Required Steel Area = 36.623

Selected bar Size = #12

Minimum spacing allowed (S_{min}) = 120.000mm

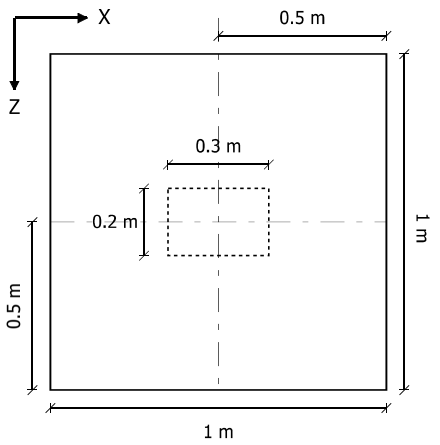
Selected spacing (S) = 150.000mm

S_{min}<= S <= S_{max} and selected bar size < selected maximum bar size...

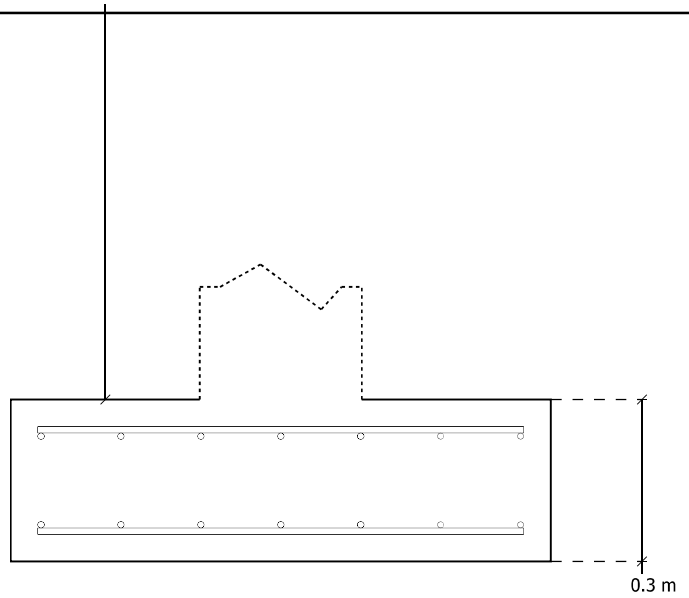
The reinforcement is accepted.

Based on spacing reinforcement increment; provided reinforcement is

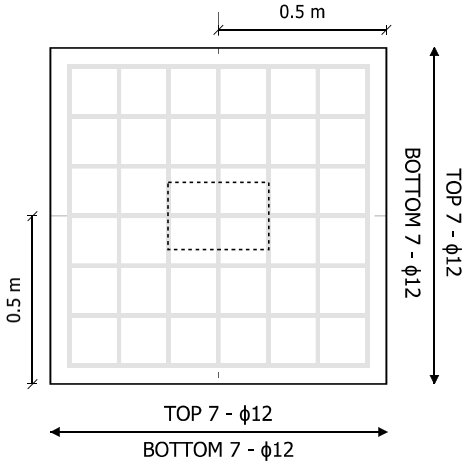
#12 @ 150mm o.c.



Plan



Elevation



Plan

	Dimensions (m)			Reinforcement			
	L	B	D	AstX(T)	AstZ(T)	AstX(B)	AstZ(B)
Foundation	1 m	1 m	0.3 m	7 - ϕ12	7 - ϕ12	7 - ϕ12	7 - ϕ12
	l	b	h	Main Reinforcement		Trans Reinforcement	
Pedestal	N/A	N/A	N/A	N/A		N/A	

Print Calculation Sheet

file:///C:/Staad%20Foundation%20Advanced%20SS3/CalcXsl/footing.xml